

Task Decomposition for Efficient Annotation of Structured Representations

Nupoor Gandhi

Guest Lecture for 11-811 10/8/26

Roadmap

- Motivating Annotation.
- Decomposition-based procedure for reducing the cost of annotation grounded in a model of inferential load.
- Grounding procedure in the literature.
- Demonstrations of procedure efficiency.
- Deciding when to decompose.
- Other considerations beyond cost.

Roadmap

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What is annotation?

- Transformation of input document(s) into a useful, narrower representation that you can do something with.
- Example transformations:

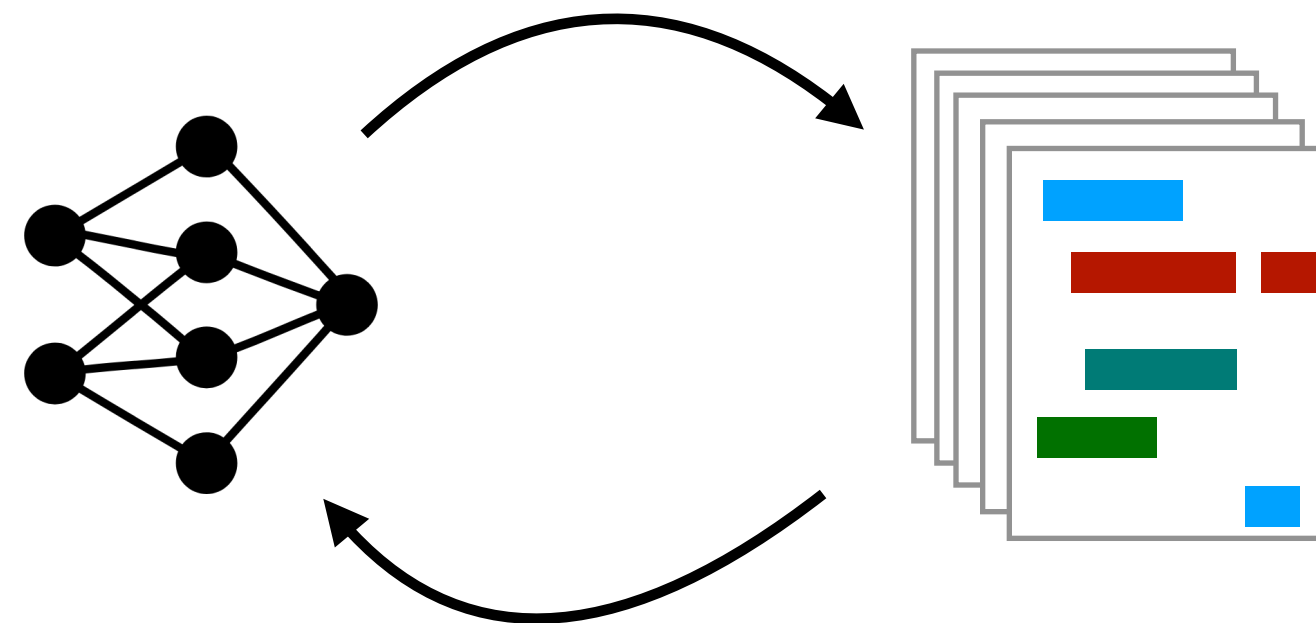
Sentence → Set of named entities

Document, question → Free-text answer

Realistic work task, computer interface → Recorded computer use trajectory

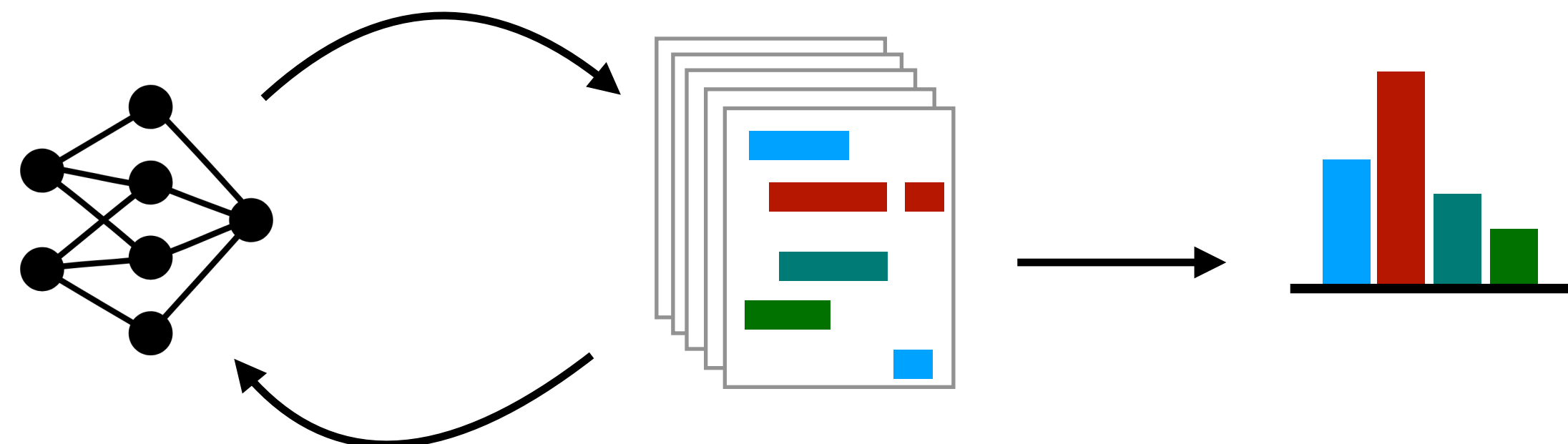
Why annotate?

- As a prerequisite for developing annotation systems
 - Improve the model by generating labels for supervision
 - Validate that the model works with an evaluation set



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- As a prerequisite for developing annotation systems
 - Improve the model by generating labels for supervision
 - Validate that the model works with an evaluation set
- For direct use in a broader pipeline or task

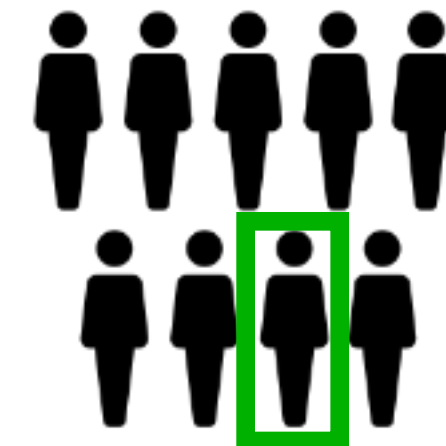


Example Annotation Use Cases

The American child welfare system faces limited resources, and ever growing demand for their services

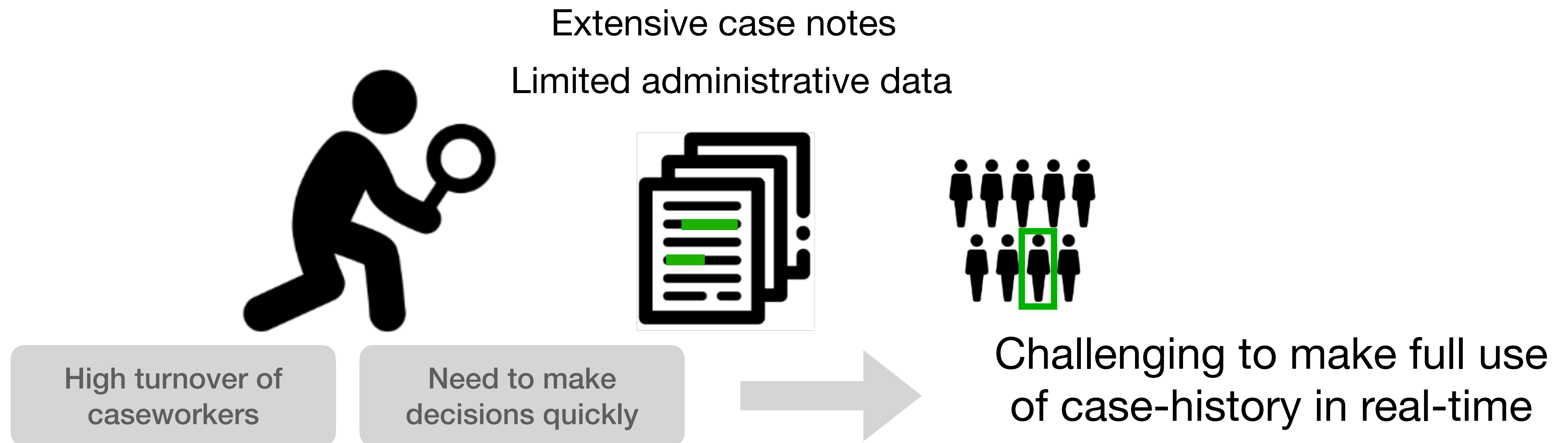
Extensive case notes

Limited administrative data



Example Annotation Use Cases

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Example Annotation Use Cases

Gray policy literature includes non-commercially distributed articles produced by governments and nonprofits.



White papers

Technical reports

Policy briefs

State planning documents

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White papers

Technical reports

Policy briefs

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Timely

Globally representative

(Yoshida et al, 2024, Lynch et al., 2021, Paez 2017)

Example Annotation Use Cases

For a given policy area, gray literature can be used to measure and characterize government action or inaction.



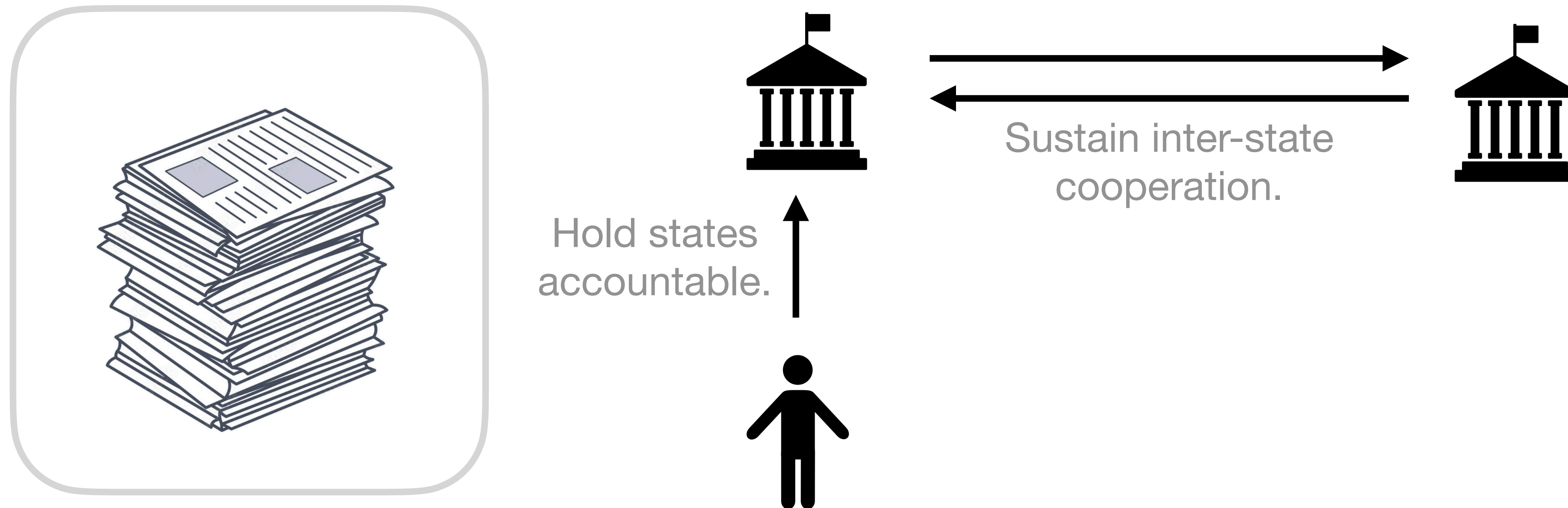
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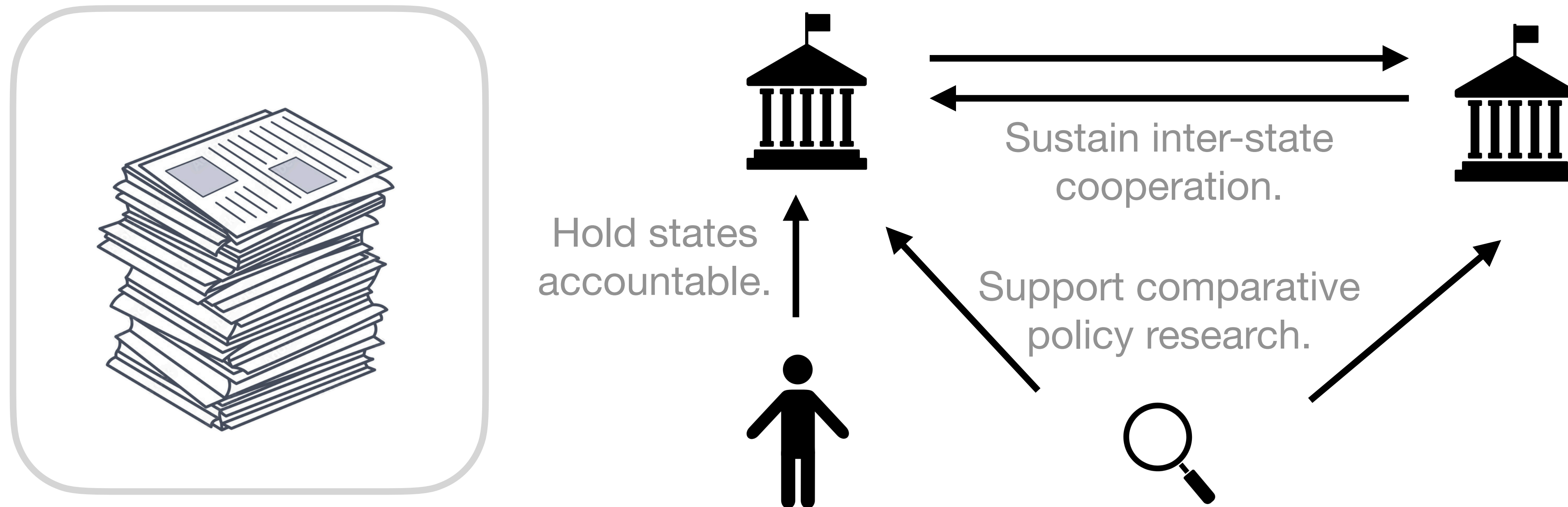
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Example Annotation Use Cases



Not widely indexed

Large volume

Complex, heterogeneous

(Lawrence et al., 2015; Rothstein and Hopewell 2009; Pandita and Singh 2011)

Example Annotation Use Cases



Gray policy literature is underutilized by experts.

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Annotation of Structured Representations

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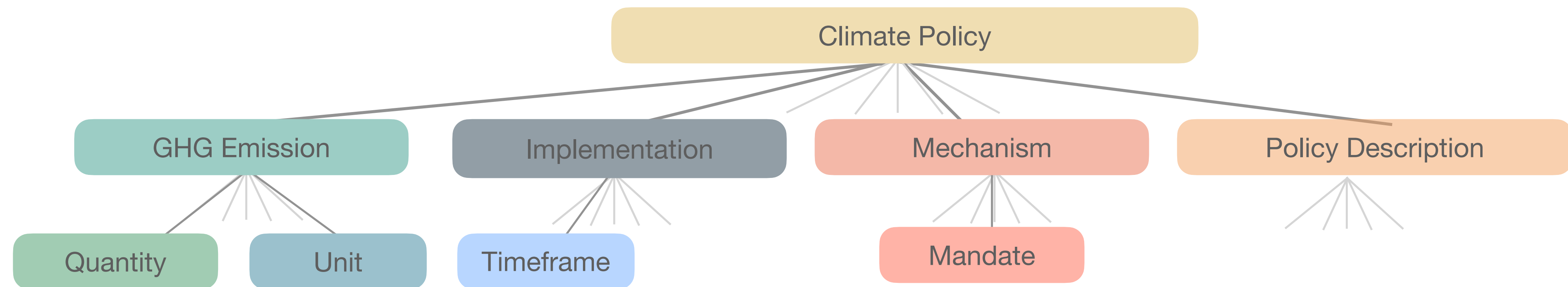
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Annotation of Structured Representations

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- Enables **both richly descriptive and controlled, fully-specified** corpus analysis and information extraction.

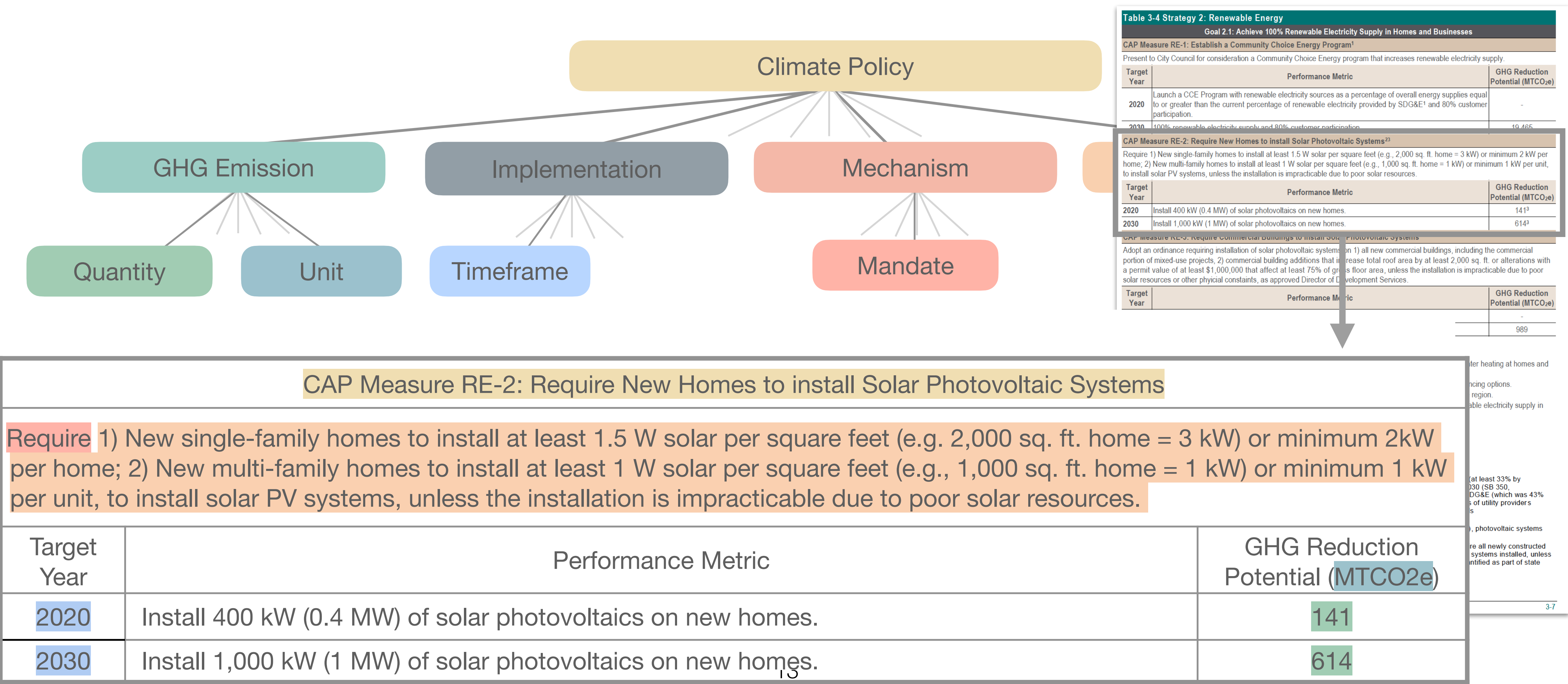
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Example set of concepts to structure climate policies.

Annotation of Structured Representations

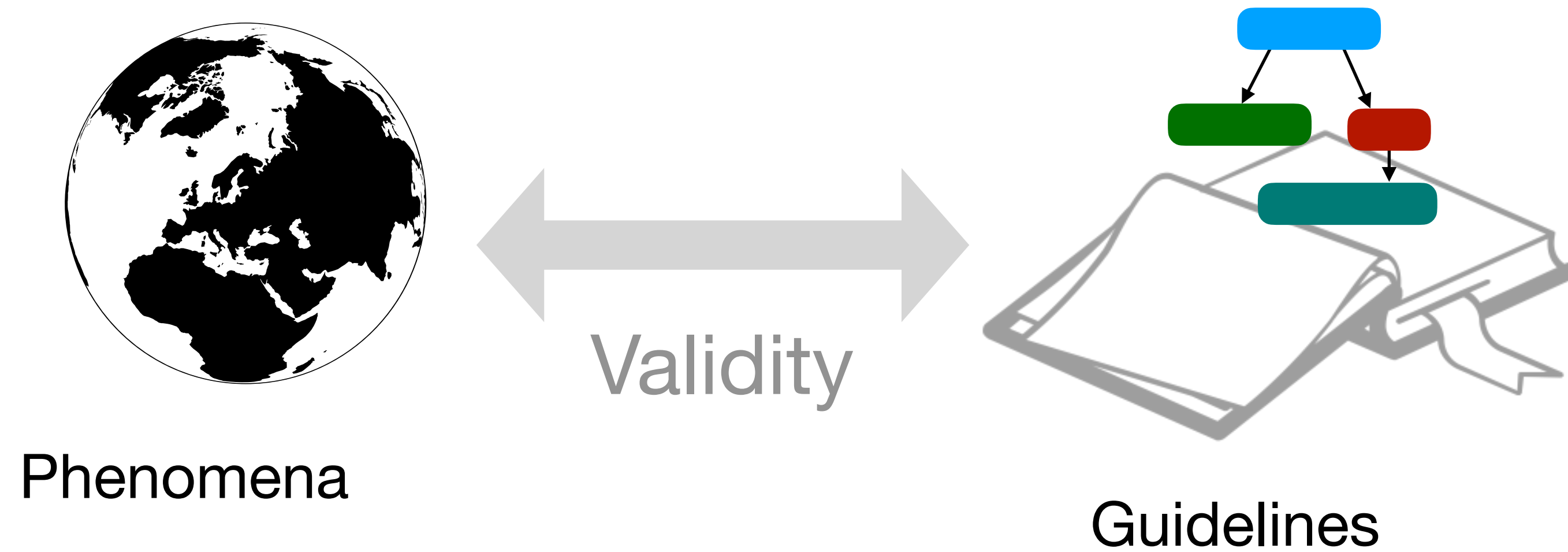


Annotation of Structured Representations

Given a structure, documents are *annotated* by marking span-bound concepts and relations.

CAP Measure RE-2: Require New Homes to install Solar Photovoltaic Systems		
Require 1) New single-family homes to install at least 1.5 W solar per square feet (e.g. 2,000 sq. ft. home = 3 kW) or minimum 2kW per home; 2) New multi-family homes to install at least 1 W solar per square feet (e.g., 1,000 sq. ft. home = 1 kW) or minimum 1 kW per unit, to install solar PV systems, unless the installation is impracticable due to poor solar resources.		
Target Year	Performance Metric	GHG Reduction Potential (MTCO ₂ e)
2020	Install 400 kW (0.4 MW) of solar photovoltaics on new homes.	141
2030	Install 1,000 kW (1 MW) of solar photovoltaics on new homes.	614

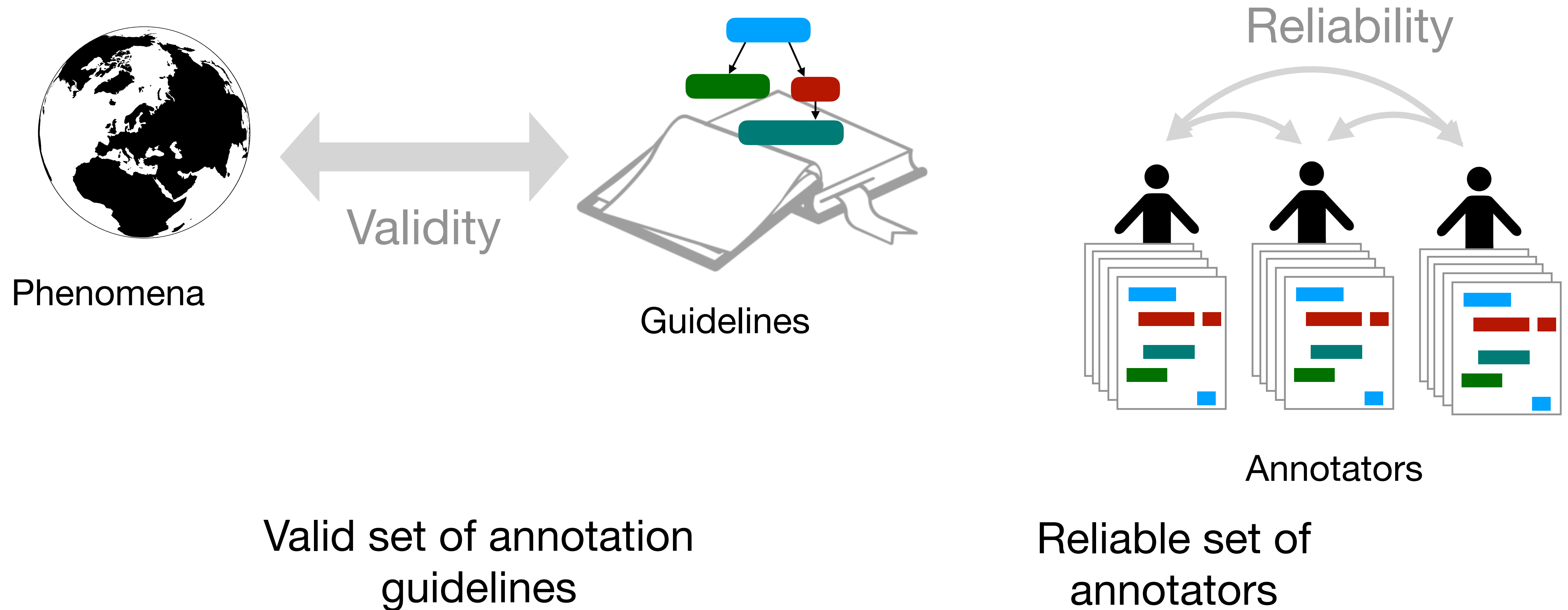
Annotation Quality



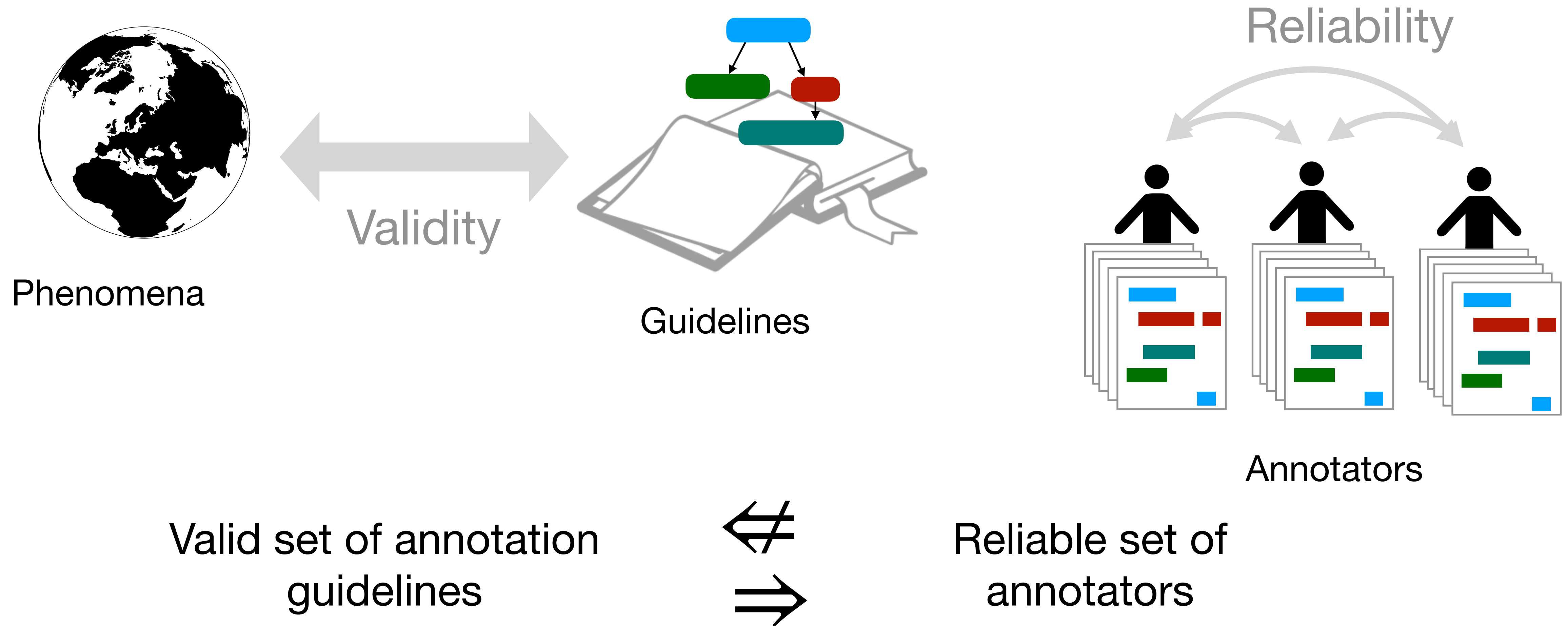
Valid set of annotation
guidelines

(Artstein and Poesio 2008)

Annotation Quality

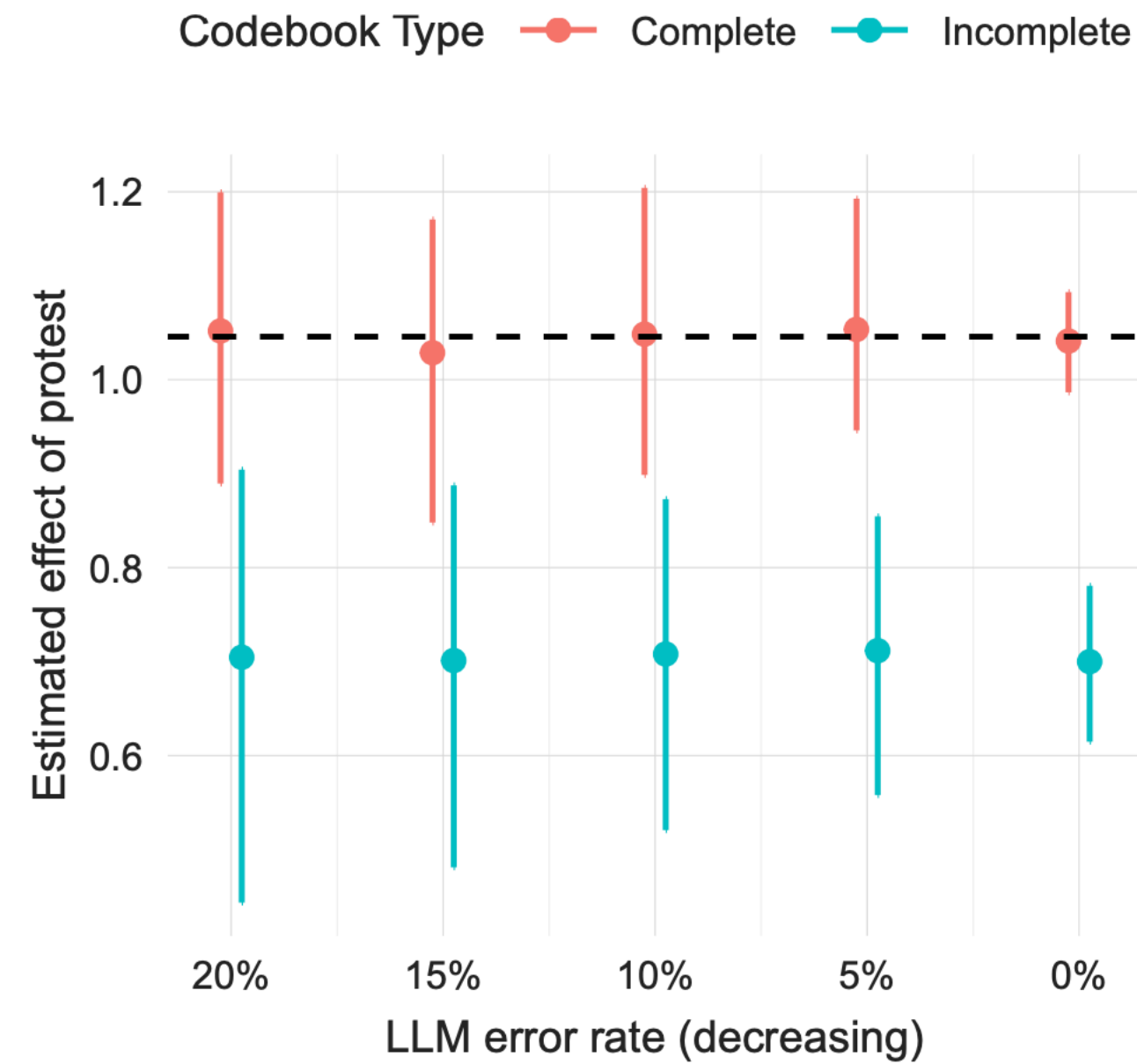


Annotation Quality



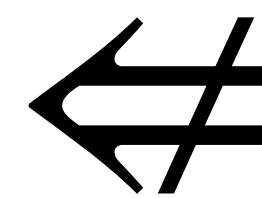
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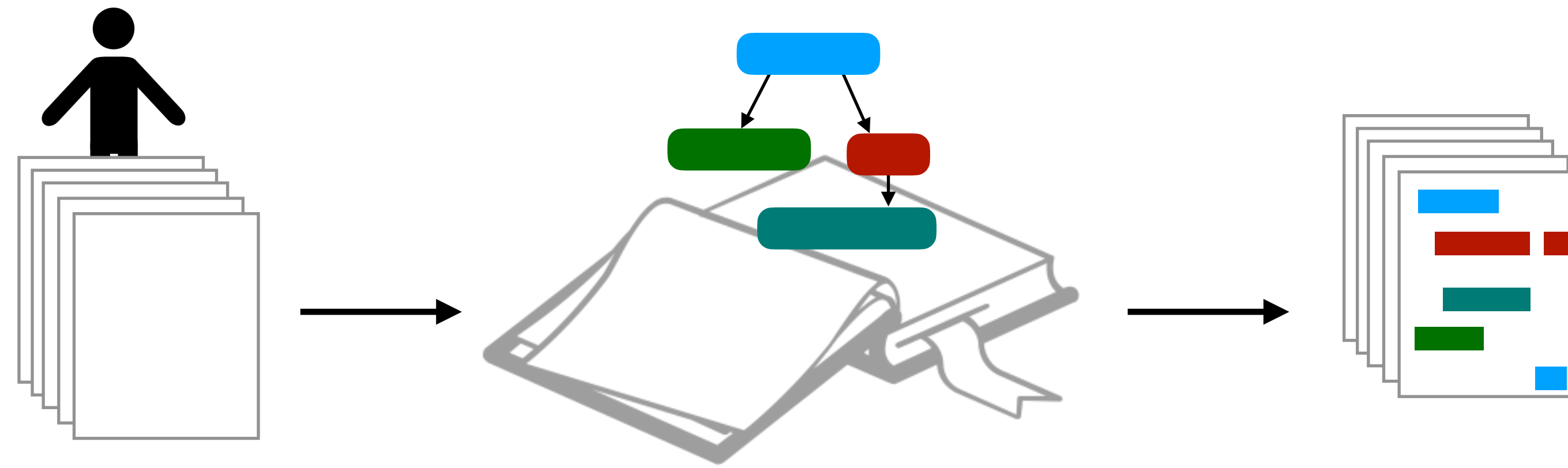
Example from Halterman and Keith 2026 illustrating that incomplete codebooks result in biased estimates.

Valid set of annotation
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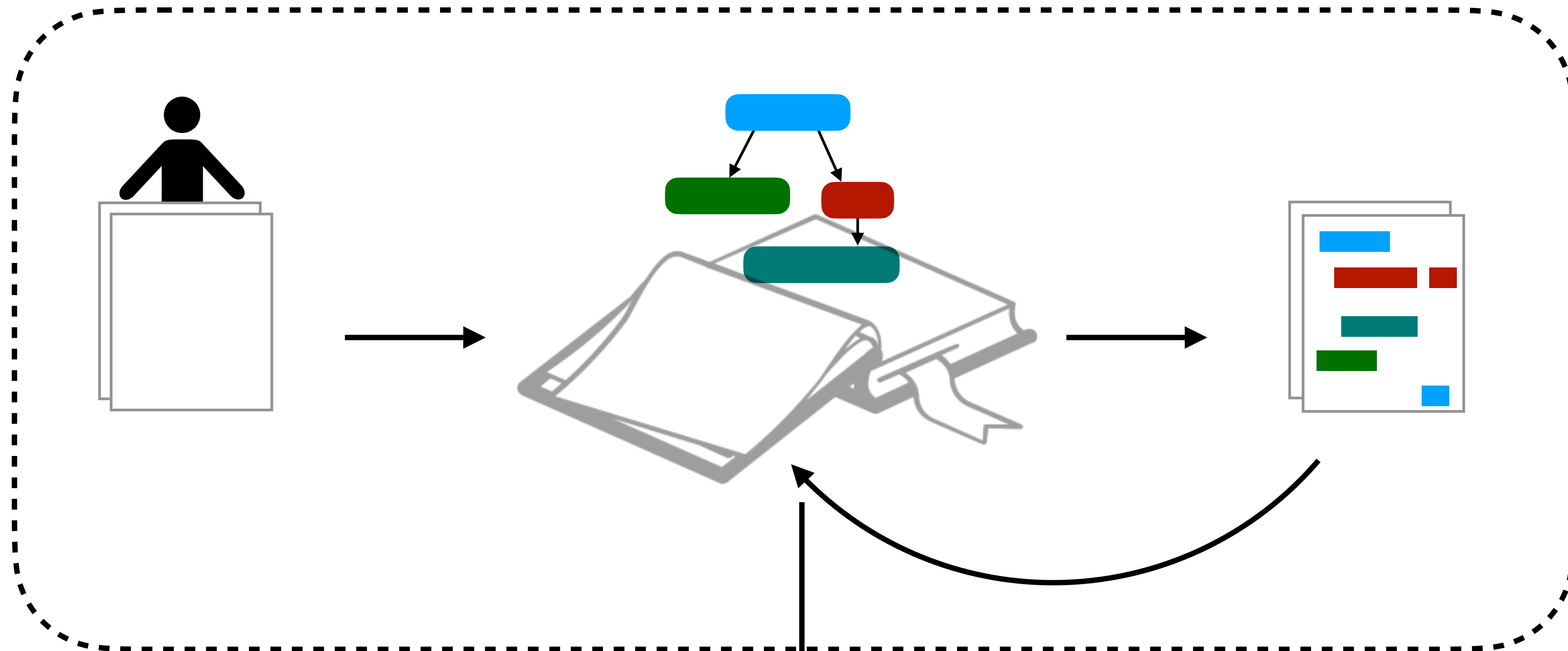


Reliable set of
annotators

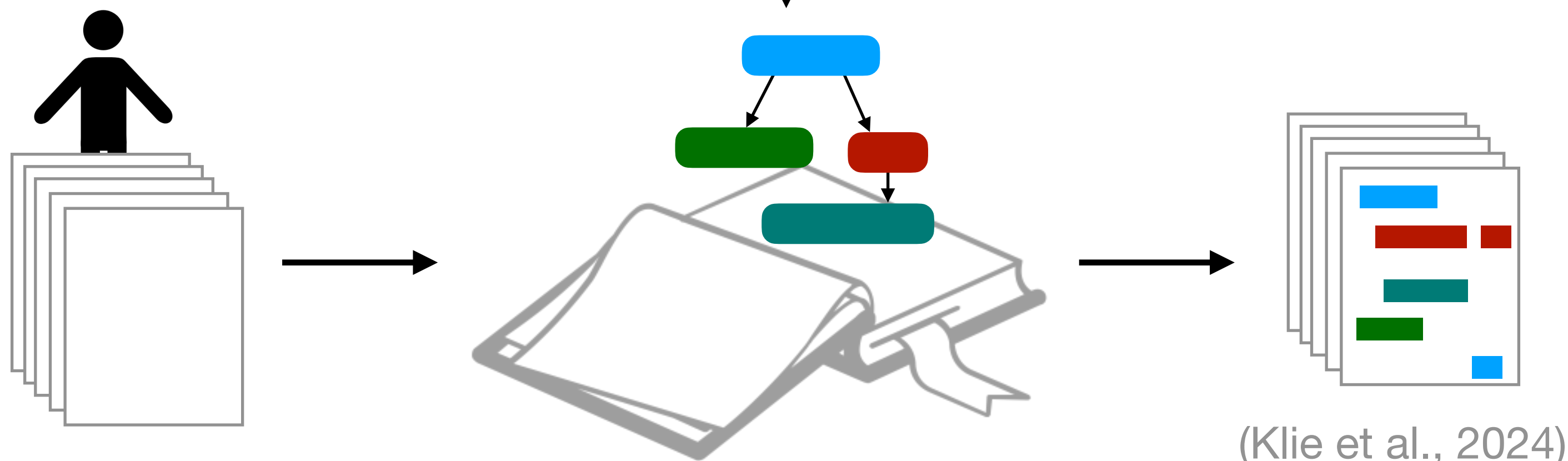
Best Practices for Annotation are Expensive



Best Practices for Annotation are Expensive

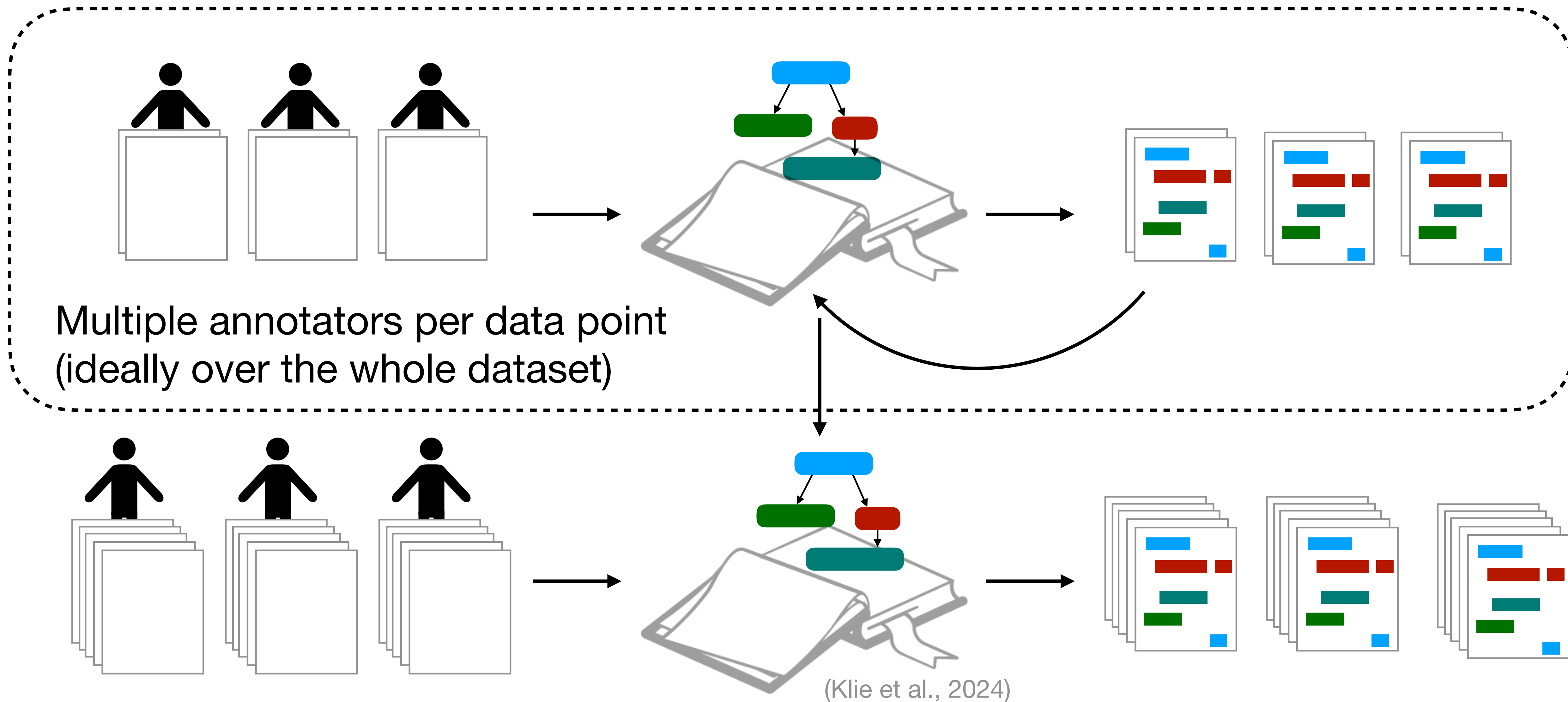


Agile corpus creation



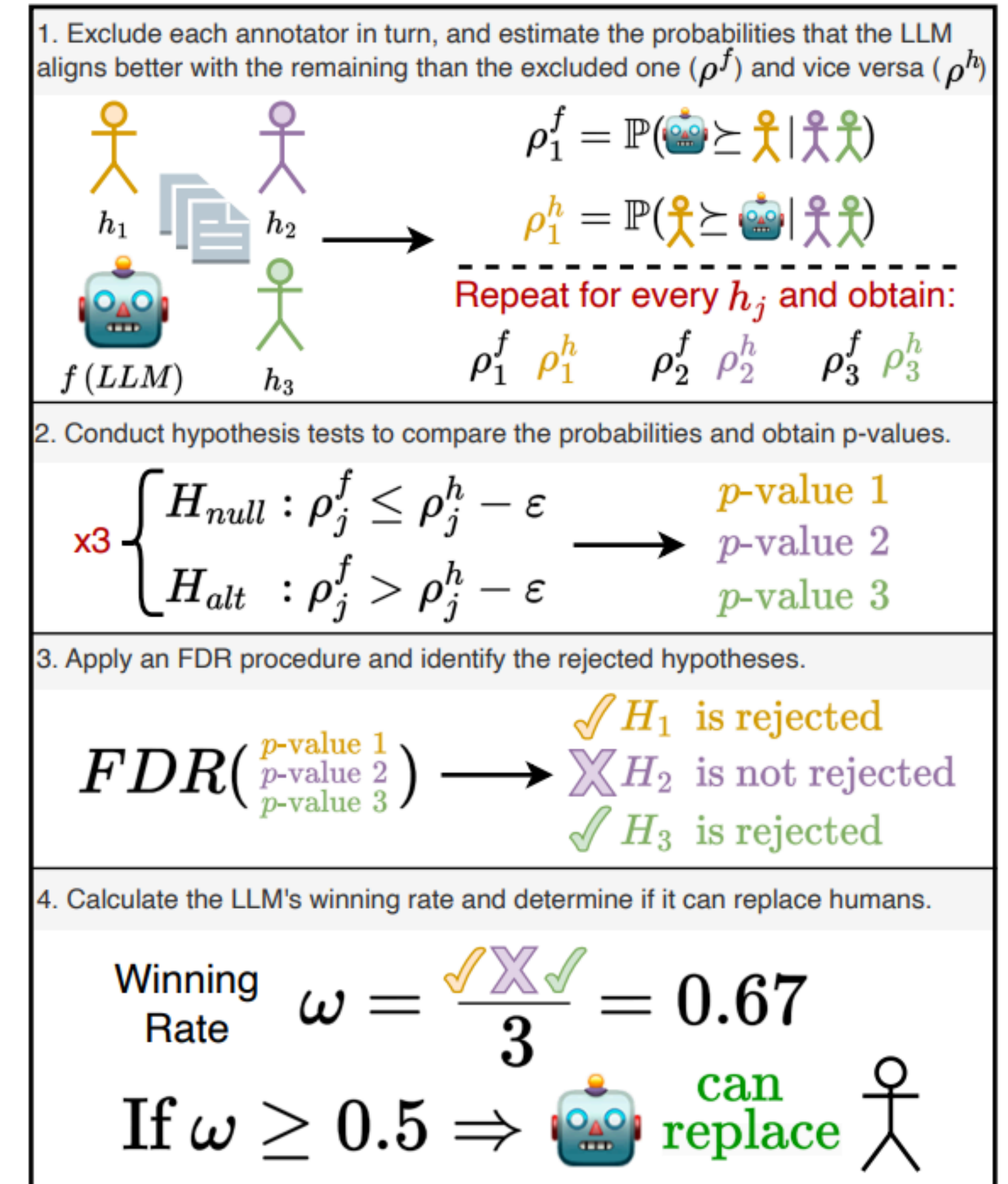
(Klie et al., 2024)

Best Practices for Annotation are Expensive



Machine-annotation alone is insufficient

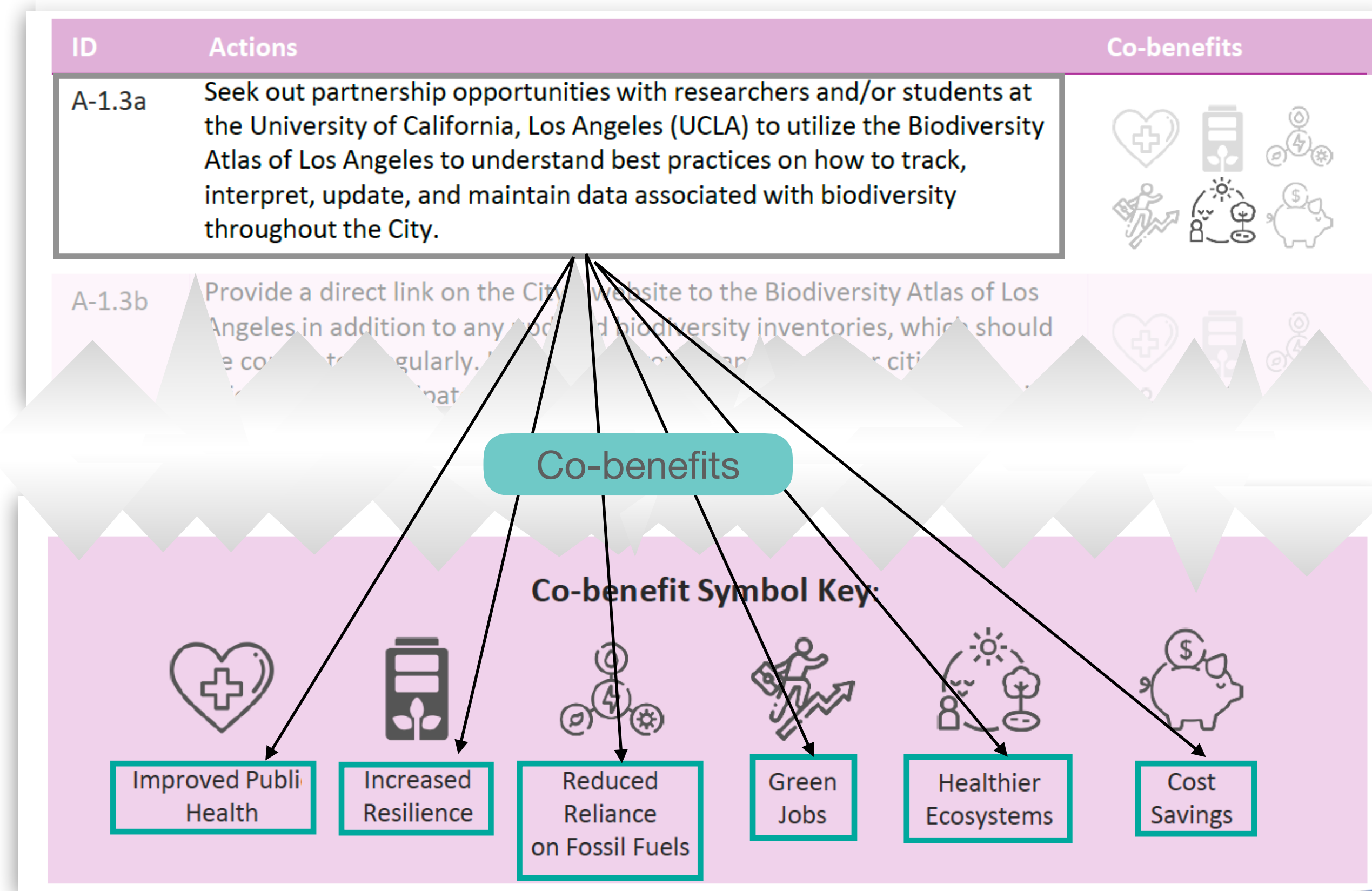
- Statistically valid machine annotation with LLM-as-a-Judge requires human annotation (Calderon et al., 2025) $\longrightarrow$
- Some domains and tasks can be challenging for machine annotation



Challenges with Machine Annotation of Gray Policy Literature

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Visually rich format

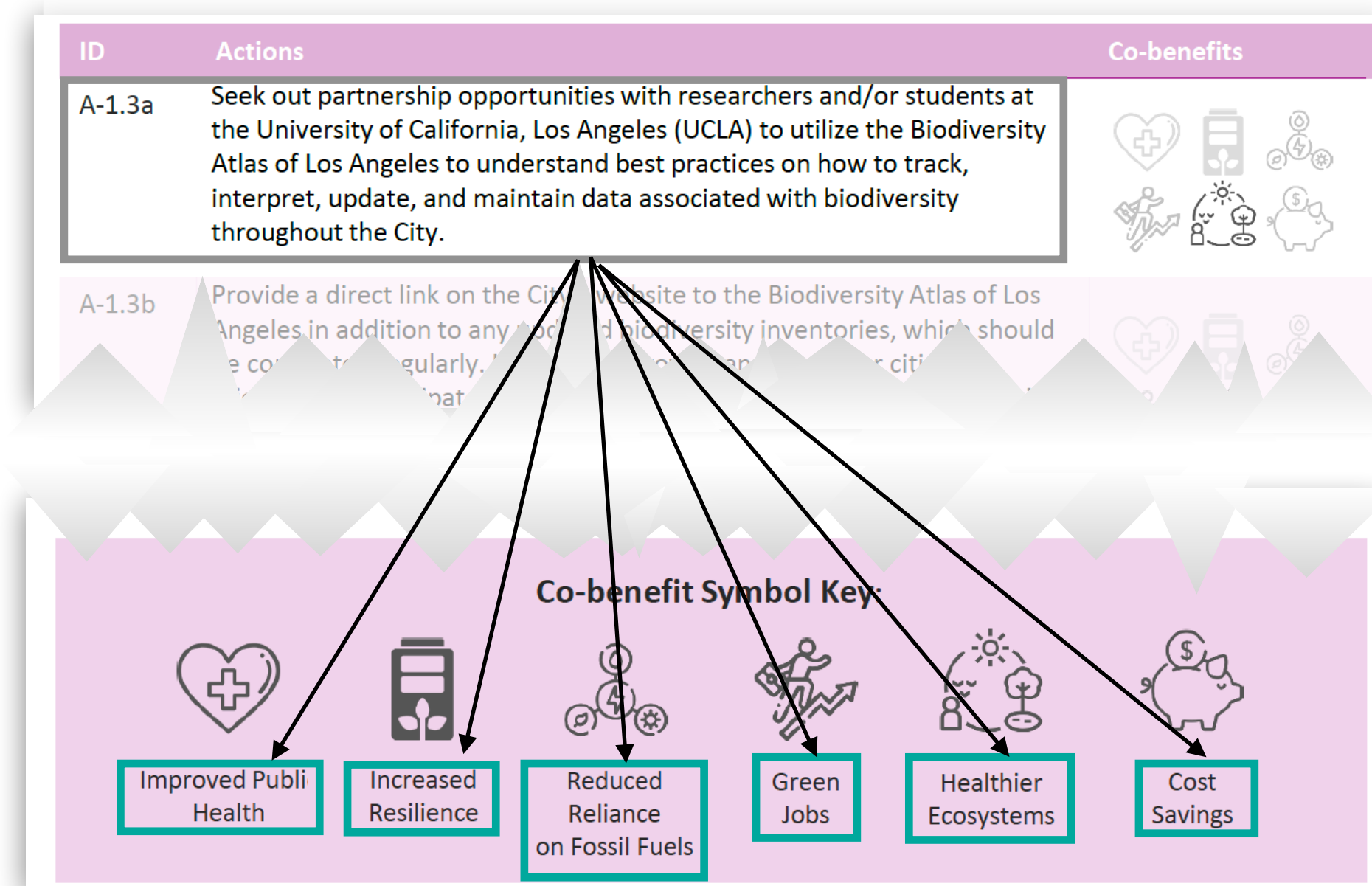


Example extraction from City of Burbank, 2022, page 104

Challenges with Machine Annotation of Gray Policy Literature

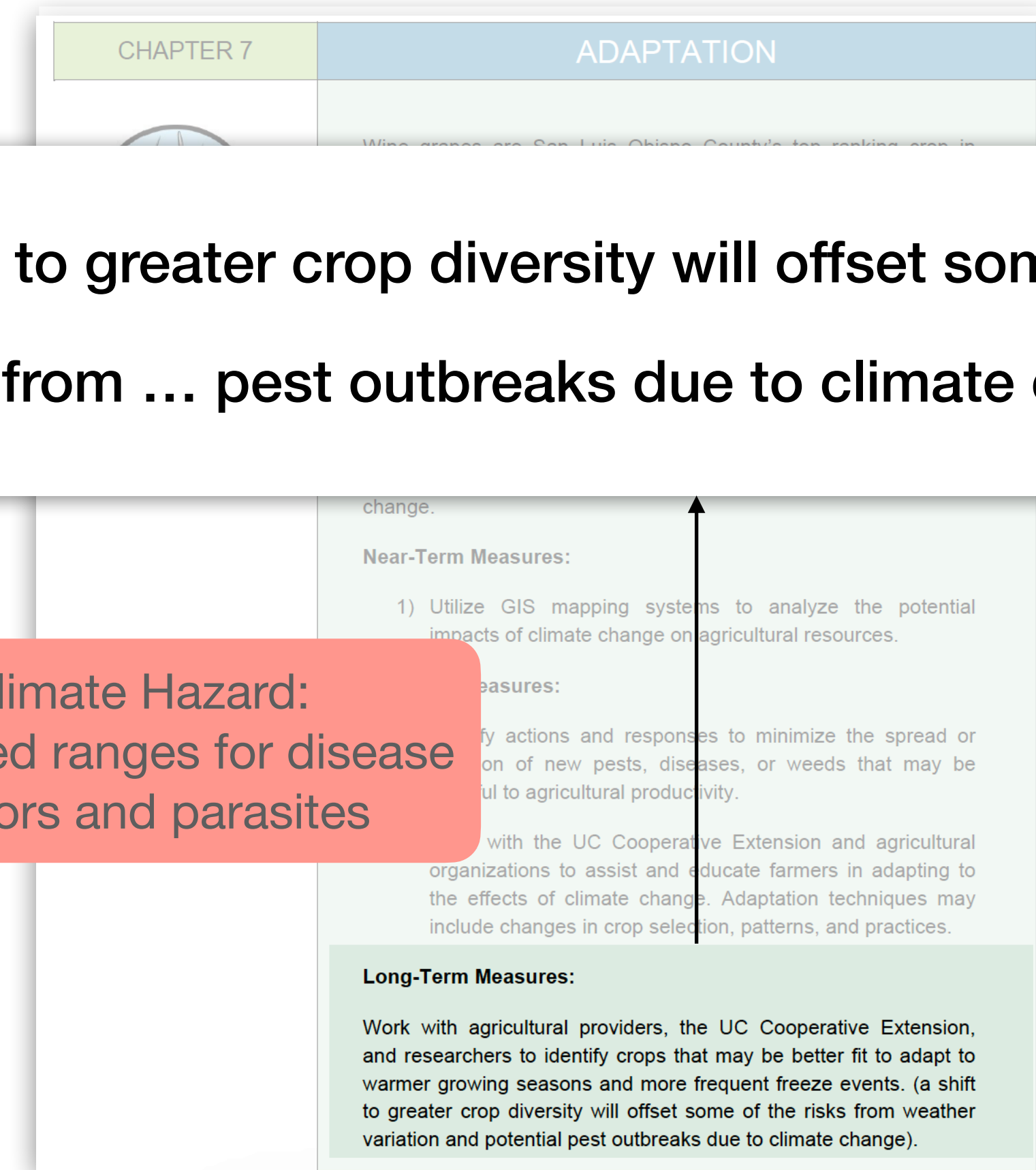
Conflating multiple concepts

Visually rich format



A shift to greater crop diversity will offset some of the risks from ... pest outbreaks due to climate change

Climate Hazard:
Expanded ranges for disease
vectors and parasites



Example extraction from City of Burbank, 2022, page 104

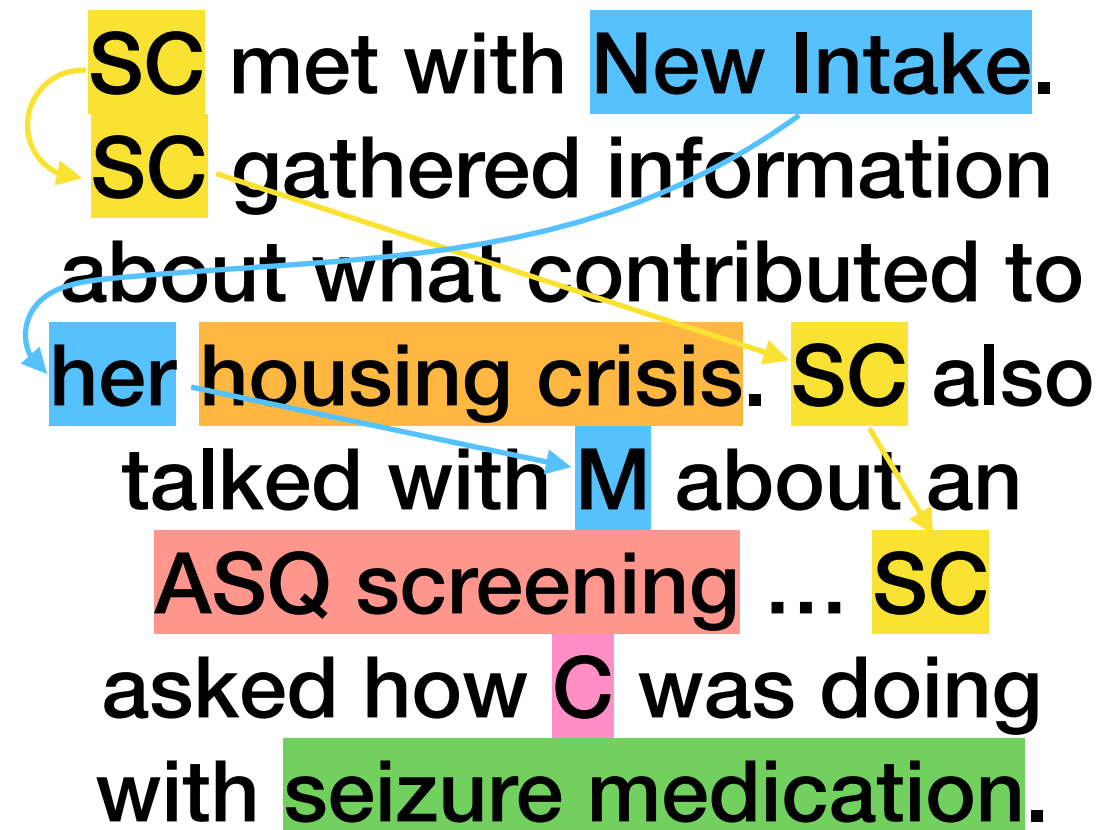
Example extraction from San Luis Obispo County, 2011, page 234

Challenges with Machine Annotation of Child Welfare Caseworker Notes (with BERT-sized models)

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Ground truth:

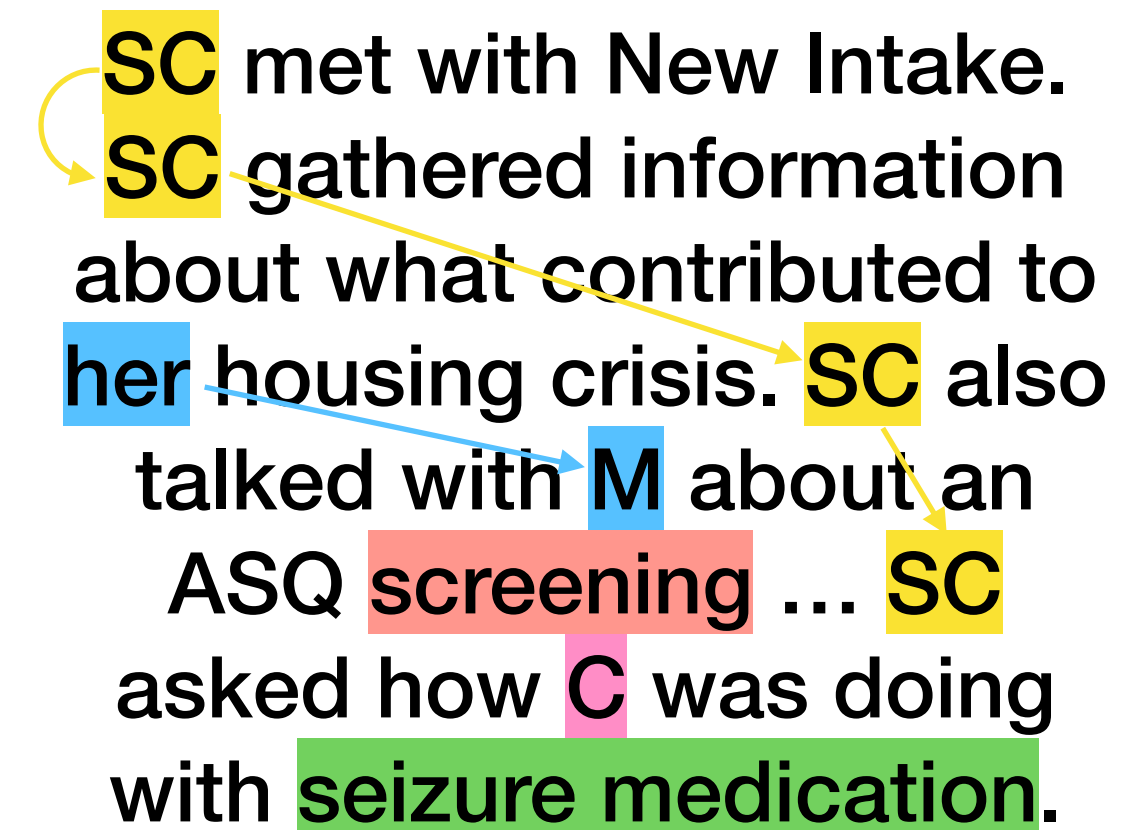
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≠

Off-the-shelf coreference model:

SC met with New Intake.
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Performance degrades with
differing data domain,
annotation scheme

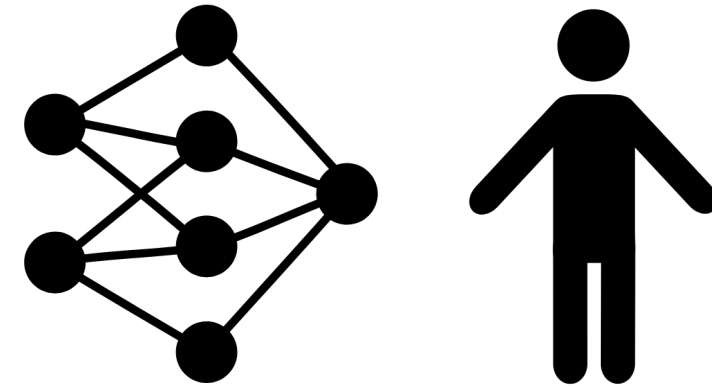
Anonymized annotated example snippet of caseworker
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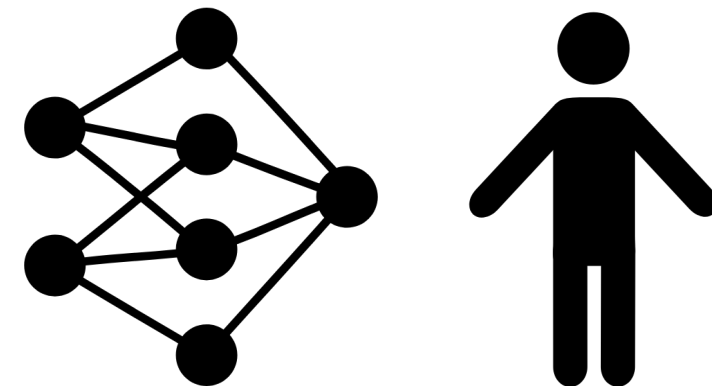
Reducing the prohibitively high cost of annotating structured representations

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Reducing the prohibitively high cost of annotating structured representations

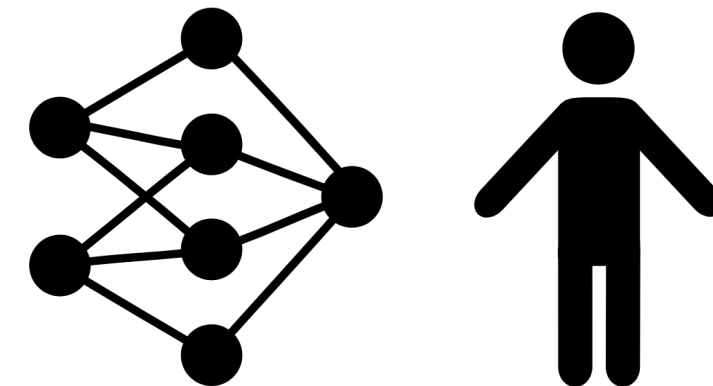
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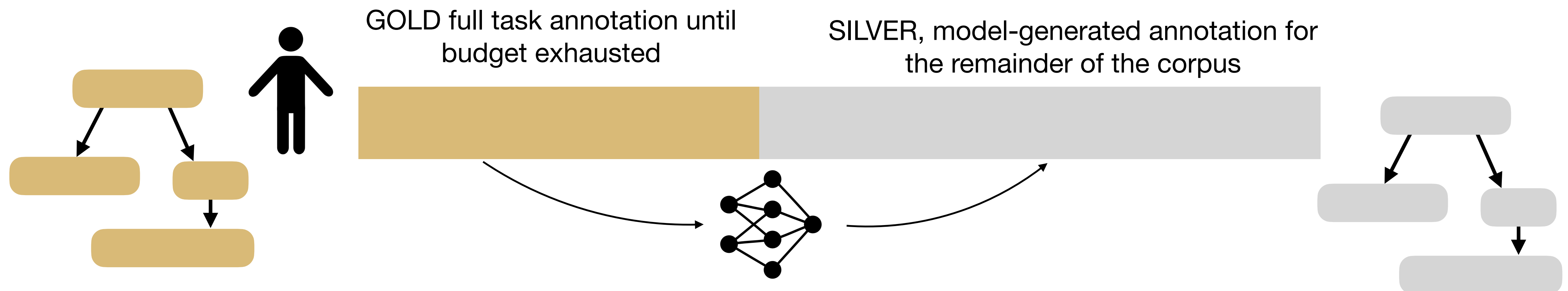
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Reducing the prohibitively high cost of annotating structured representations

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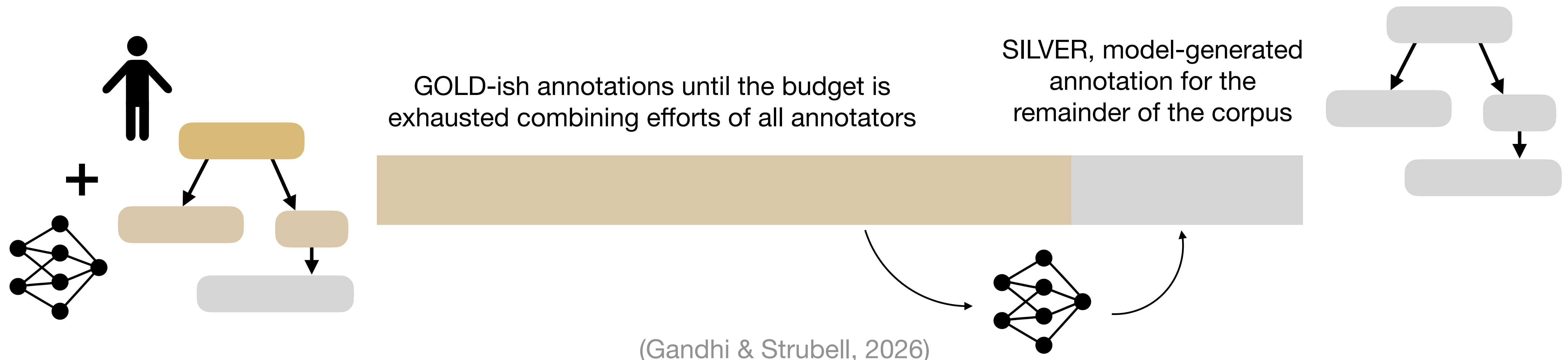
- Given a heterogeneous set of annotators and a limited annotation budget, we aim to reduce costs and maximize the resulting annotation quality.
- Baseline Annotation Procedure:



Reducing the prohibitively high cost of annotating structured representations

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- Decomposing structured annotation tasks can (1) reduce the aggregate annotation complexity and (2) allow for complementarity —> resulting in a **more efficient use of resources**.
- Proposed Annotation Procedure:



Model of inferential load

- Inferential load: complexity of decisions that annotator considers
- We should decompose task t into sub-tasks $\{t_1, t_2, \dots, t_k\}$ when

$$\text{Inferential Load}(\{t_1, t_2, \dots, t_k\}) \leq \text{Inferential Load}(t)$$

Model of Inferential Load

Set of entities realized by annotation



$$\text{INFERENTIAL LOAD}(\{t_1, \dots, t_k\}) = \sum_{j=1}^k \frac{\text{CENTERS INTRODUCED}(t_j)}{+ \text{OUTPUT SPACE}(t_j) \cdot w_o} \cdot w_{\text{center}} \\ + \text{INPUT SPACE}(t_j) \cdot w_i \\ + \text{MAP SPACE}(t_j) \cdot w_m$$

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Output space of all valid annotations



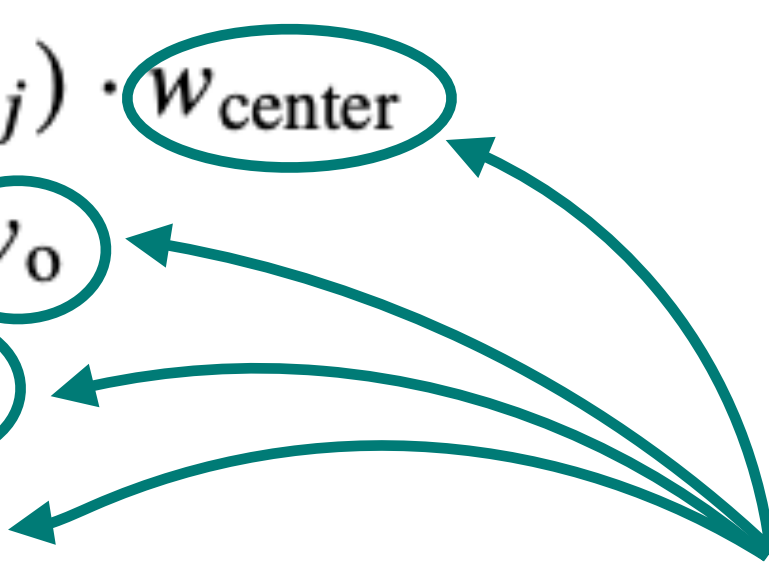
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Input space and
mapping structure

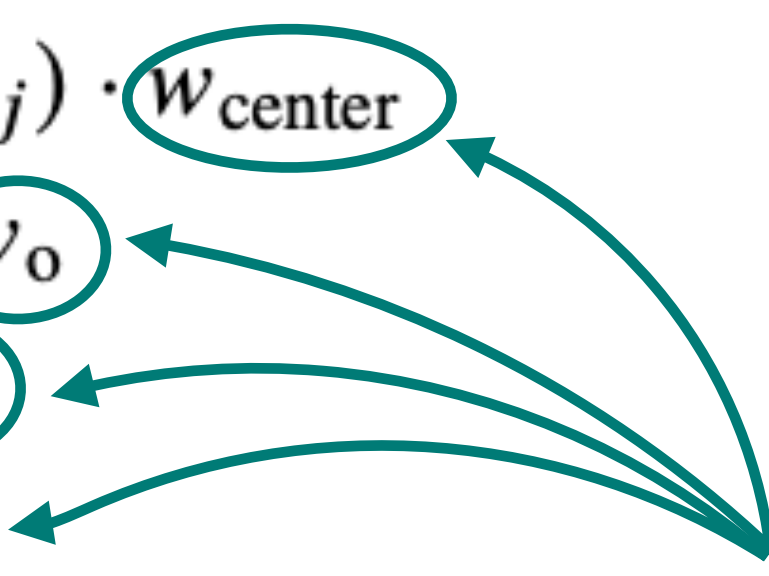
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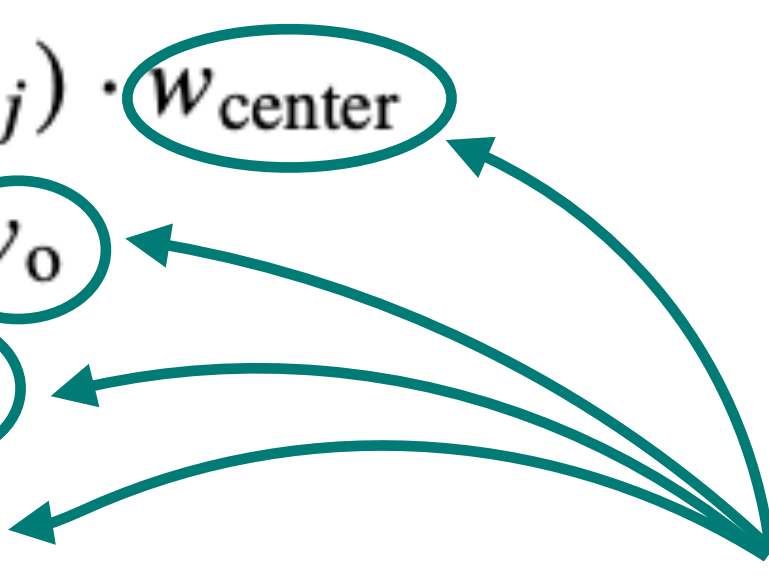
Annotator sensitivity to individual components

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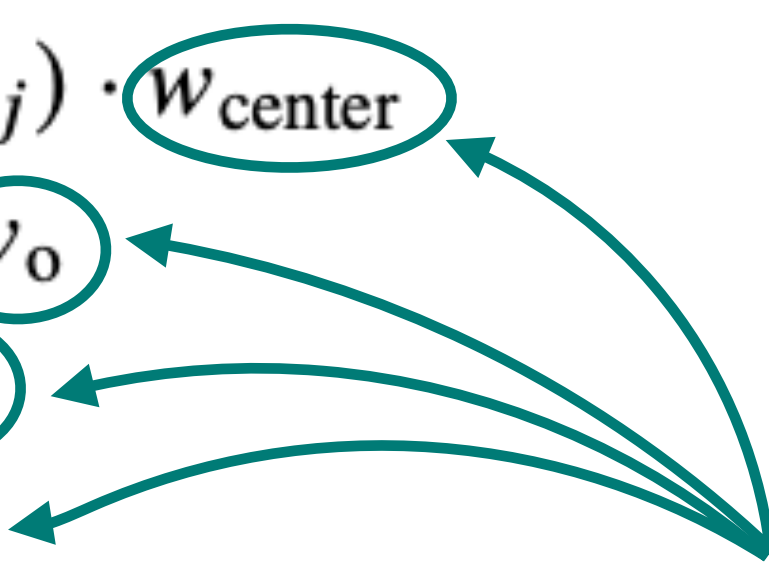
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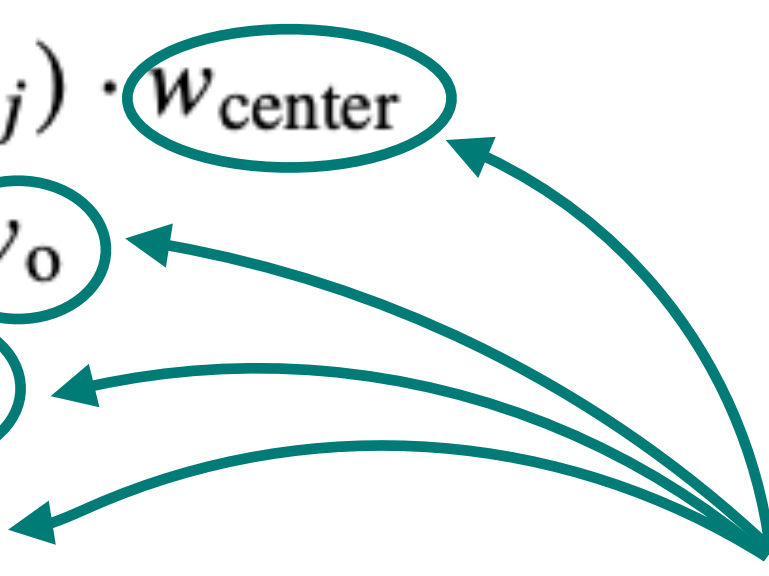
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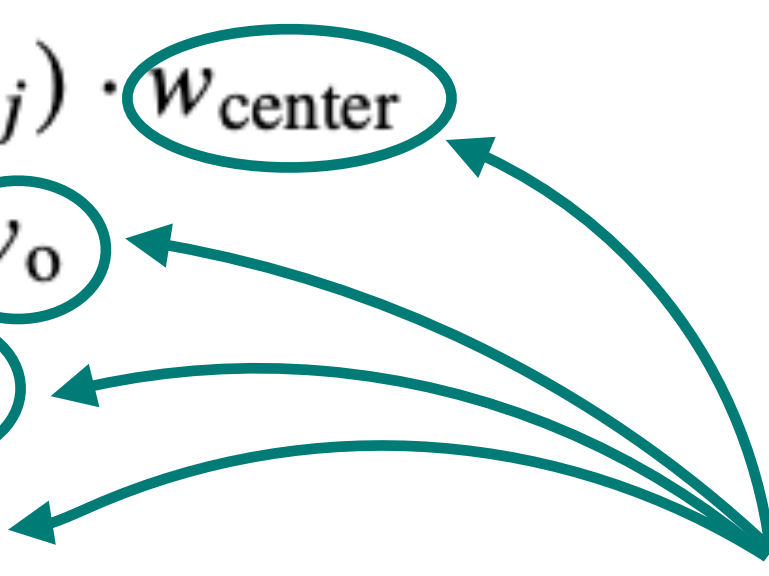
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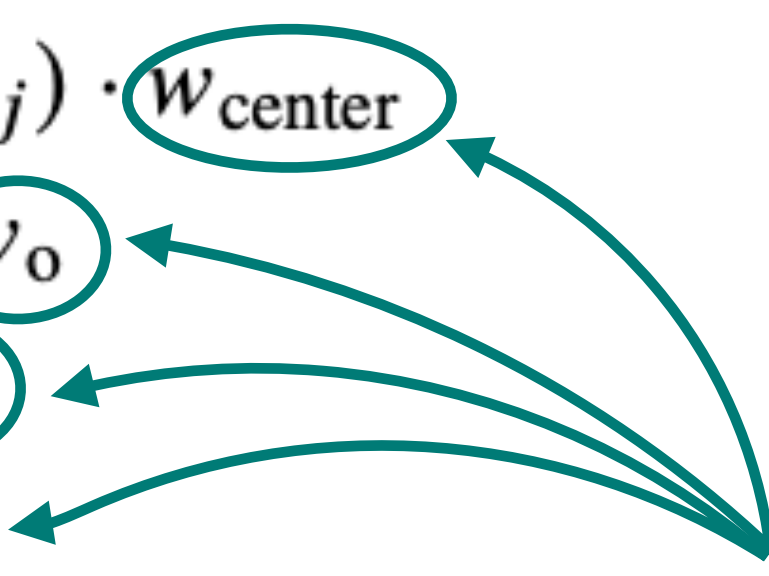
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Annotator sensitivity to individual components

- Our model of inferential load is dominated by the output space of valid annotations for a given task.

Example: Decomposition for Coreference

SC met with New Intake.
SC gathered information
about what contributed to
her housing crisis. SC also
talked with M about an
ASQ screening ... SC
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Anonymized annotated example snippet of caseworker
note from dataset developed in Gandhi et al., 2023.

Example: Decomposition for Coreference

In coreference resolution, we jointly identify mentions of relevant entities and link mentions with their antecedents.

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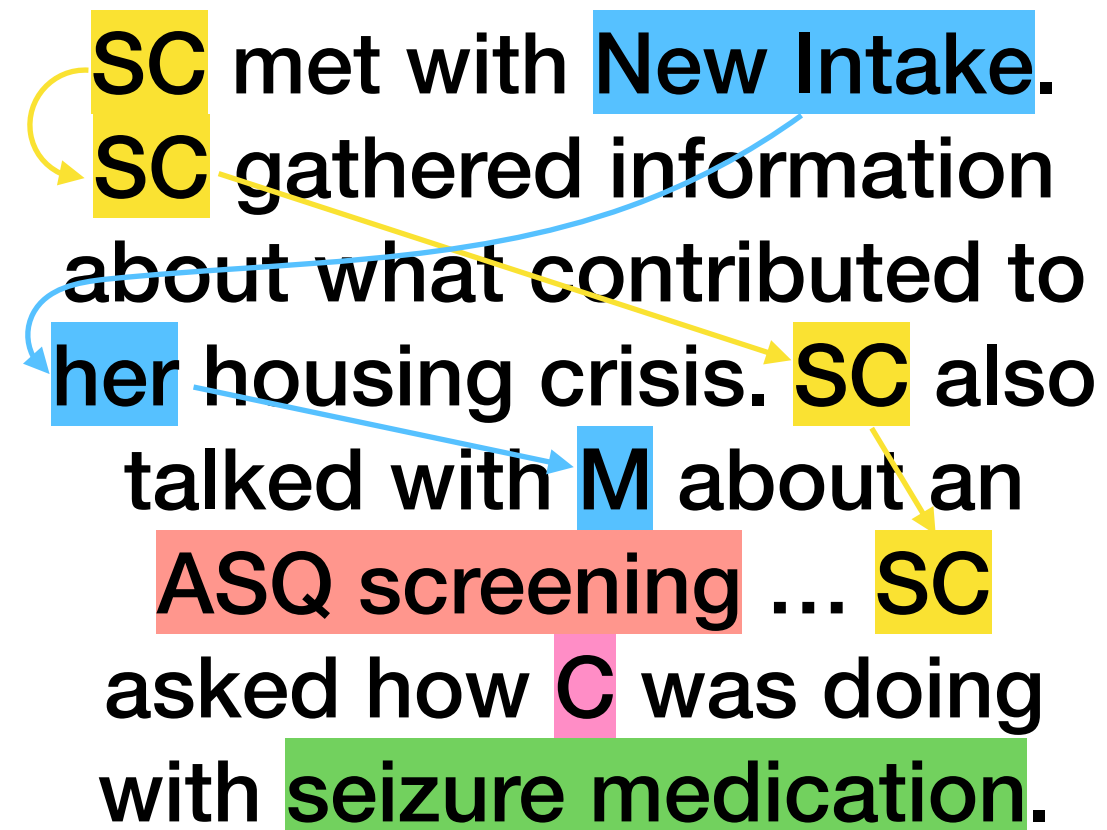
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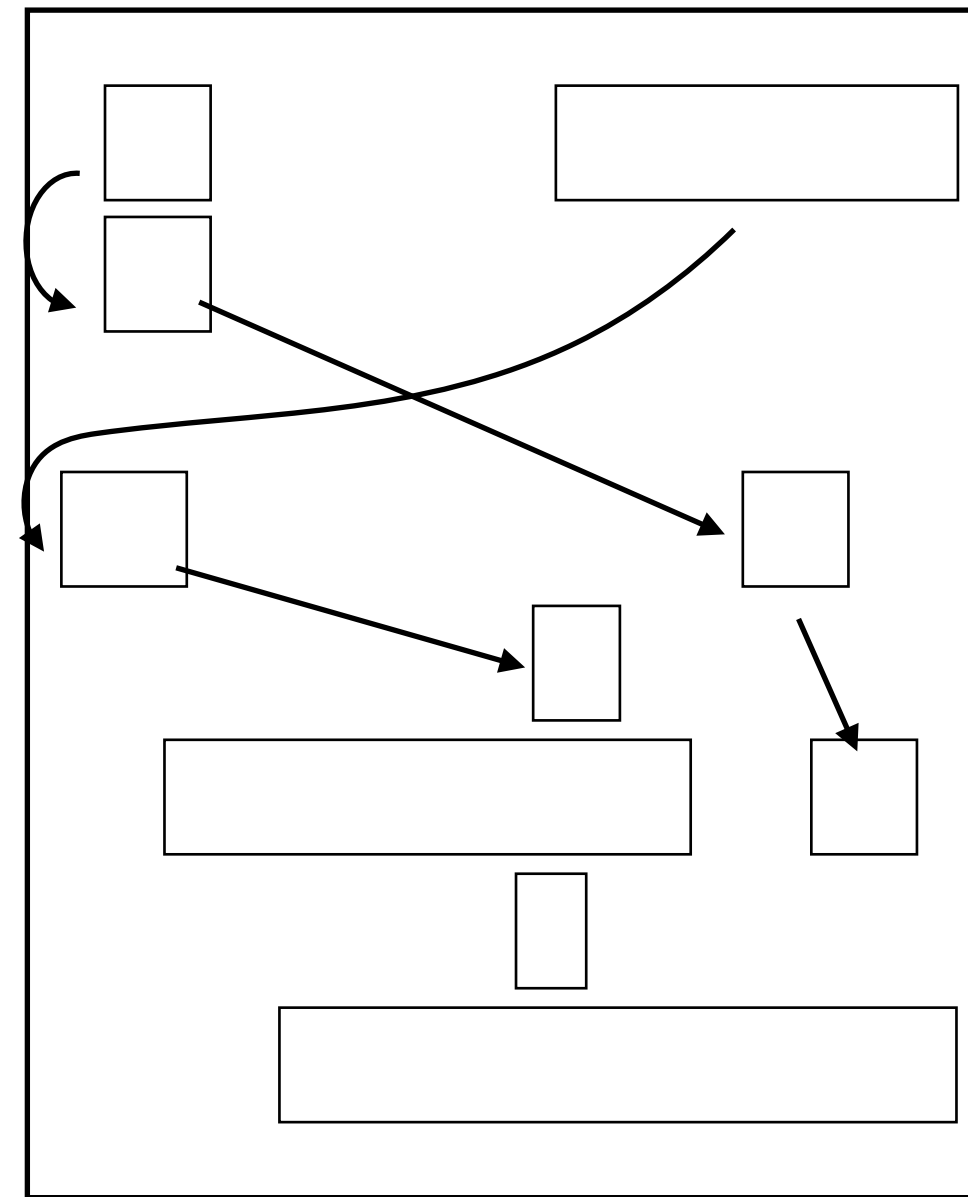
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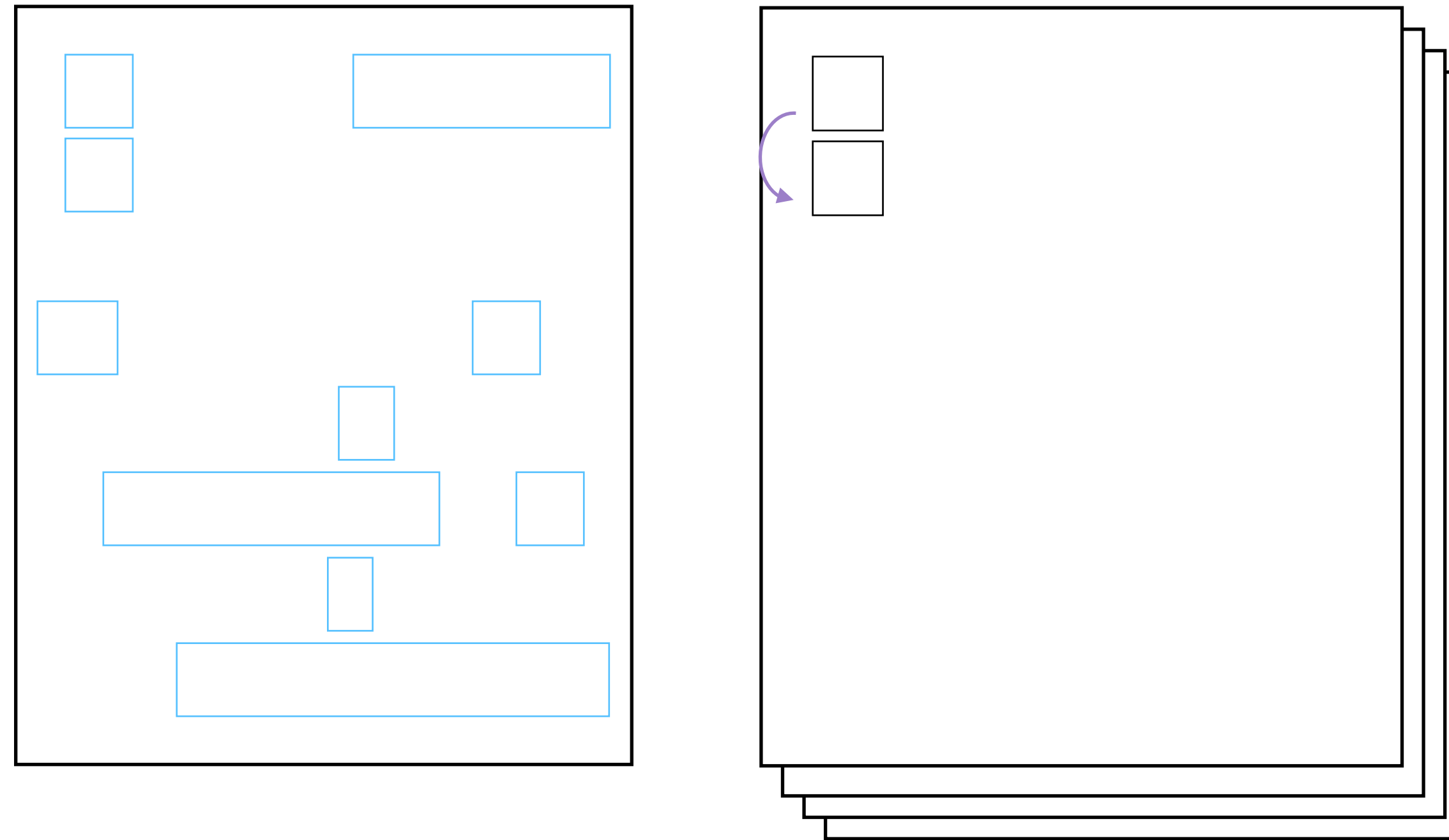
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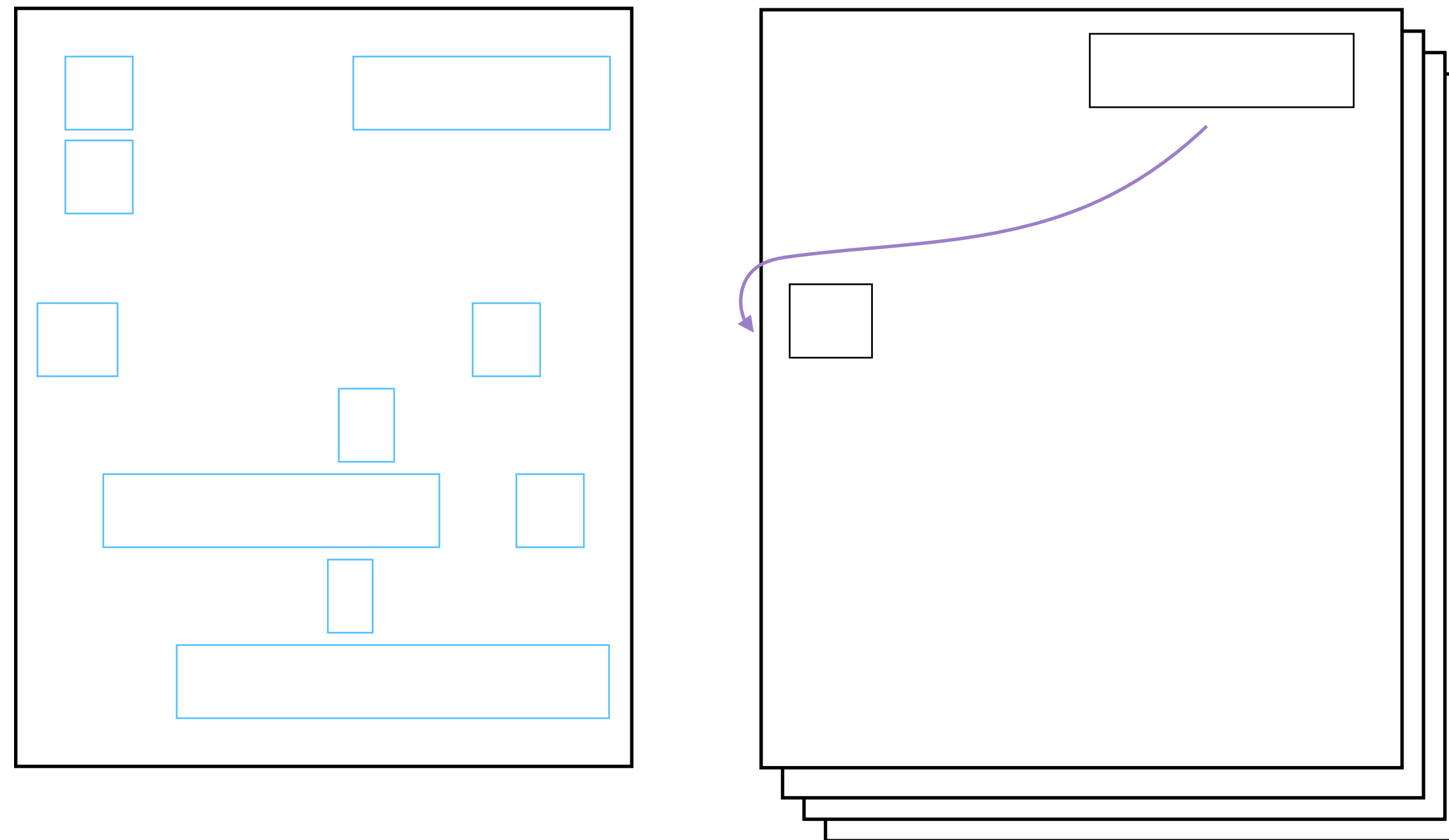
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Coreference resolution can be decomposed to mention detection and antecedent linking.



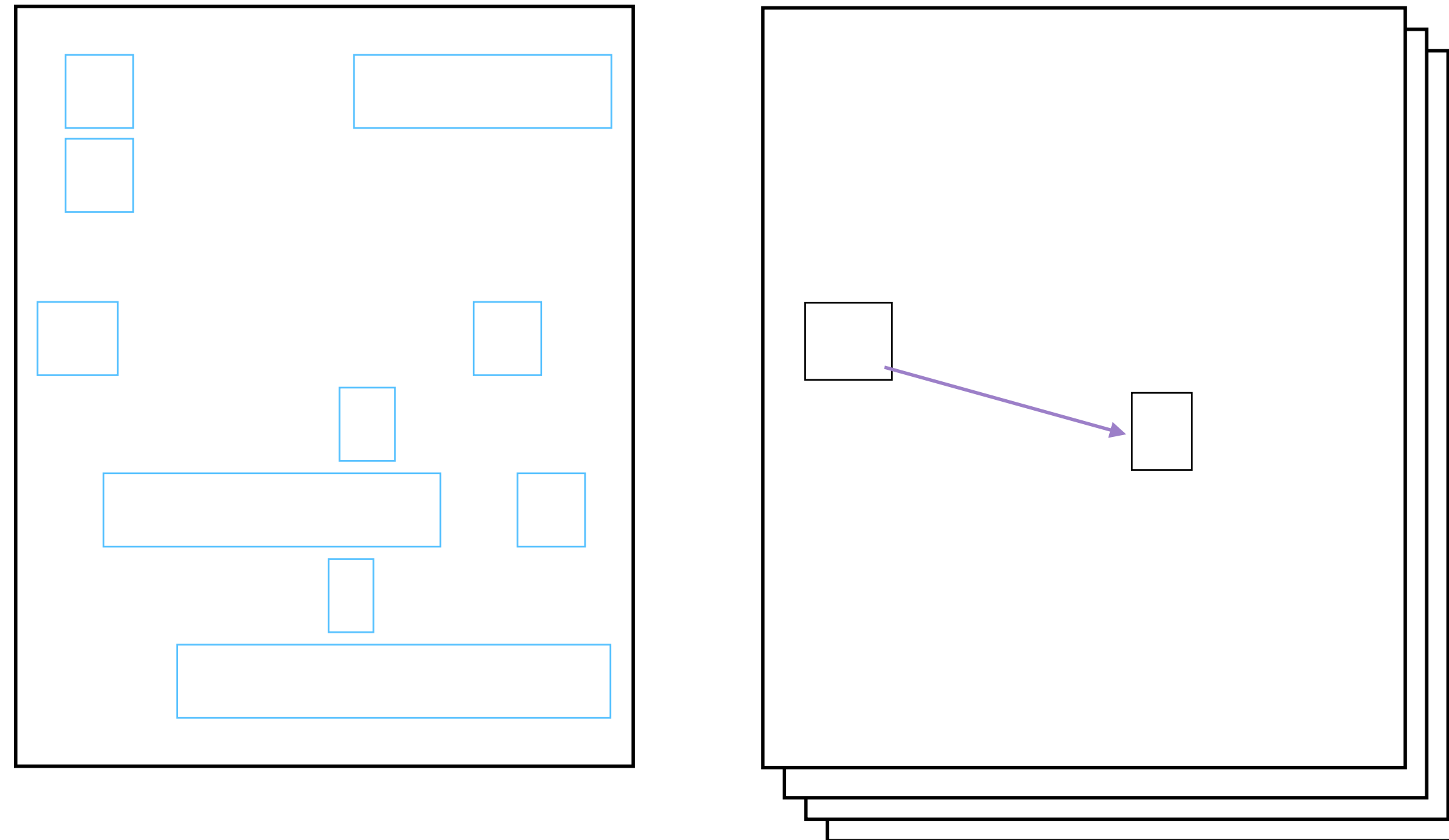
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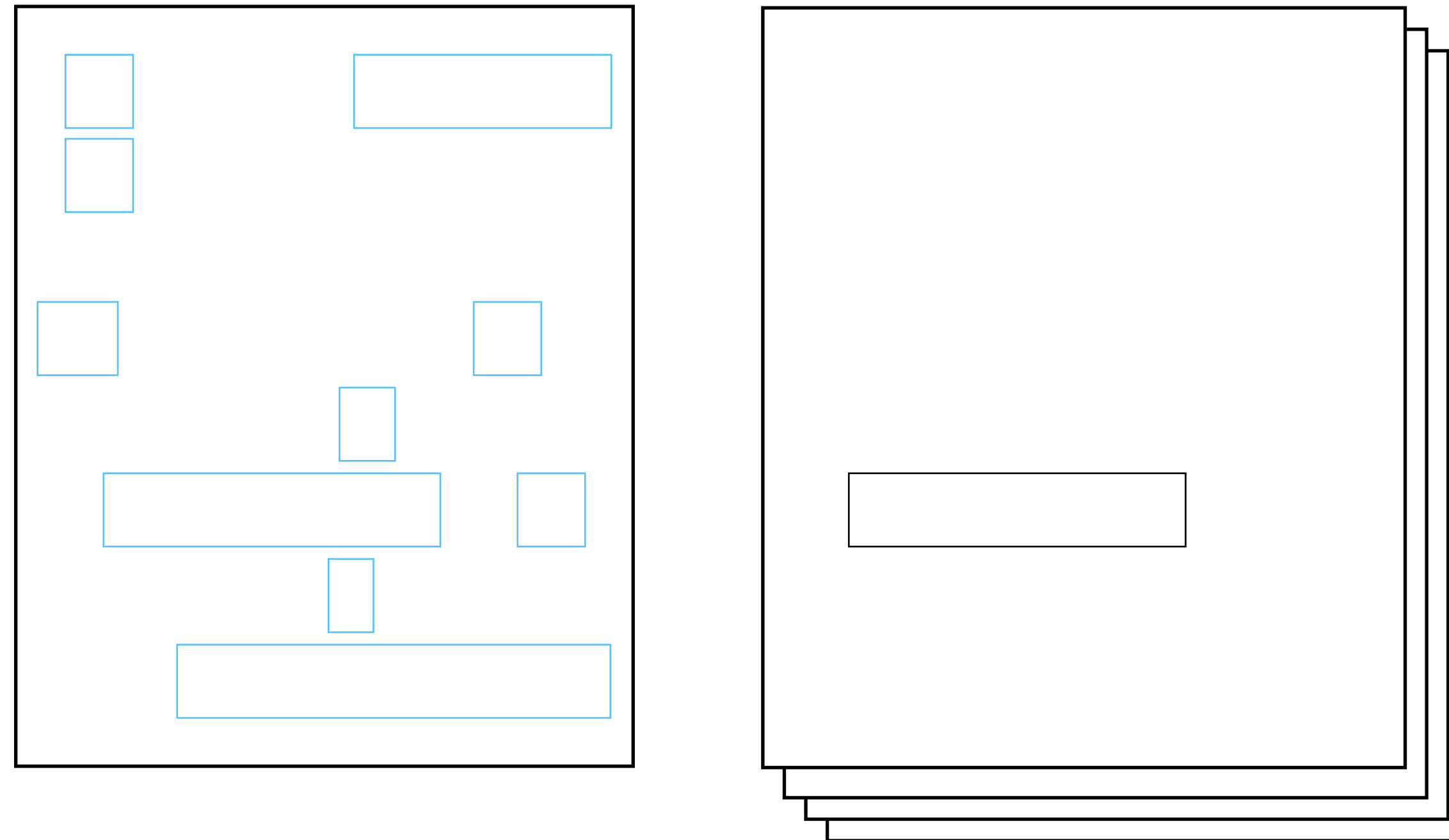
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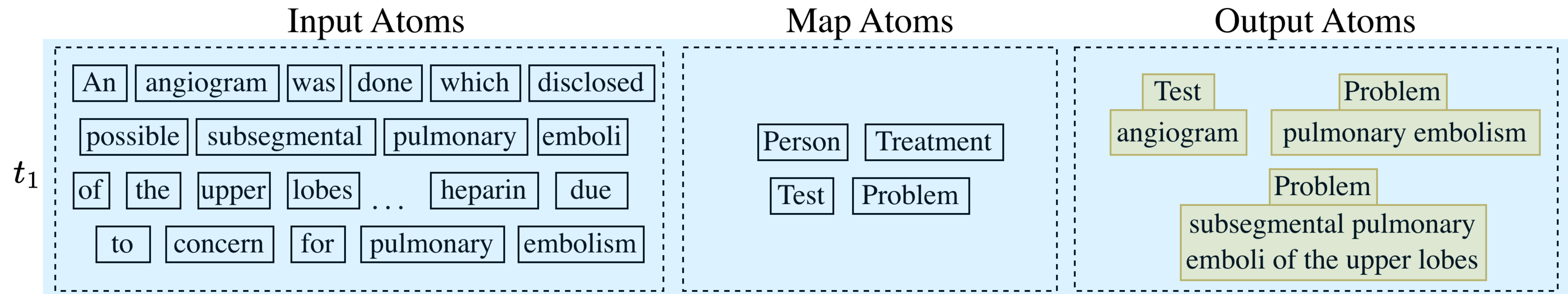
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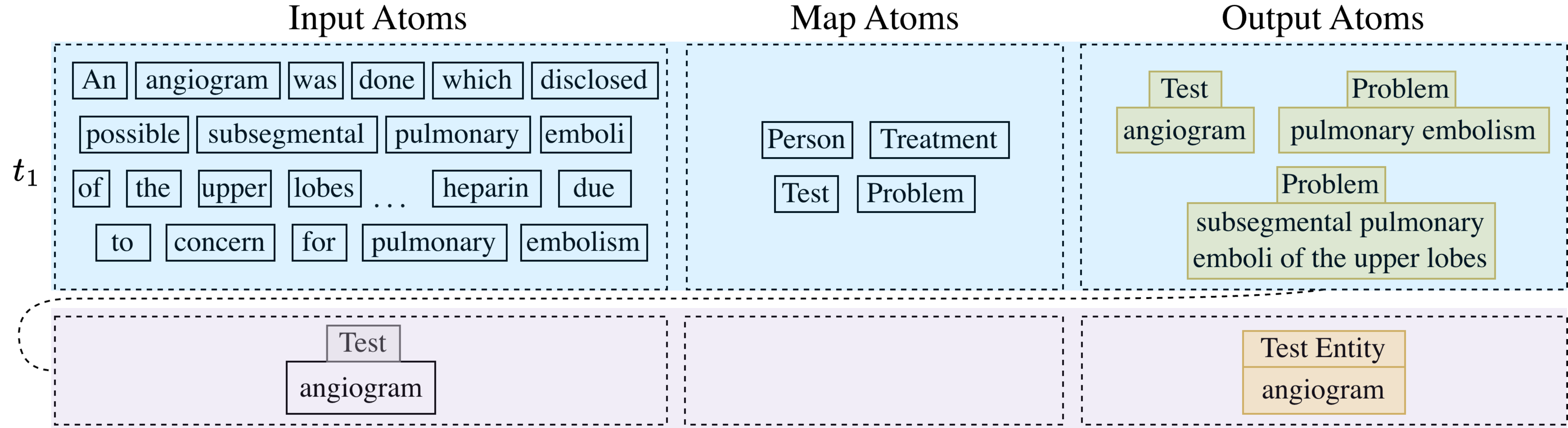


Example: Decomposition for Coreference

Mention Detection

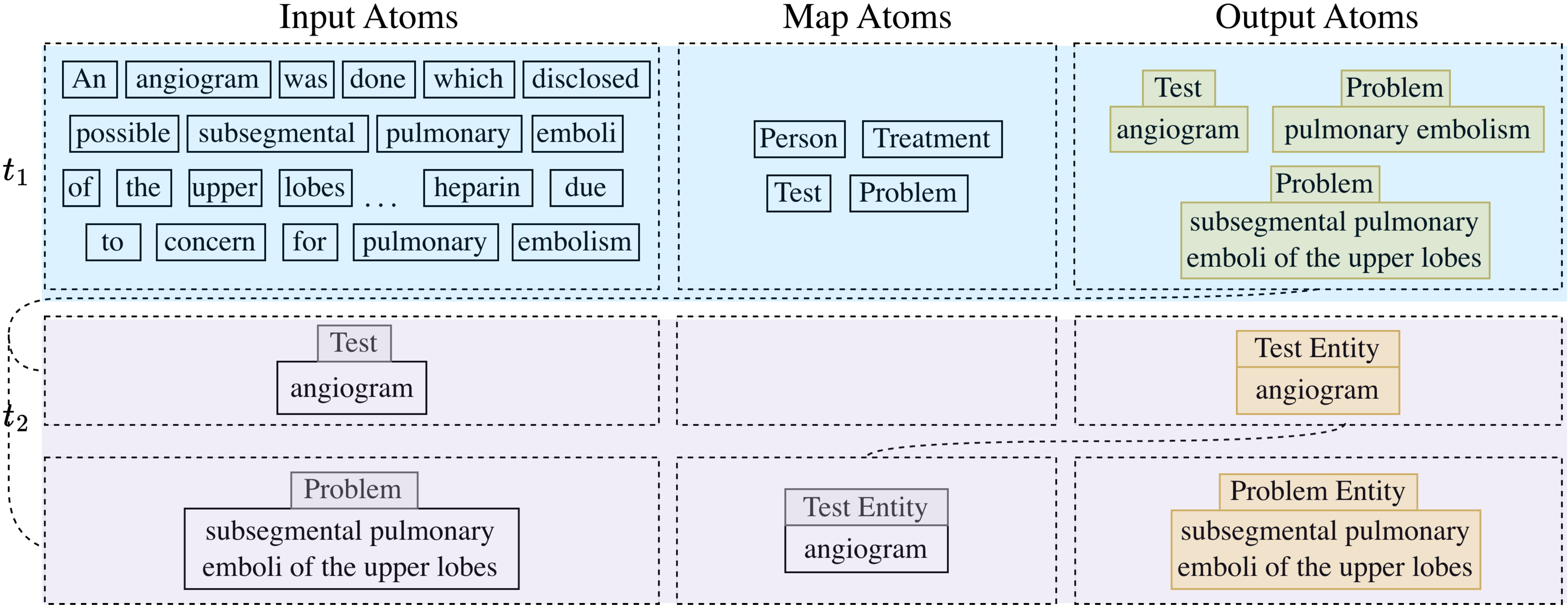


Example: Decomposition for Coreference



Antecedent
Linking

Example: Decomposition for Coreference



Antecedent
Linking

Input Atoms

Map Atoms

Output Atoms

t_1

An angiogram was done which disclosed possible subsegmental pulmonary emboli of the upper lobes ... heparin due to concern for pulmonary embolism

Person Treatment
Test Problem

Test Problem
angiogram pulmonary embolism
Problem
subsegmental pulmonary emboli of the upper lobes

t_2

Test
angiogram

Test Entity
angiogram

Problem
subsegmental pulmonary emboli of the upper lobes

Test Entity
angiogram

Problem Entity
subsegmental pulmonary emboli of the upper lobes

Problem
pulmonary embolism

Test Entity
angiogram

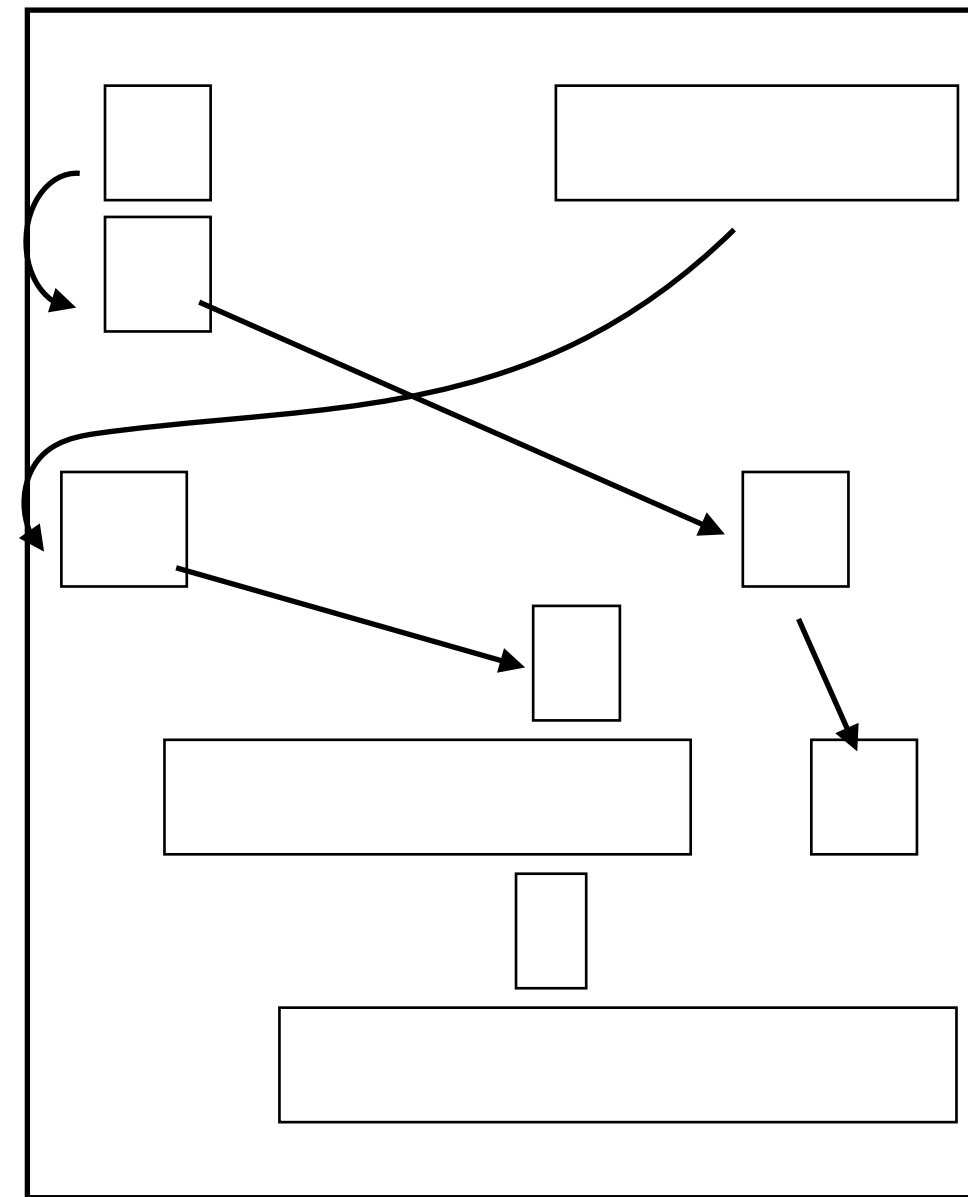
Problem Entity
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Test Entity
angiogram

Problem Entity
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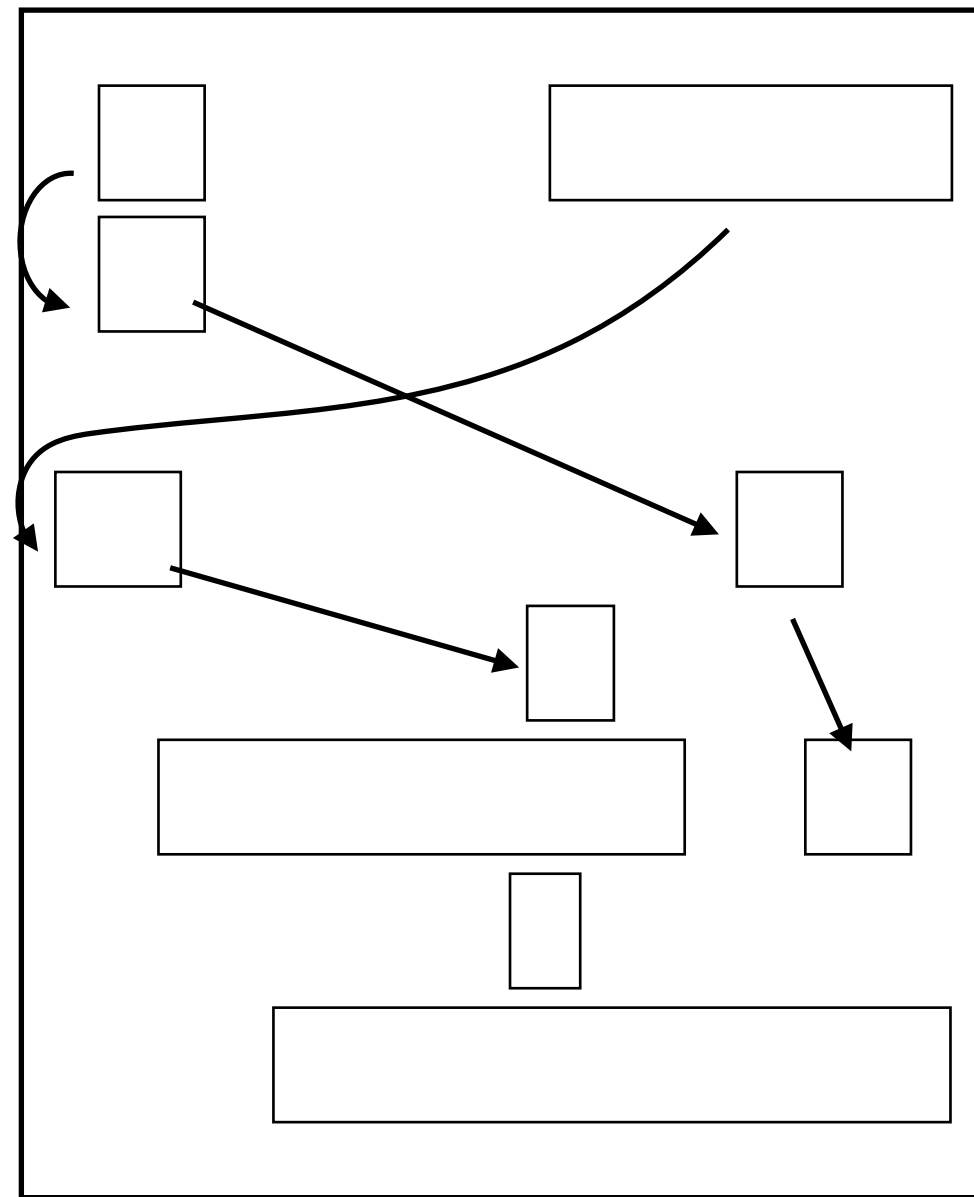
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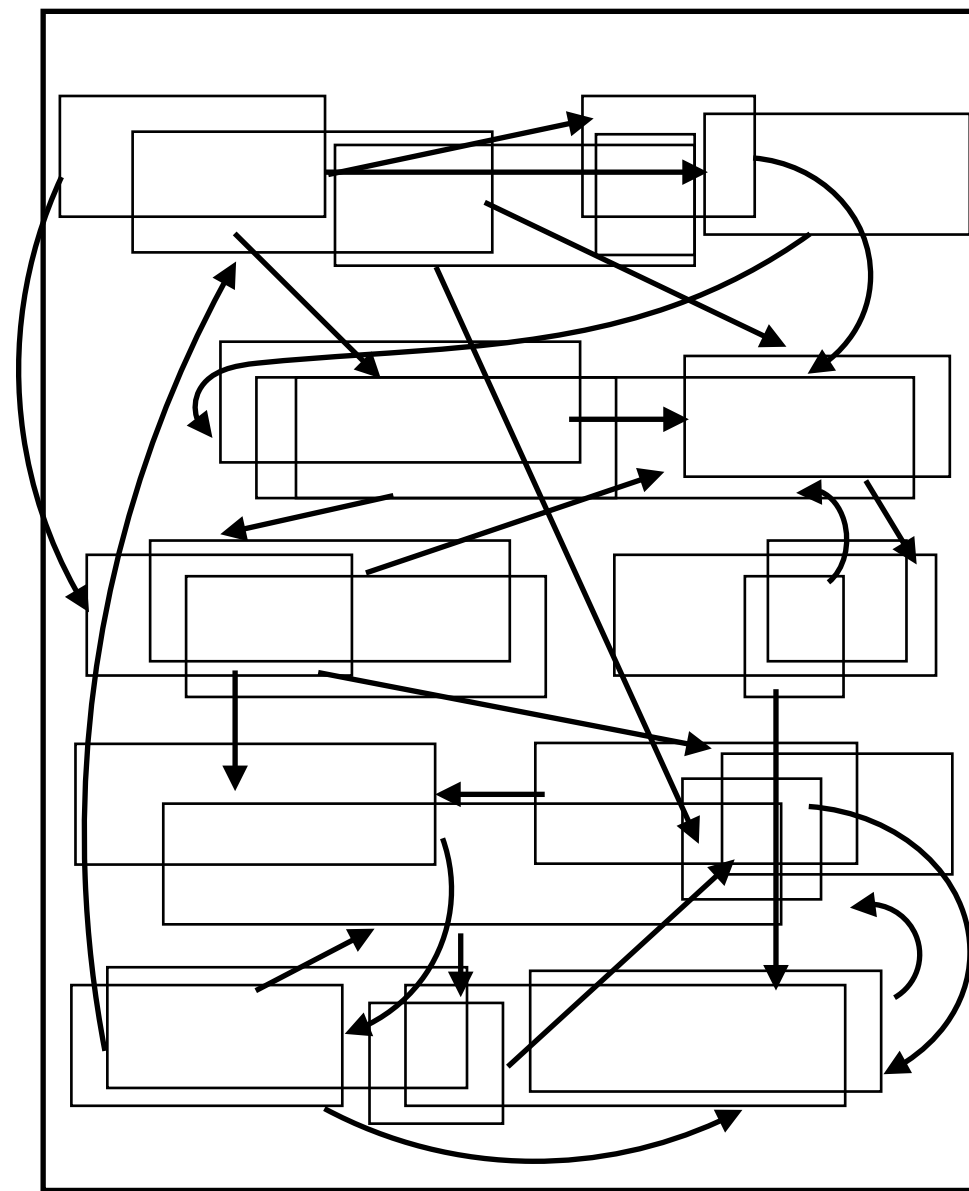
We will focus on a dominant component of inferential load: output space, which is the number of valid annotations for a task.

Example: Decomposition for Coreference

In coreference resolution, we jointly identify mentions of relevant entities and link mentions with their antecedents



Output space of valid annotations:

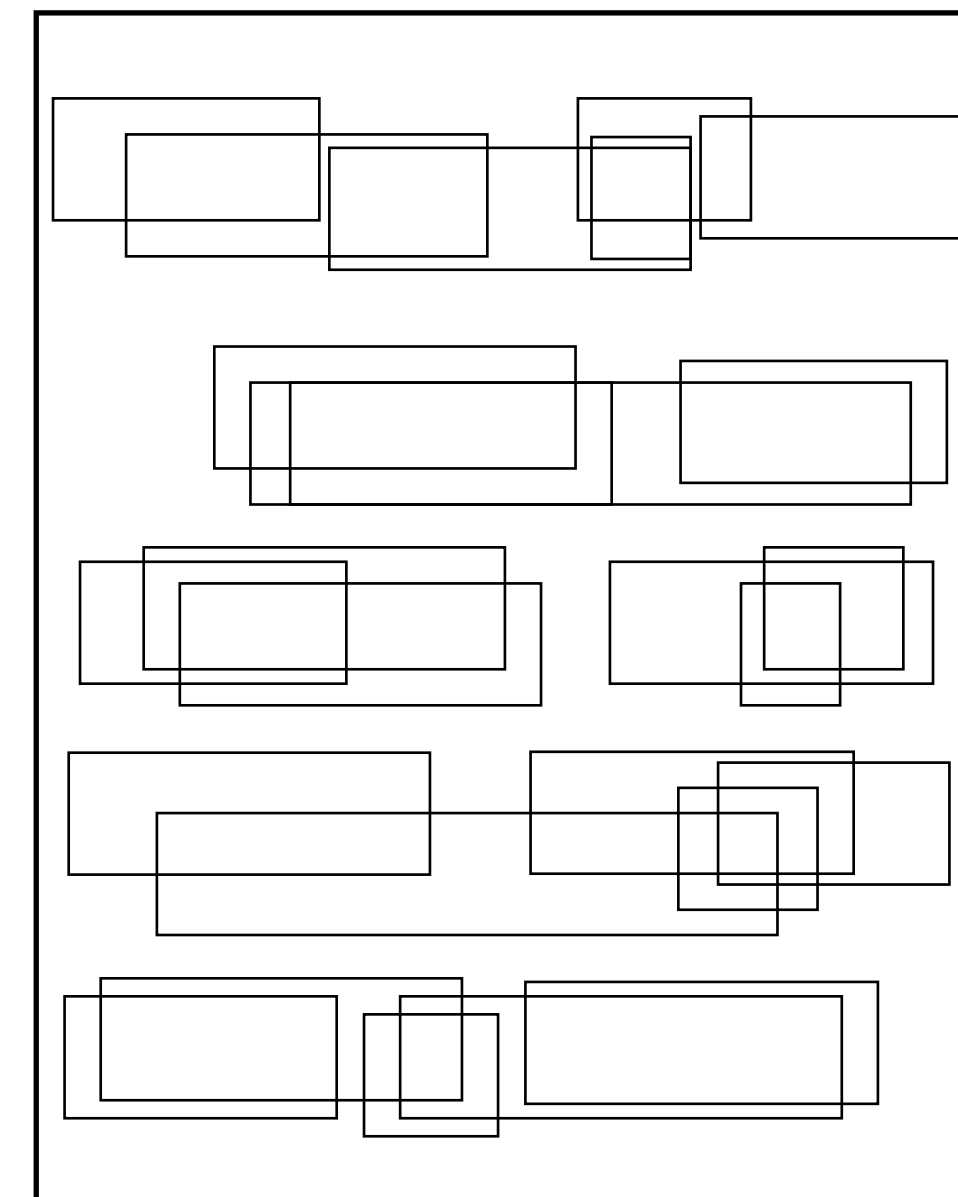
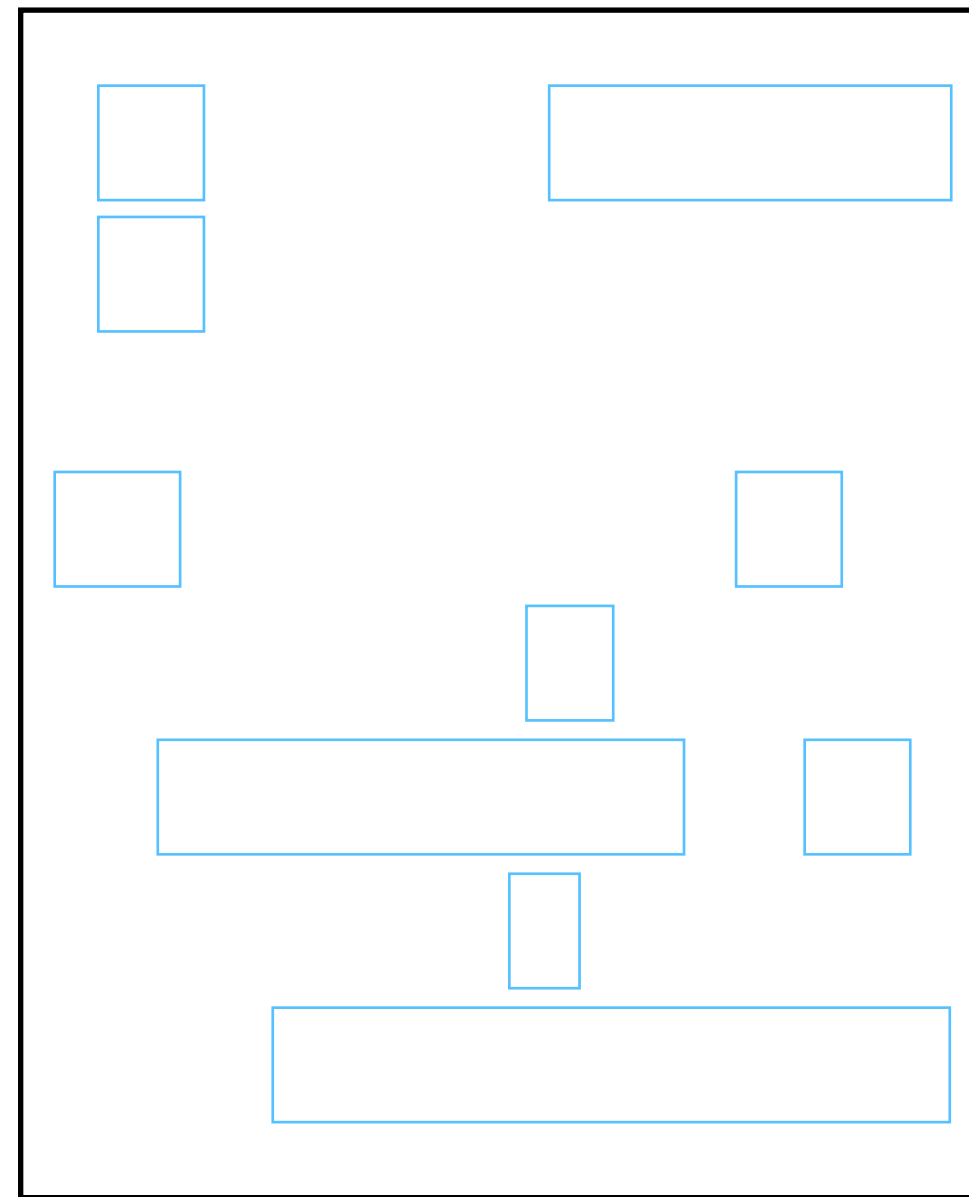


Large output space for complete task t

Example: Decomposition for Coreference

Coreference resolution can be decomposed to mention detection and antecedent linking.

Output space of valid annotations:

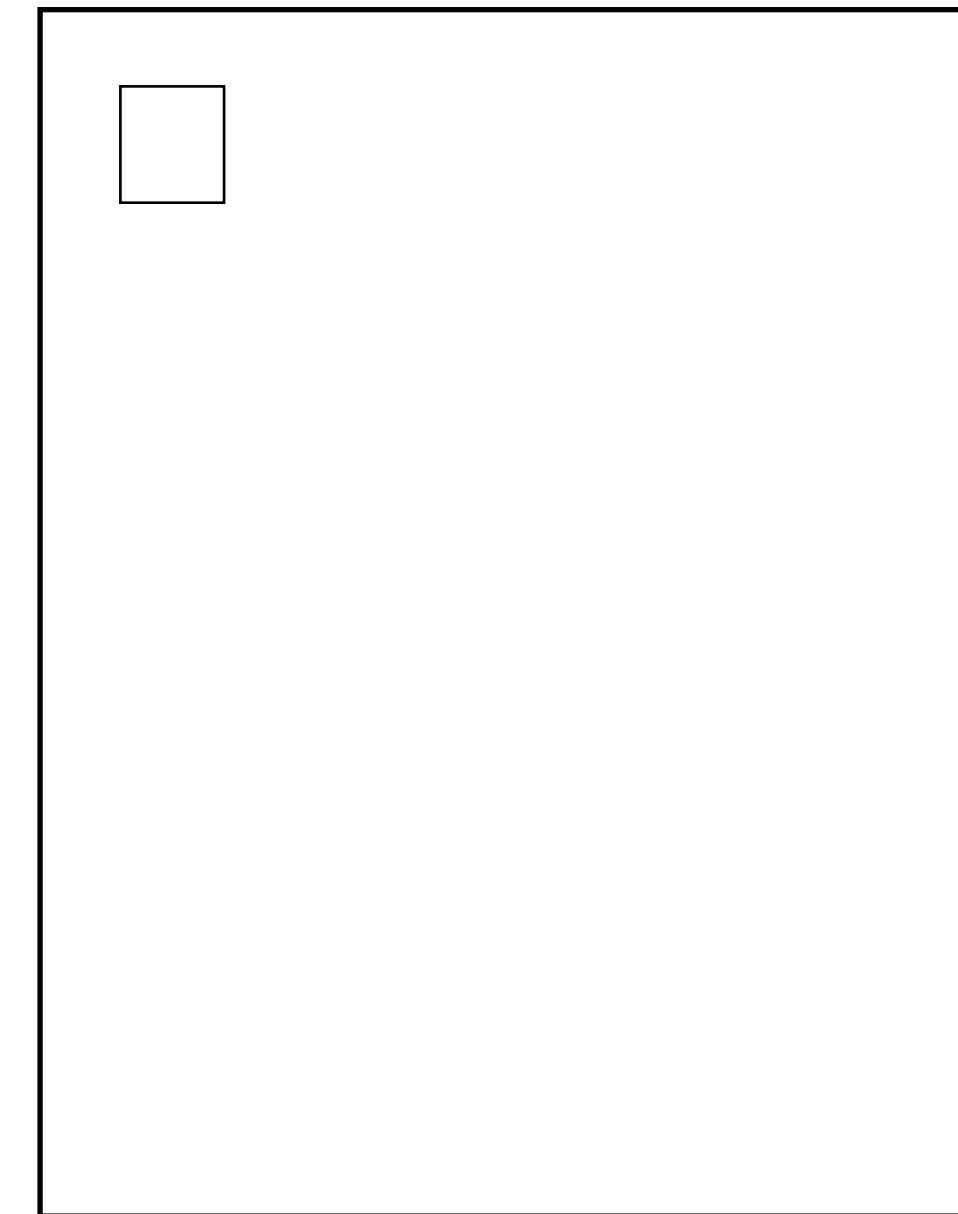
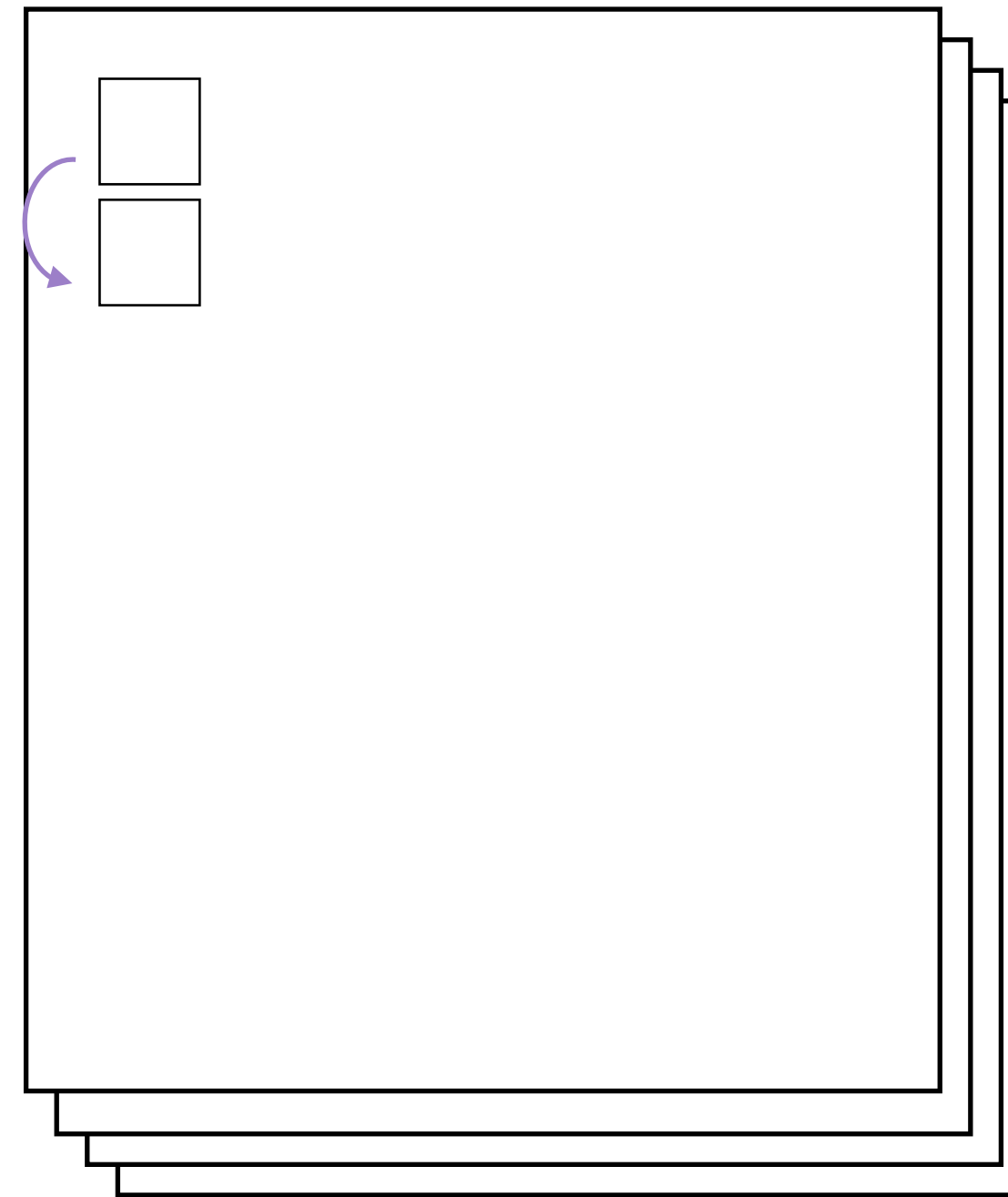


Smaller output space for
mention detection task t_1

Example: Decomposition for Coreference

Coreference resolution can be decomposed to mention detection and antecedent linking.

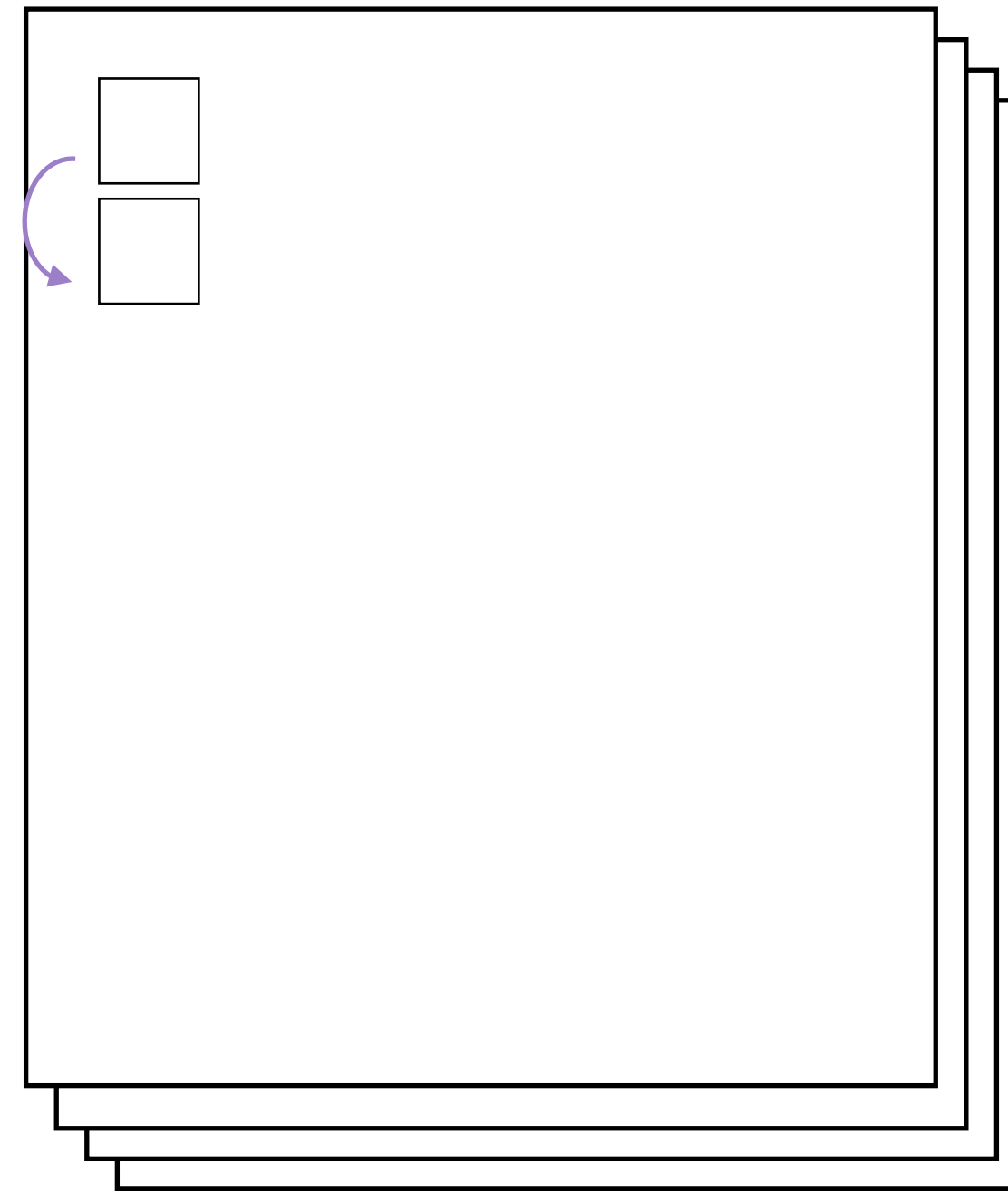
Output space of valid annotations:



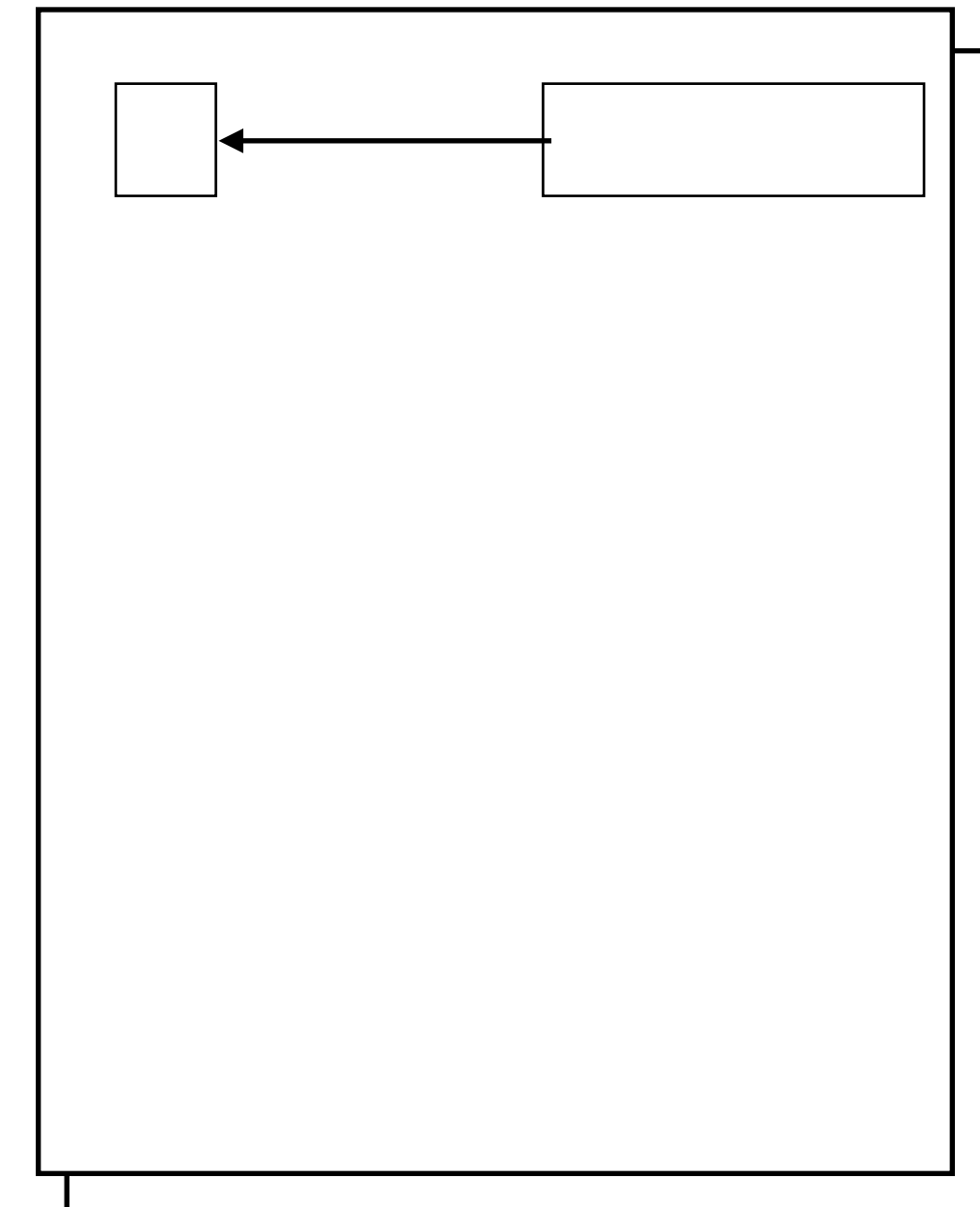
Smaller output space for
antecedent linking task t_2

Example: Decomposition for Coreference

Coreference resolution can be decomposed to mention detection and antecedent linking.



Output space of valid annotations:

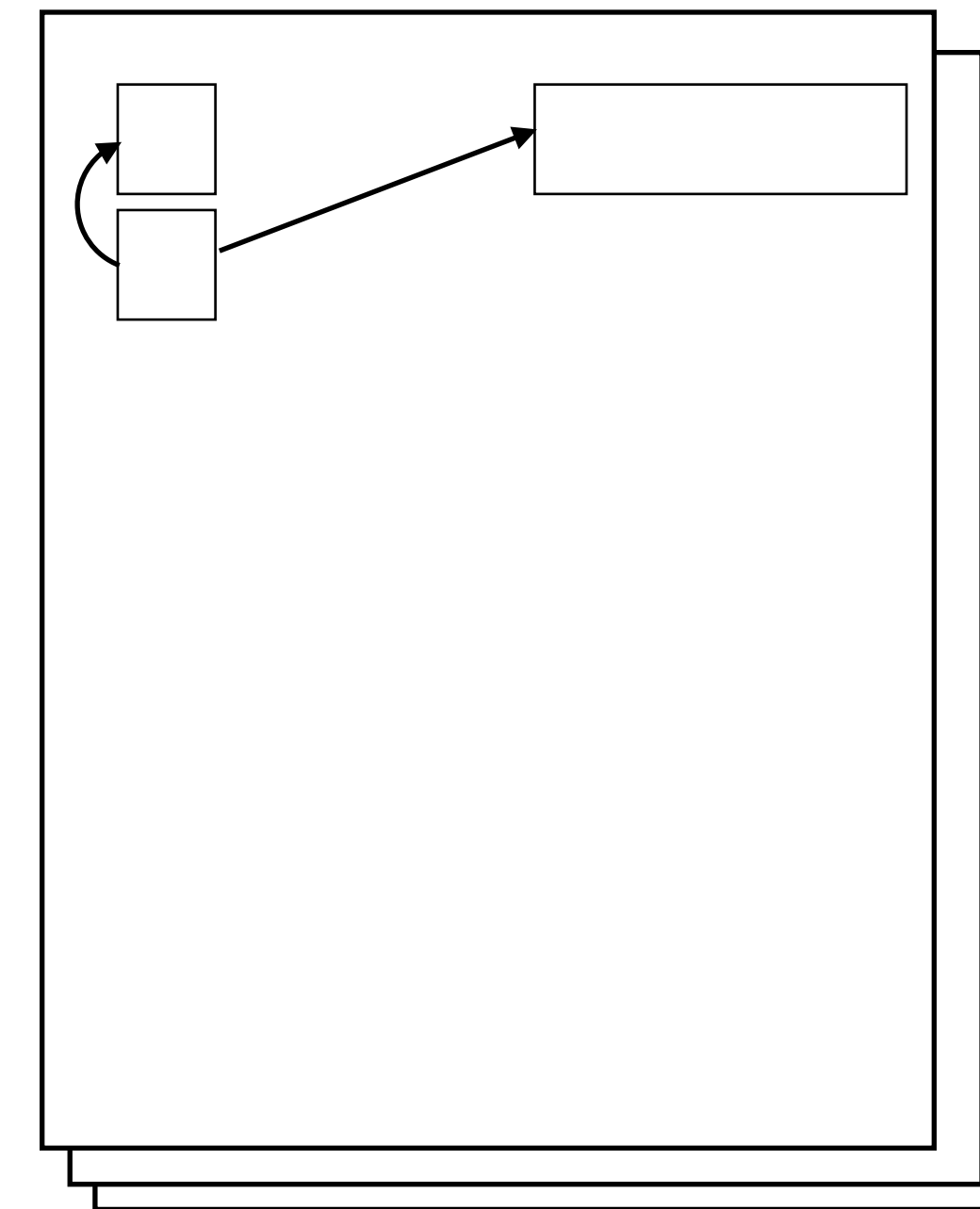
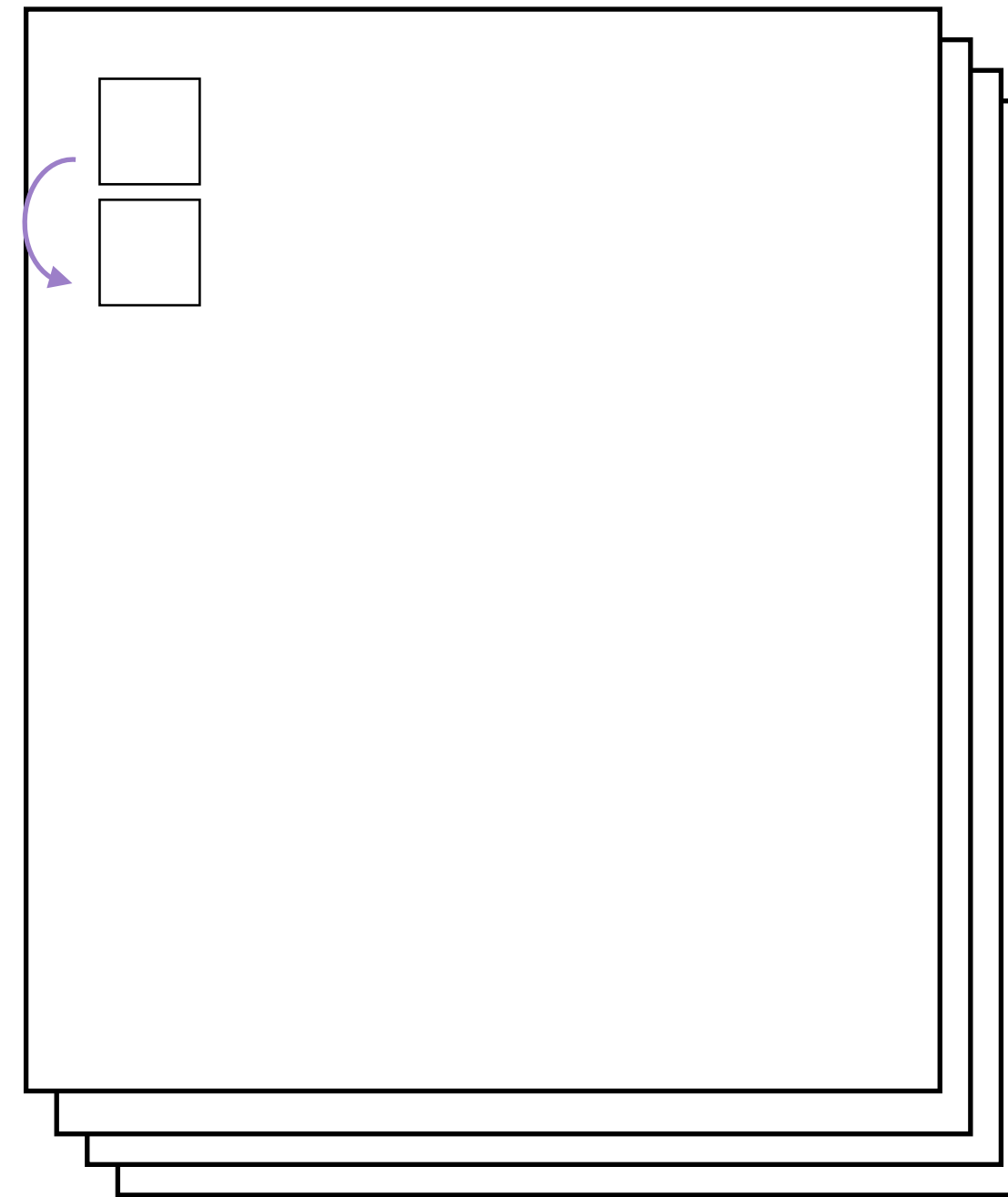


Smaller output space for
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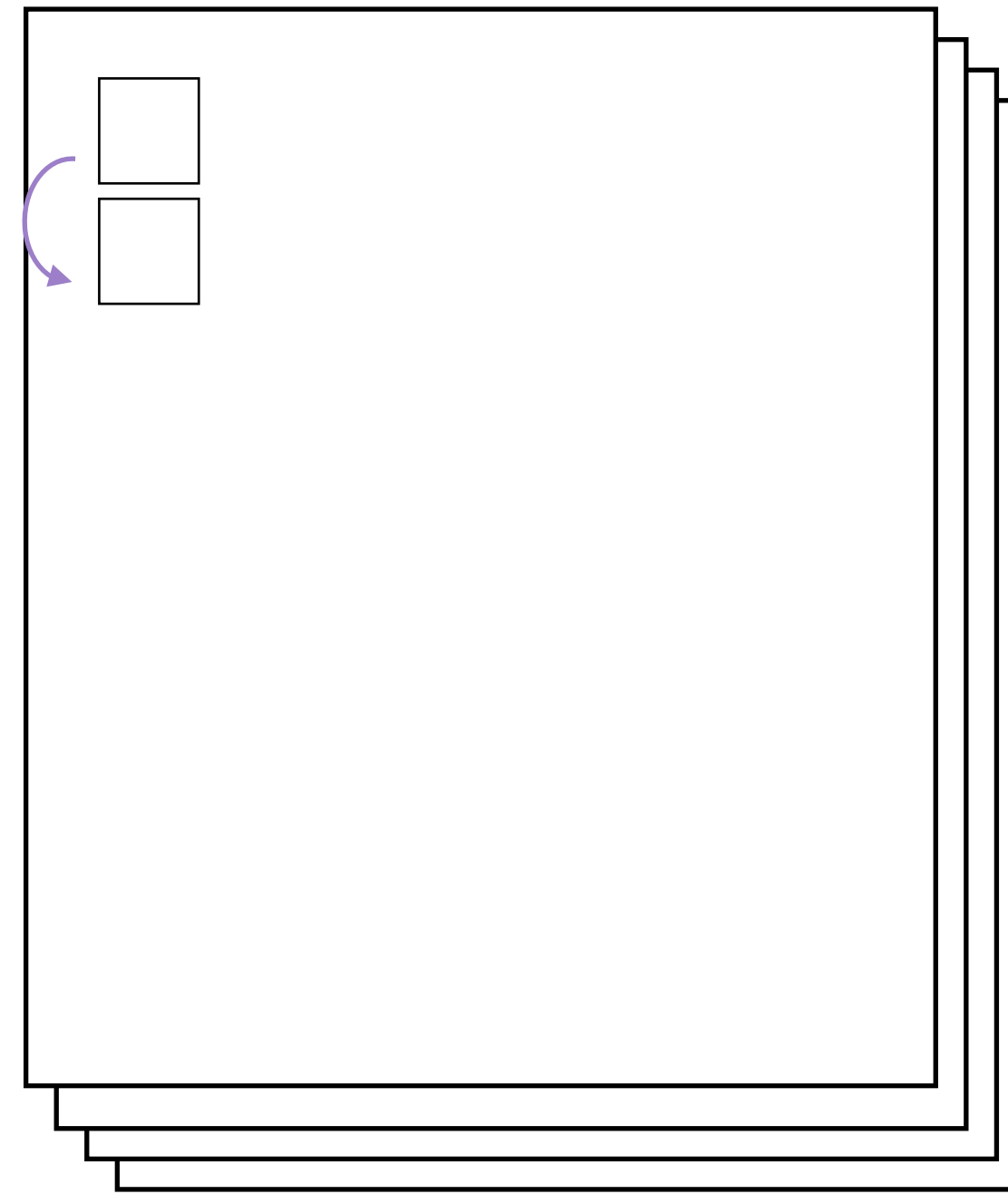
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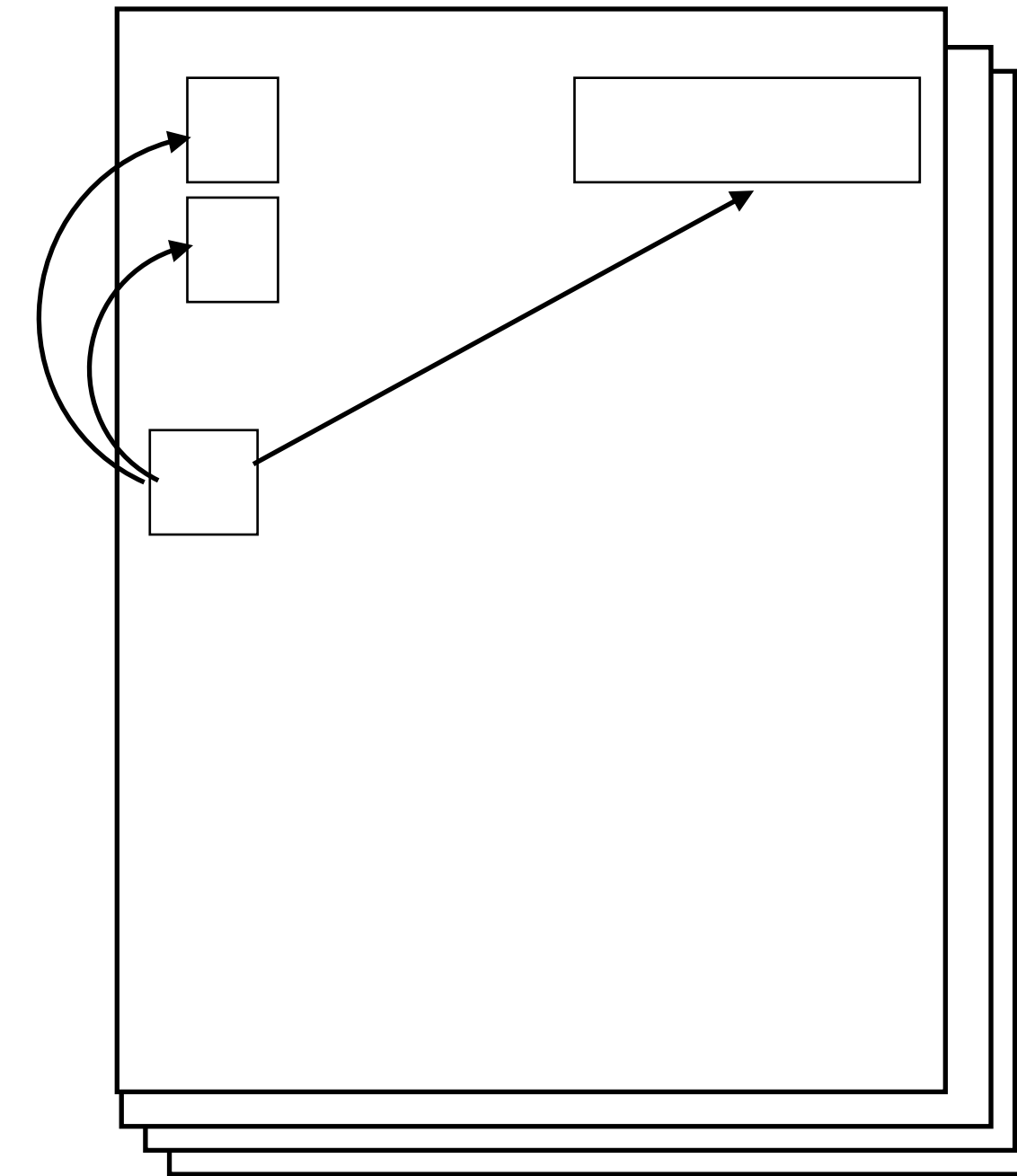
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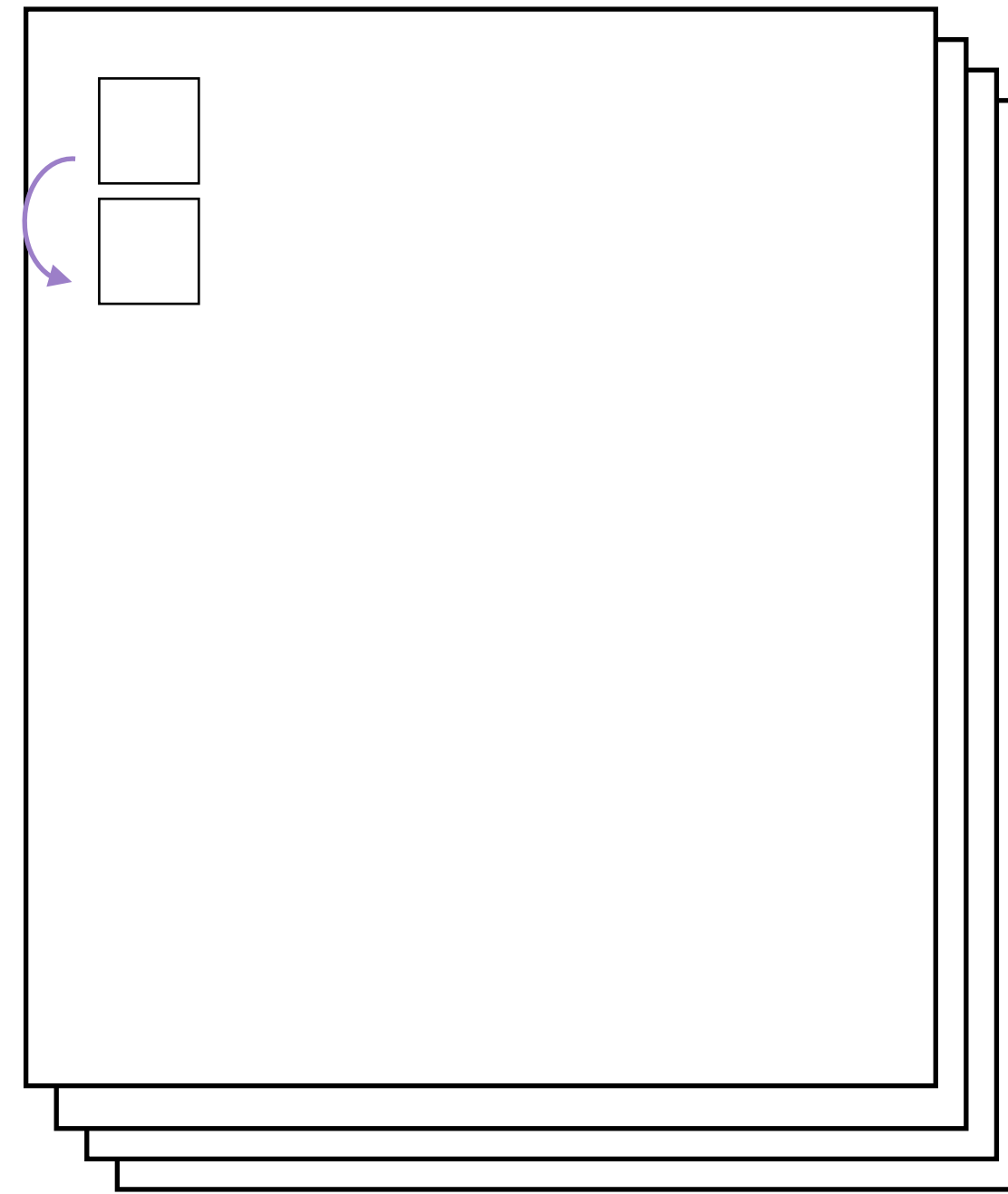
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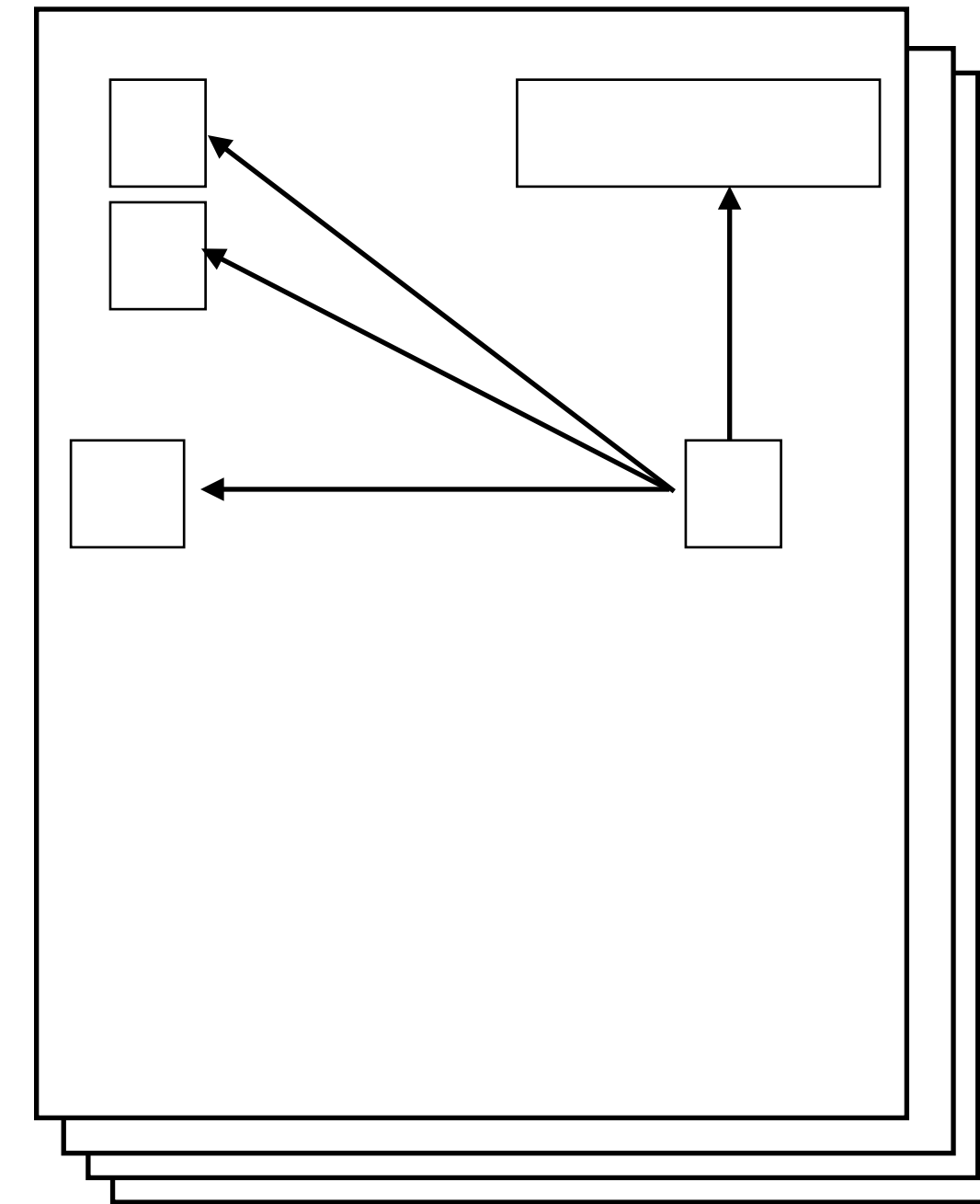
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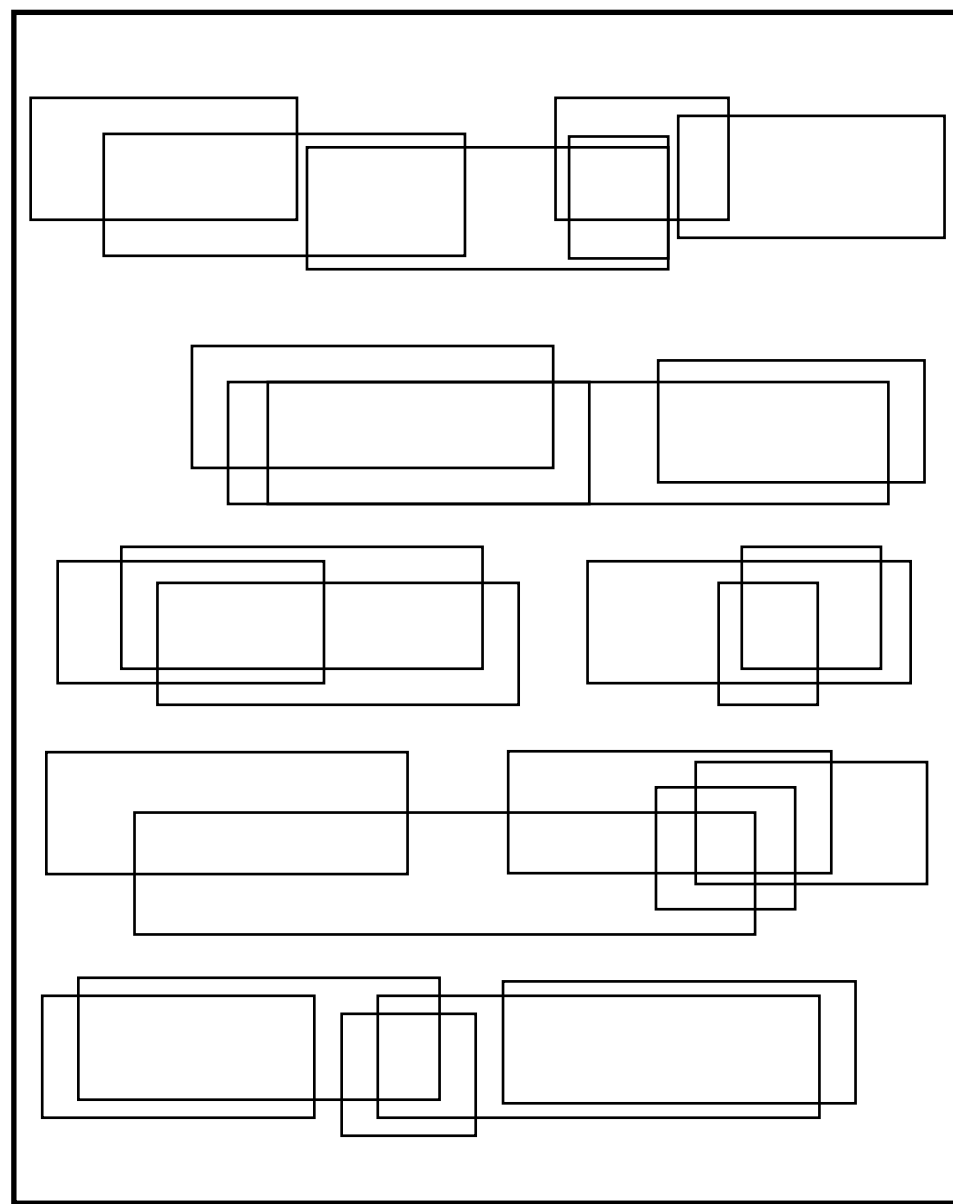
Output space of valid annotations:



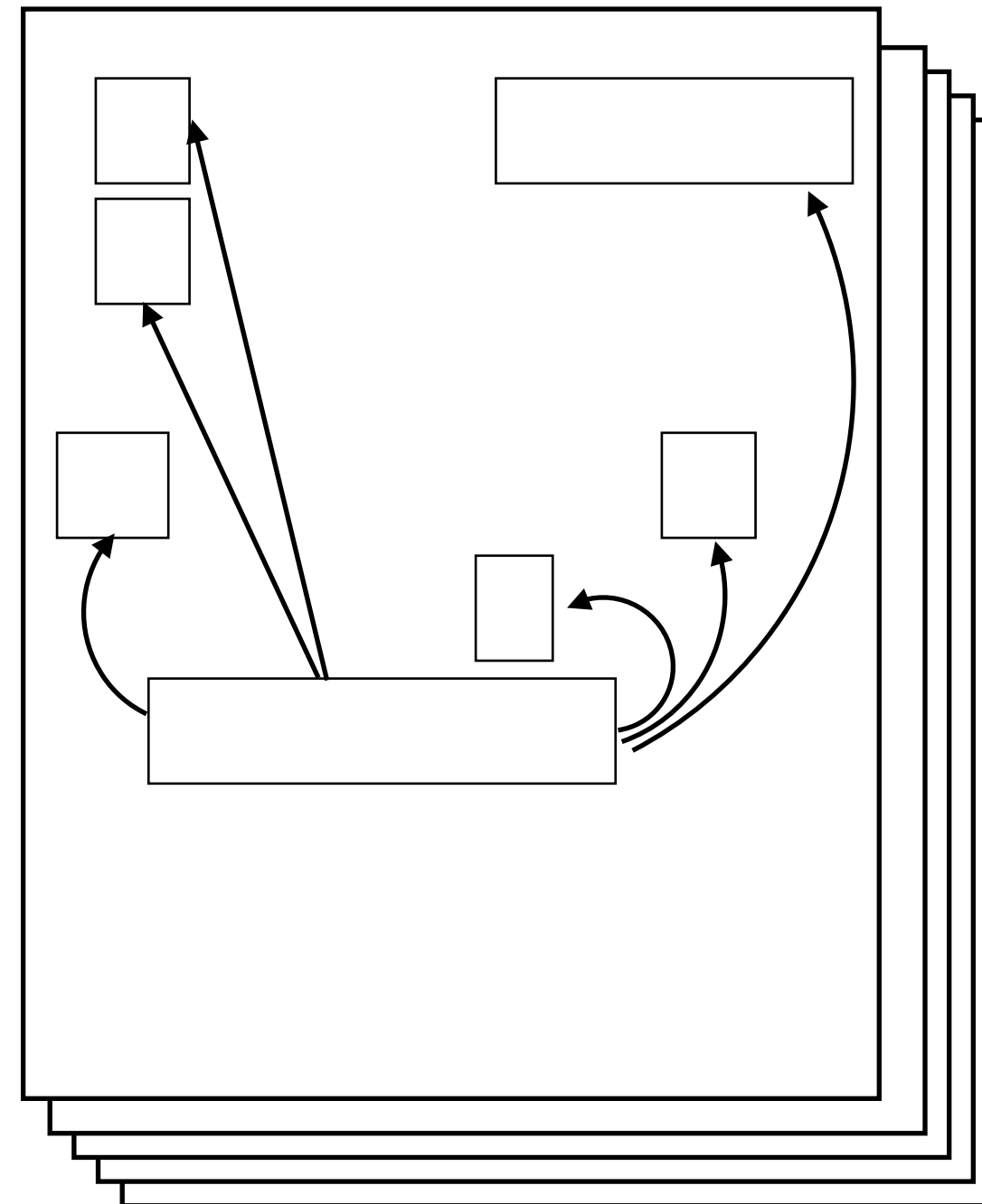
Smaller output space for
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Example: Decomposition for Coreference

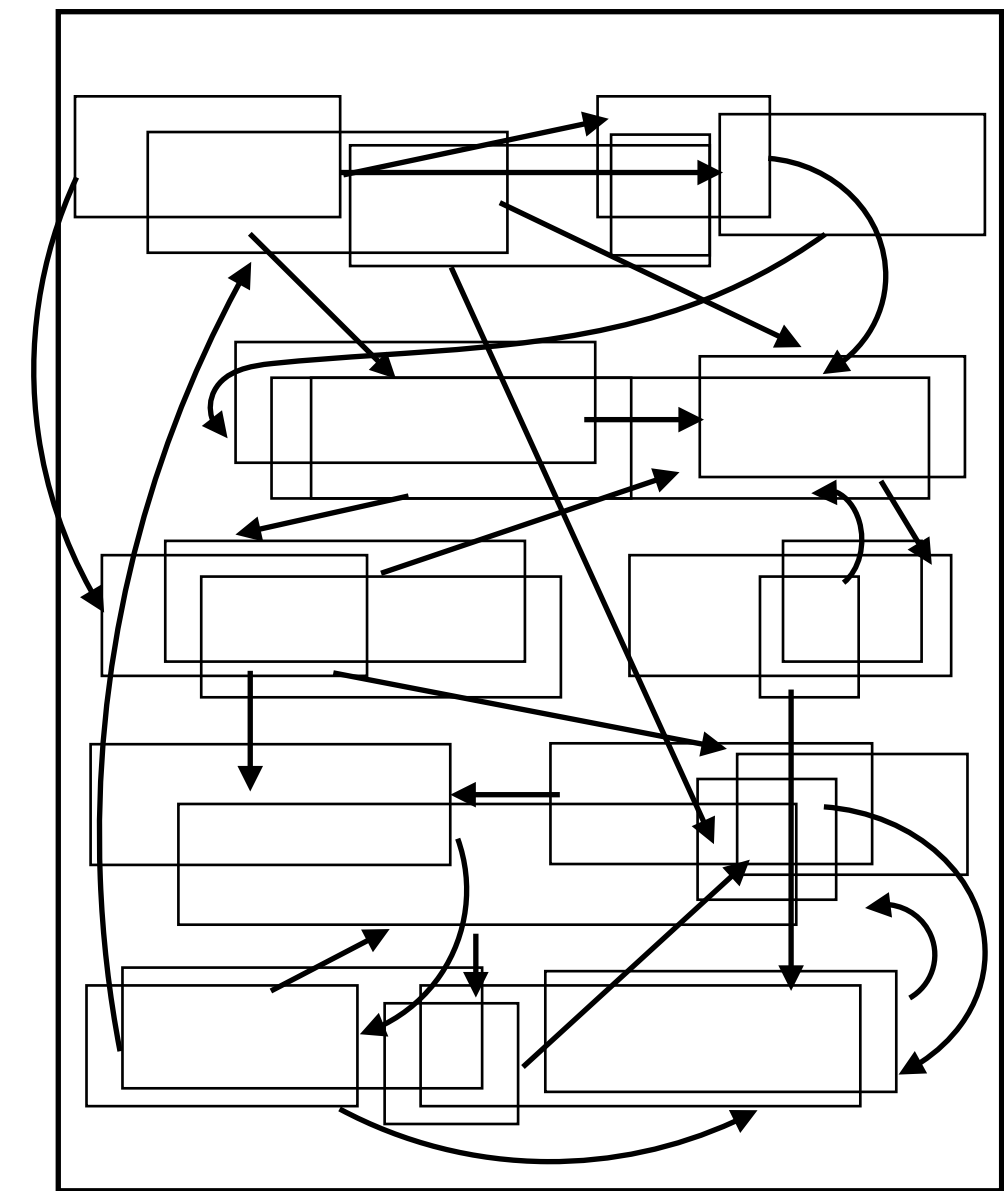
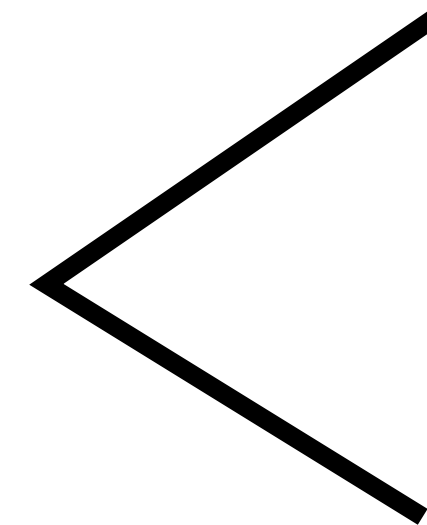
Coreference resolution can be decomposed to mention detection and antecedent linking, resulting in a smaller aggregate inferential load.



Output space for **mention**
detection task t_1



Output space for **antecedent**
linking task t_2



Output space for complete task t

Formalization of Annotation Tasks

We formalize annotation tasks to clarify the relationship between inferential load and task decomposition.

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- Adapted from centering theory (Grosz et al., 1995a)

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 SC gathered information
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 her housing crisis. SC also
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 ASQ screening ... SC
 asked how C was doing
 with seizure medication.

Centers for Coreference Resolution:

Supports Coordinator

Mother

Ages and Stages Questionnaire

Child

Seizure Medication

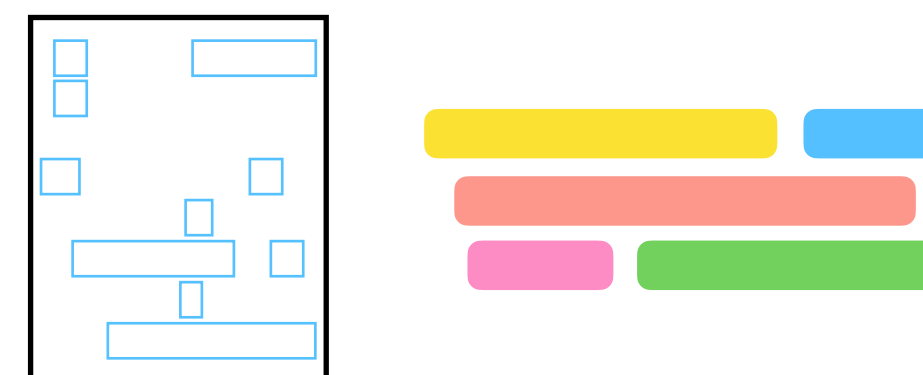
Formalization of Annotation Tasks

For a given task, we either:

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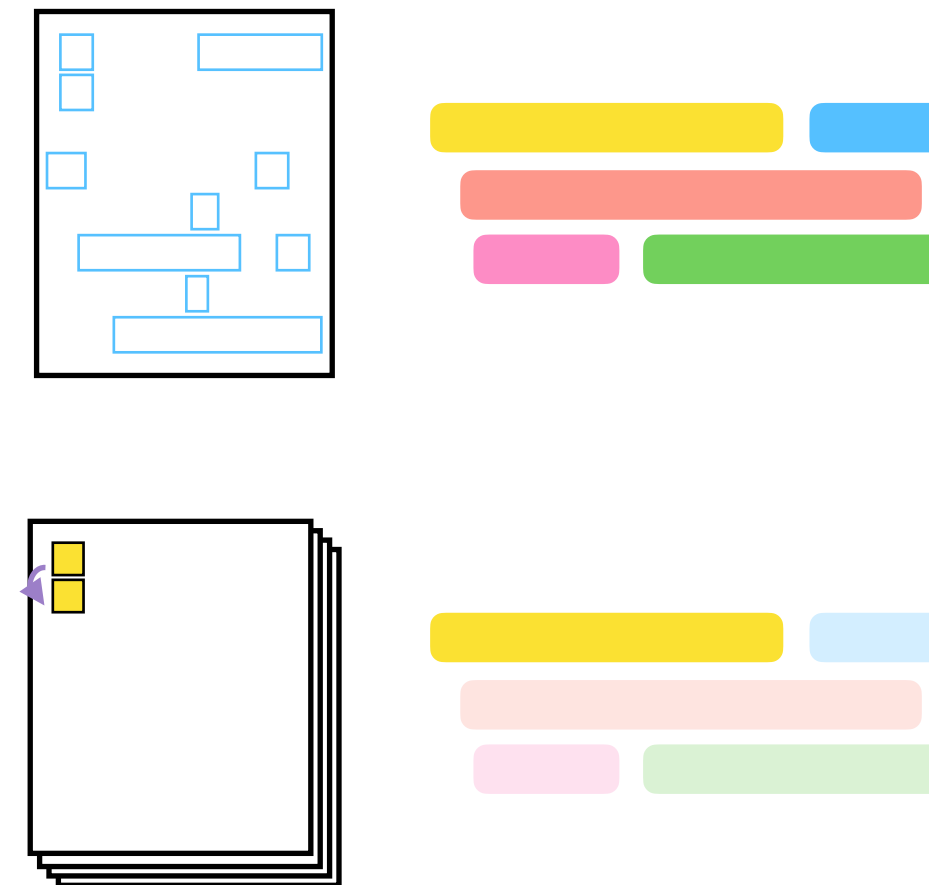
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Identify a discourse entity from the **input**



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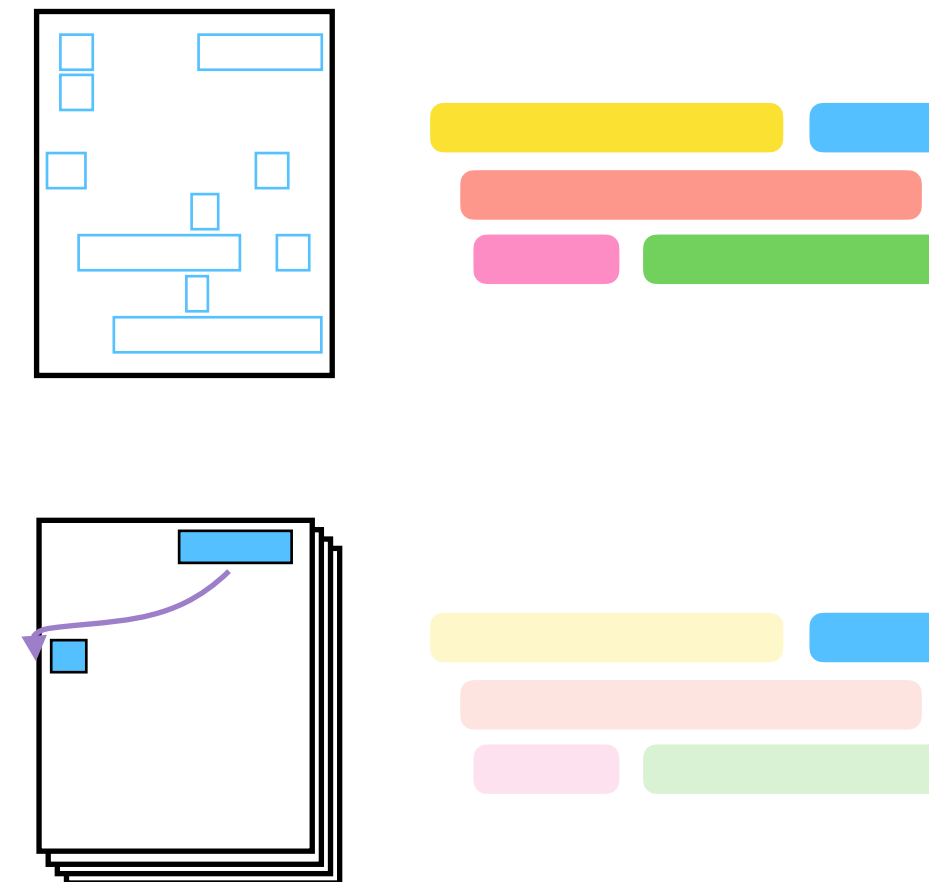
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From a set of **mapping concepts**,
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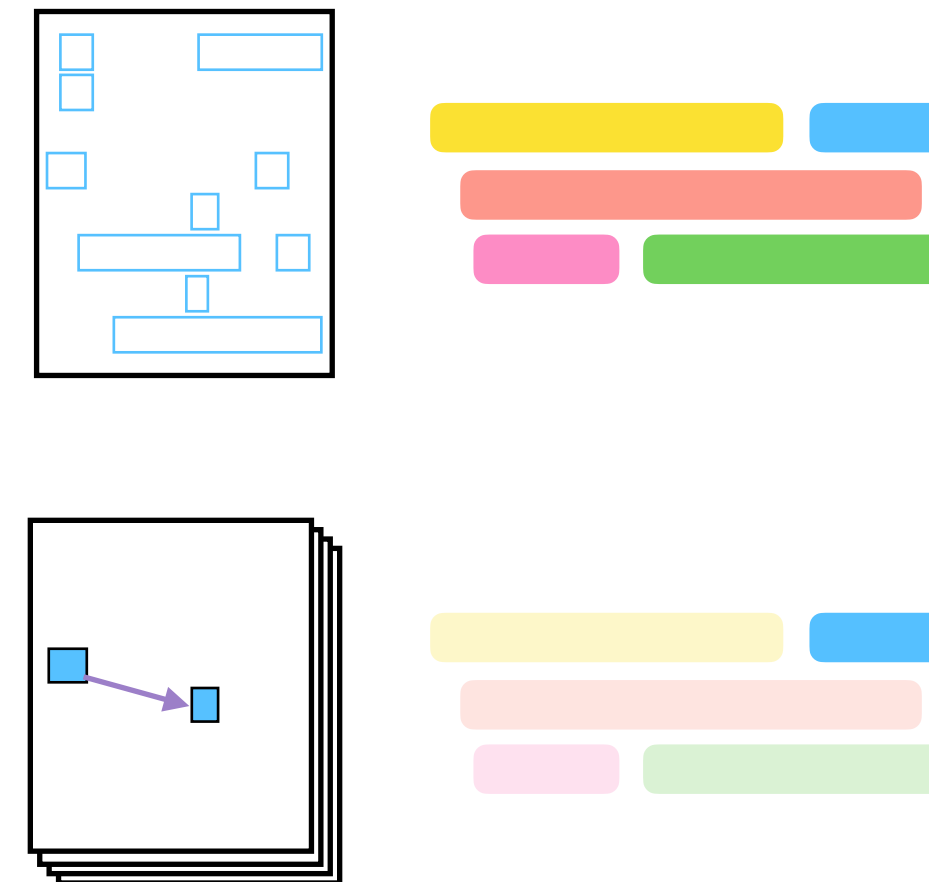
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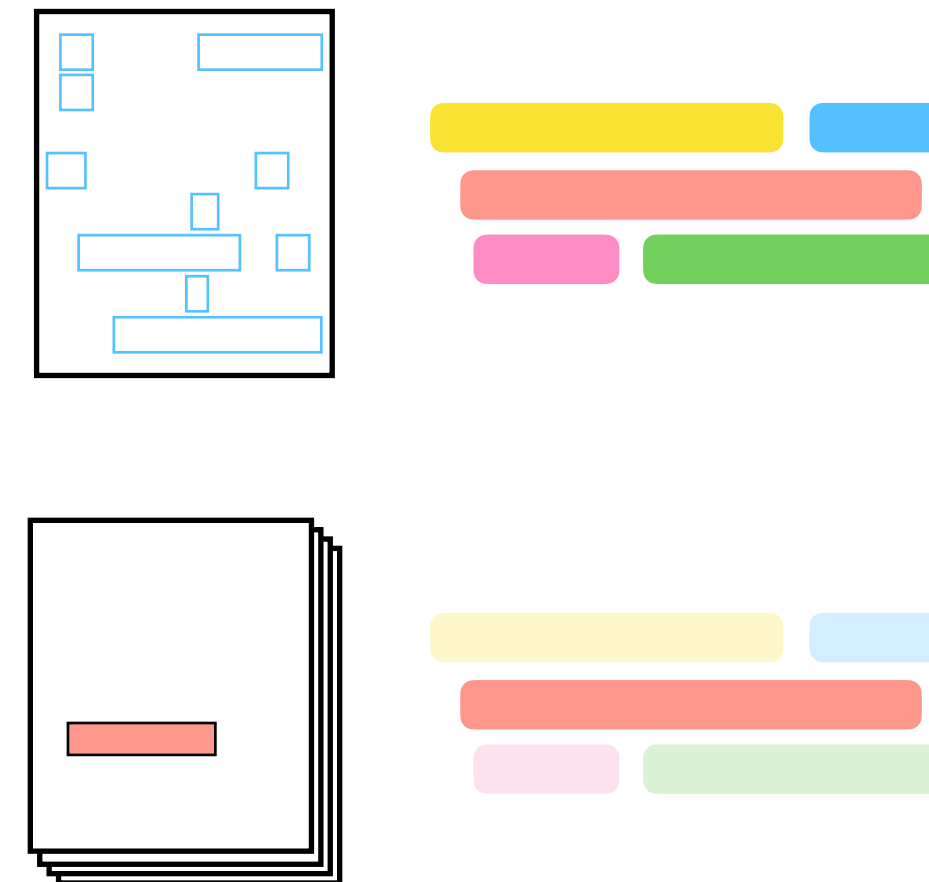
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Formalization of Annotation Tasks

We formalize annotation tasks to clarify the relationship between inferential load and task decomposition.

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- For a given task, we
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Identify a discourse entity from the **input**
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 - Jointly identify and enrich centers



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Output space of valid annotations:

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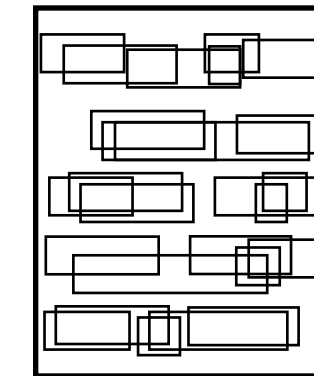
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Large

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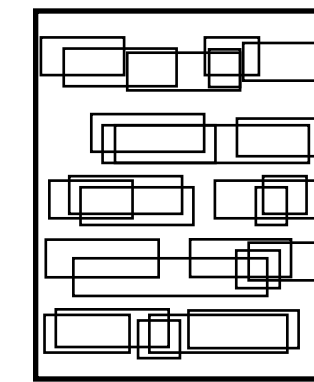
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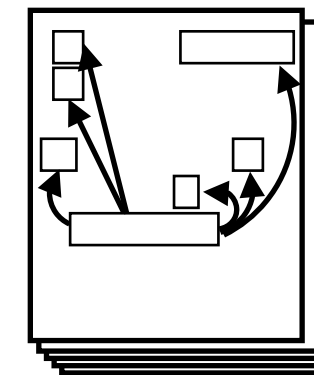
Output space of valid annotations:

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Large

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Small

- Jointly identify and enrich centers

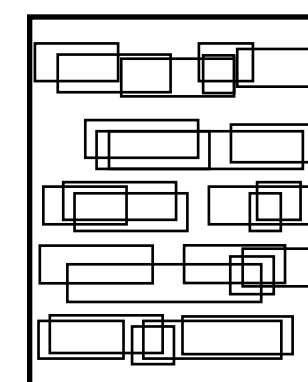
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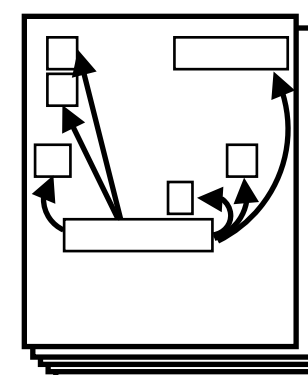
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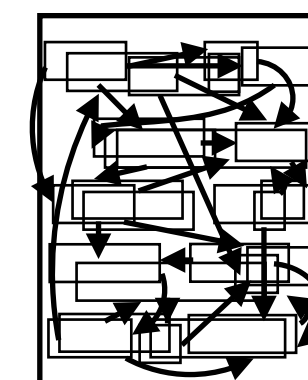
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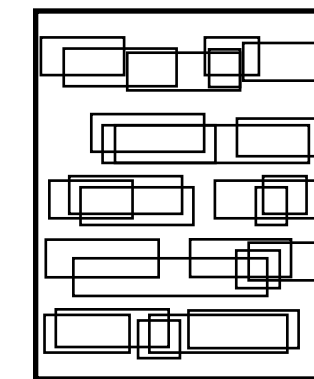
Very large

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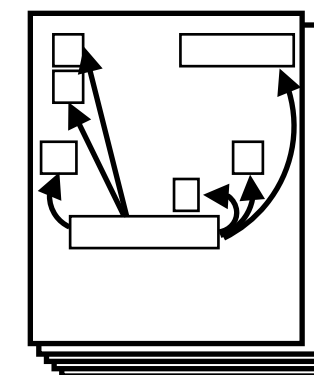
Output space of valid annotations: (Map Space)^{Input Space}

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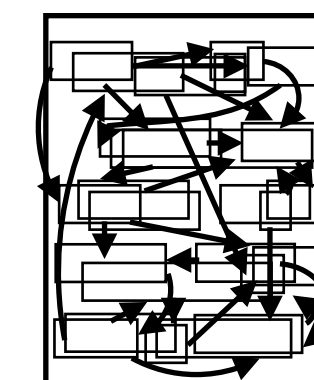
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- Enrich centers:
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Small

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Very large

Characteristics of Efficient Decompositions

Efficient task decompositions that reduce the output space: (Map Space)^{Input Space}

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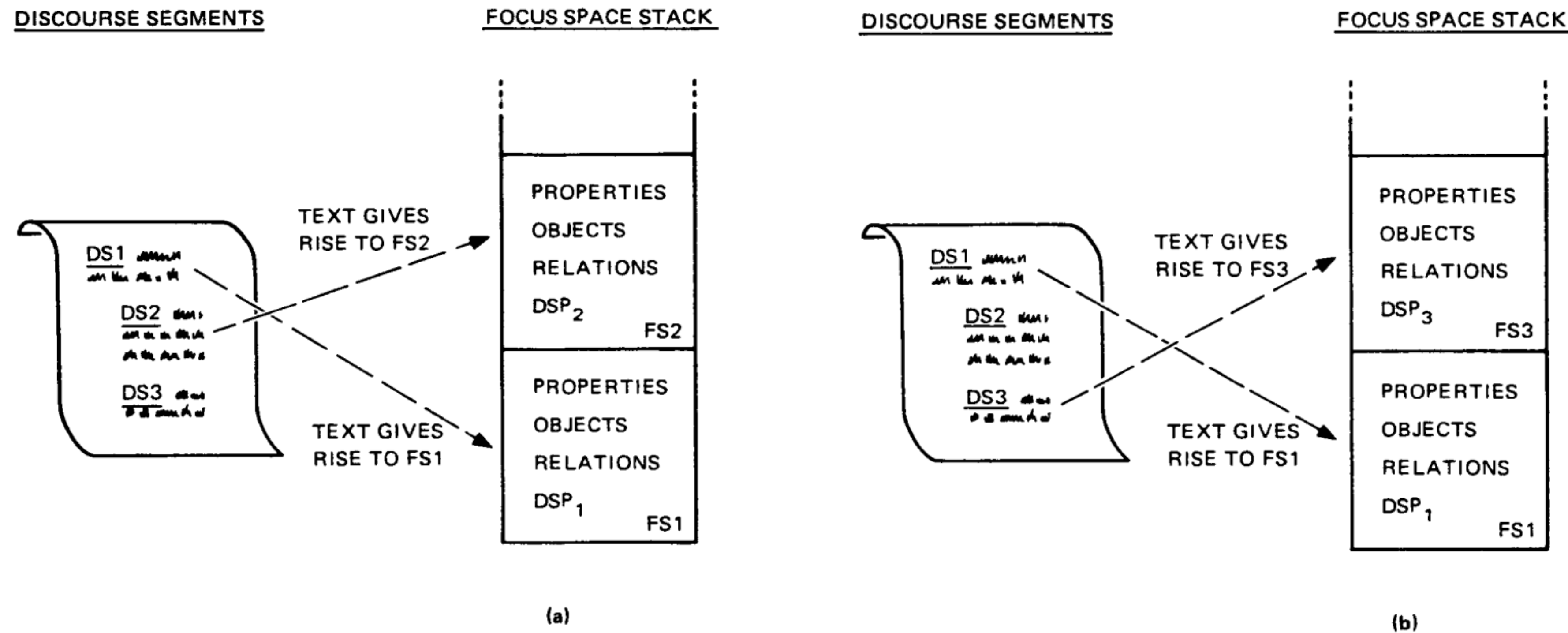


Roadmap

- Motivating Annotation.
- Decomposition-based procedure for reducing the cost of annotation grounded in a model of inferential load.
- **Grounding procedure in the literature.**
- Demonstrations of procedure efficiency.
- Deciding when to decompose.
- Other considerations beyond cost.

Consistency with Centering Theory

In discourse theory, attentional states can be modeled with a stack.



Shifts in attention/focus spaces are costly (Grosz and Sidner 1986).

Consistency with Centering Theory

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- Under our model:
 - Center introduction > Center enrichment
Shifts the annotator's focus

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Smaller output space



Lower inferential load

Consistency with Centering Theory

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Shifts the annotator's focus Attention fixed to the current
focus space

↓
Larger output space

↓
Smaller output space

↓
Higher inferential load

↓
Lower inferential load

Consistency with Centering Theory

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Fewer candidates to consider



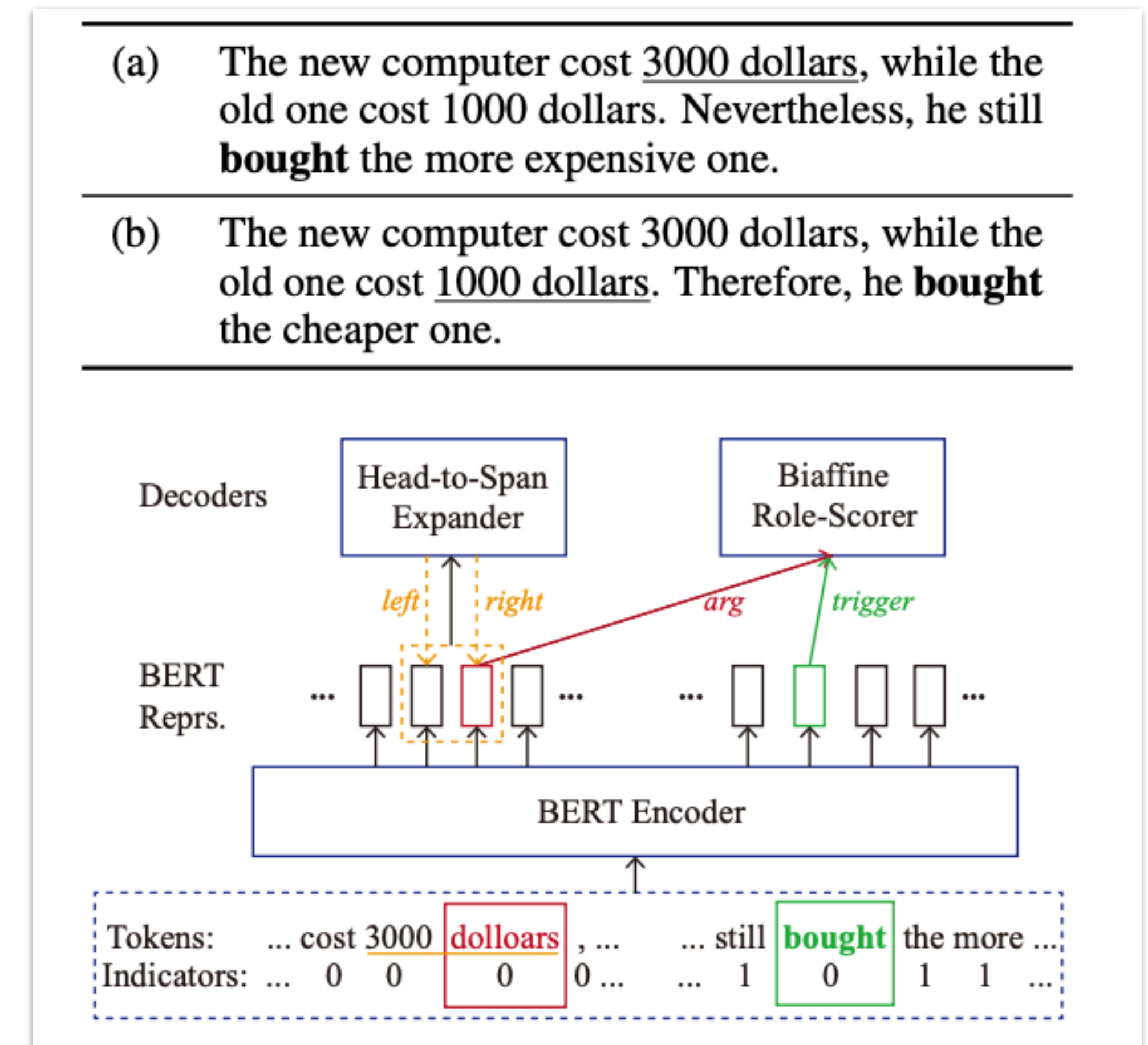
Fewer potential shifts in focus



Lower inferential load

Center-based Decomposition in Prior Work

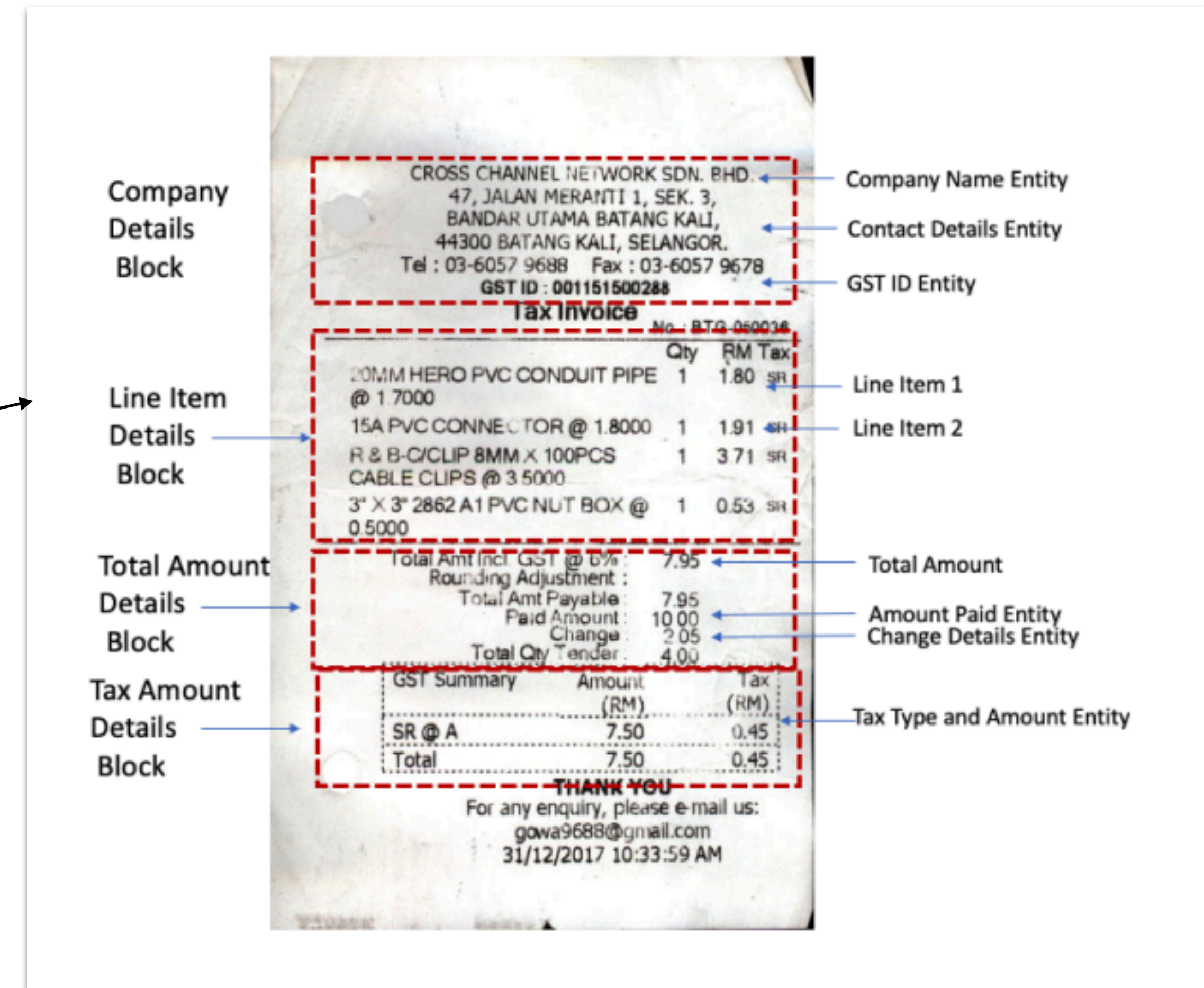
- Reducing aggregate output space is common for both machine-only and human annotation
- Larger atomic space associated with higher annotation time for humans (Wei et al., 2018, Fort et al., 2012) — number of words, dependency distance, number of entities



Decomposition example from Zhang et al., 2020 for event argument detection

Center-based Decomposition in Prior Work

- Task decomposition can make smaller models more competitive with larger models
- Bhattacharyya et al. 2025 decompose visually rich documents into semantic blocks such that 32B models perform at least as well as 200B models



Center-based Decomposition in Prior Work

- Center annotations as anchors to constrain downstream complexity
 - Michael 2023 advocates constructing complex annotations by bootstrapping from simple, narrowly scoped representations.
 - QA-SRL fixes part of the structure to simplify the remaining structure (He et al., 2015)

Temple also said Sea Containers' plan raises numerous legal, regulatory, financial and fairness issues, but didn't *elaborate*.

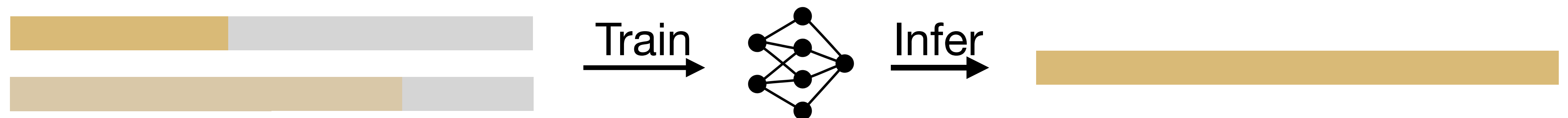
Q:	What didn't <i>elaborate</i> ?
[1] ****	Temple
[2] *	Sea Containers' plan
[3]	None of the above.

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Validating our Model of Inferential Load

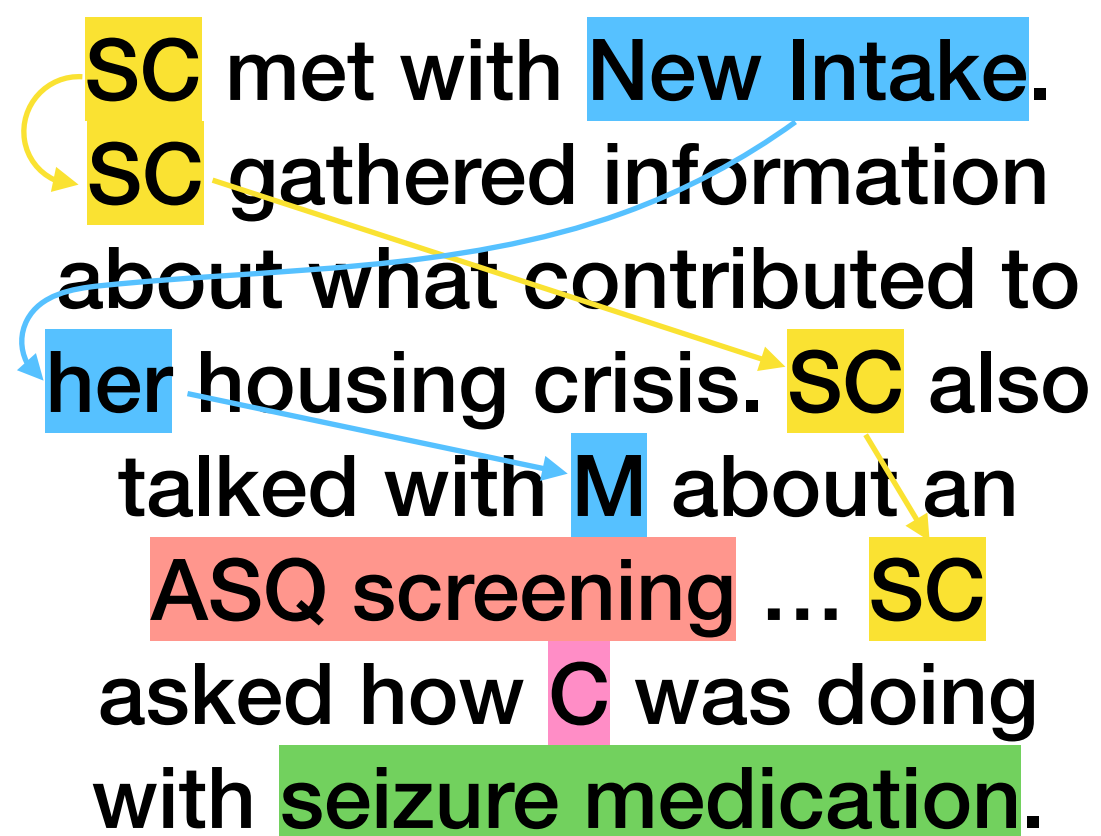
- To validate our model of aggregate inferential load, we **inspect annotation time efficiency gains under decomposition** for benchmark datasets.
- We simulate both annotation procedures over a dataset for a fixed budget, measuring the resulting utility of the dataset with discrepancy between inferred and gold annotations over held-out data.



Train a model with the dataset and evaluate over some held-out GOLD annotations

Example: Decomposition for Coreference

In coreference resolution, we jointly identify mentions of relevant entities and link mentions with their antecedents.



SC met with New Intake.
SC gathered information
about what contributed to
her housing crisis. SC also
talked with M about an
ASQ screening ... SC
asked how C was doing
with seizure medication.

The diagram illustrates coreference resolution on a text snippet. Entities are highlighted with colored boxes: 'SC' (yellow), 'New Intake' (blue), 'her' (blue), 'M' (blue), 'ASQ screening' (red), 'C' (pink), and 'seizure medication' (green). Arrows indicate coreference links: a yellow arrow from the first 'SC' to the second 'SC'; a blue arrow from 'New Intake' to 'her'; a blue arrow from 'her' to 'SC'; a blue arrow from 'SC' to 'C'; and a yellow arrow from 'SC' to 'seizure medication'.

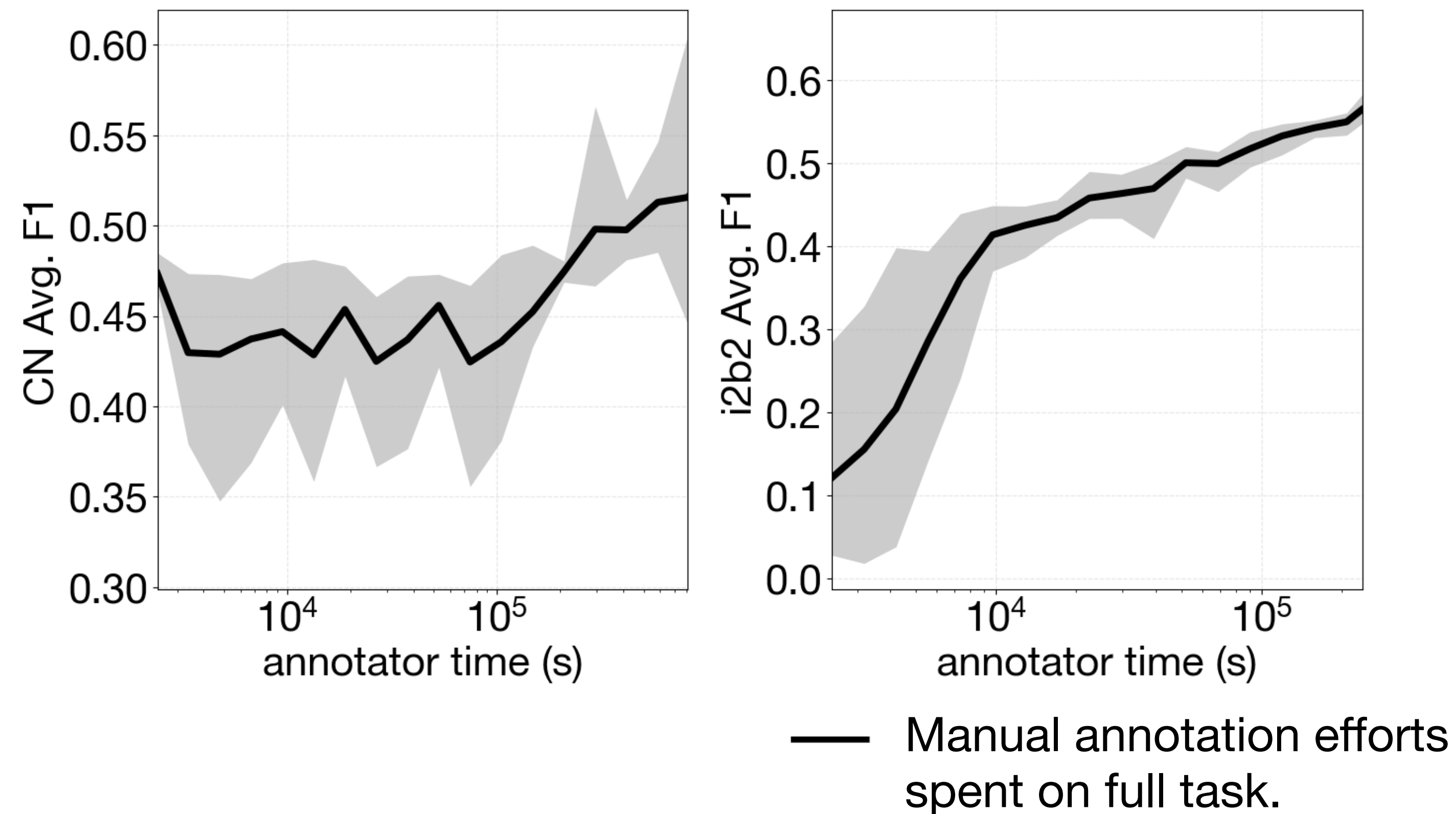
Anonymized annotated example snippet
of caseworker note from CN dataset.

We experiment with coreference datasets:

- Clinical notes from the i2b2/VA Shared Task (**i2b2**) (Uzuner et al., 2012).
- Child Welfare Caseworker Notes (**CN**) from Allegheny County DHS which we annotated (Gandhi et al., 2023).

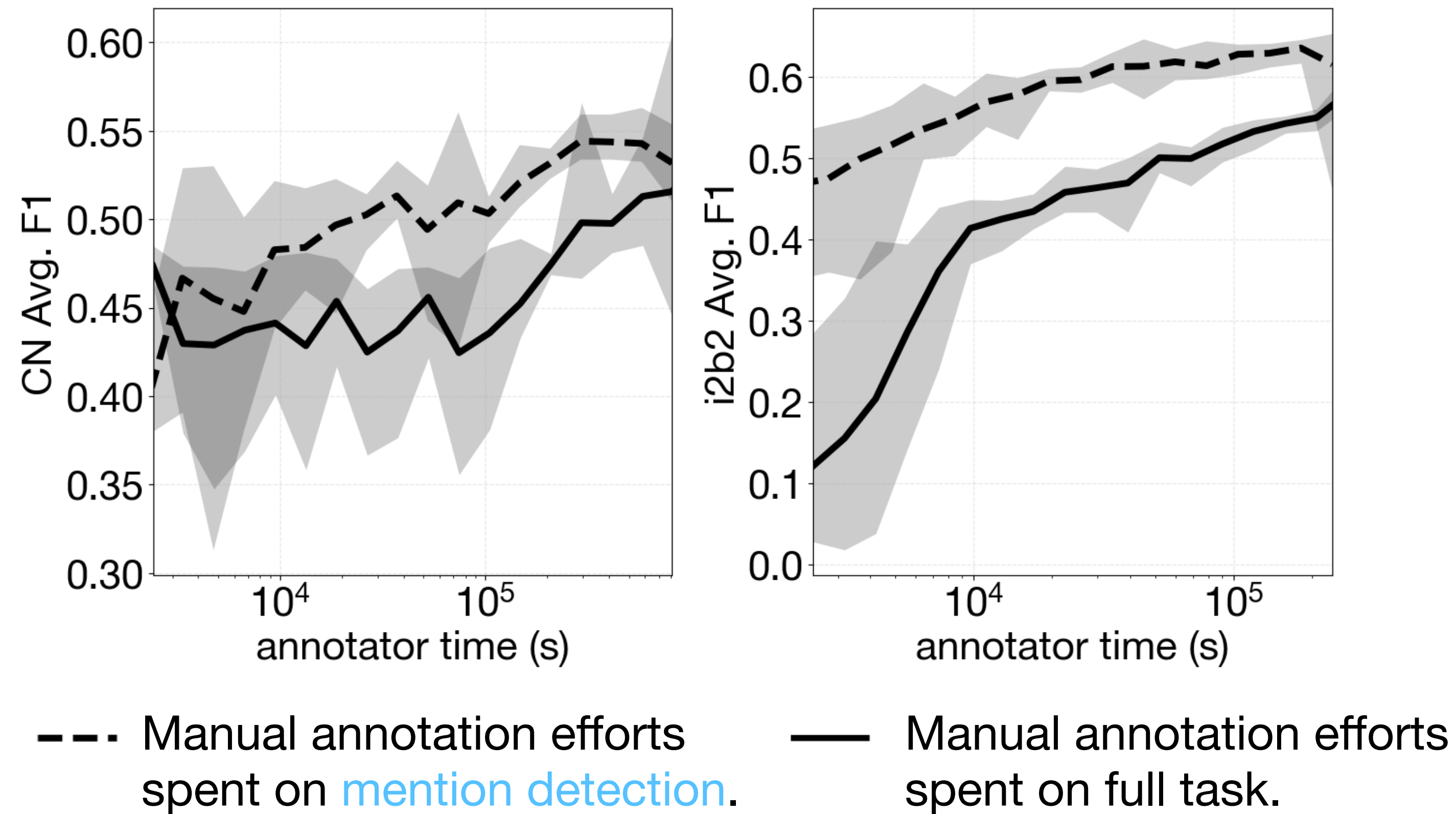
Example: Decomposition for Coreference

More efficient to concentrate manual annotation effort on identifying mentions.



Example: Decomposition for Coreference

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Example: Decomposition for Span-bound Concept Annotation

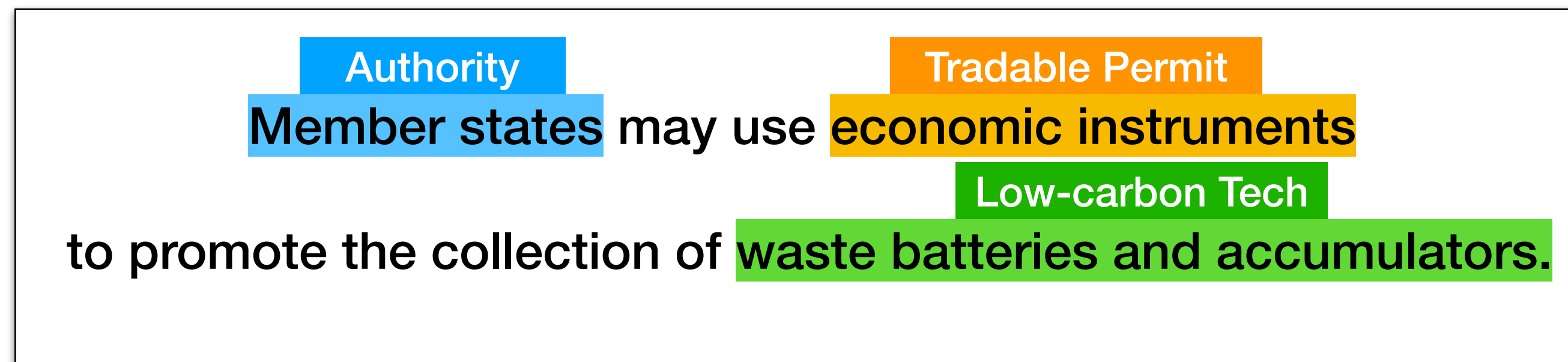
In concept annotation, we jointly identify spans and span types from a fixed set of concepts.

Authority
Member states may use Tradable Permit
economic instruments
Low-carbon Tech
to promote the collection of waste batteries and accumulators.

Example sentence with concept annotations from POLIANNA dataset.

Example: Decomposition for Span-bound Concept Annotation

In concept annotation, we jointly identify spans and span types from a fixed set of concepts.



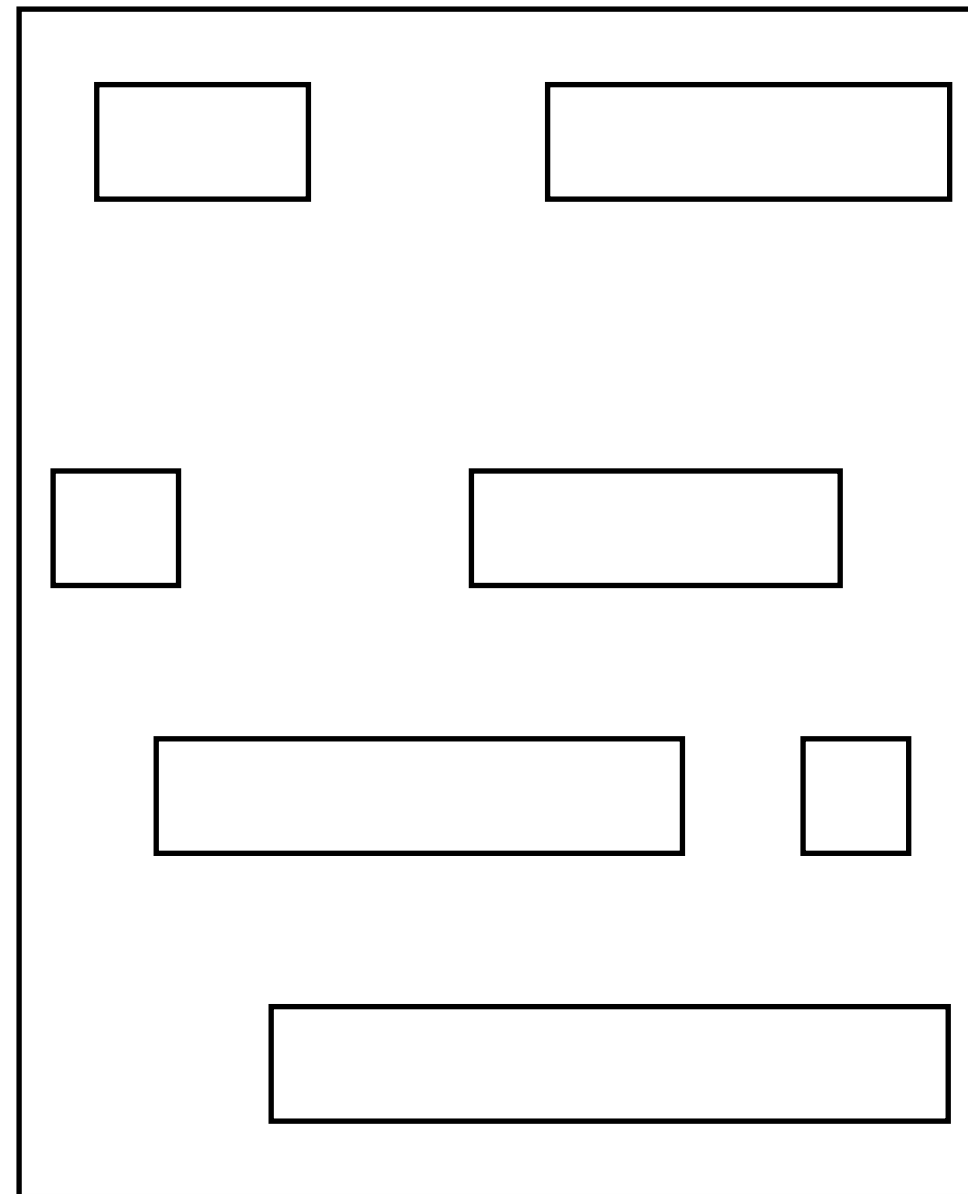
Example sentence with concept annotations from POLIANNA dataset.

We experiment with span-bound concept annotation datasets:

- A climate focused Policy-Design Corpus (**POLIANNA**) annotated with 42 span types (Sewerin et al., 2023).
- The Colorado Richly Annotated Full-Text Corpus (**CRAFT**) containing biomedical articles with hundreds of span types (Bada et al., 2012).

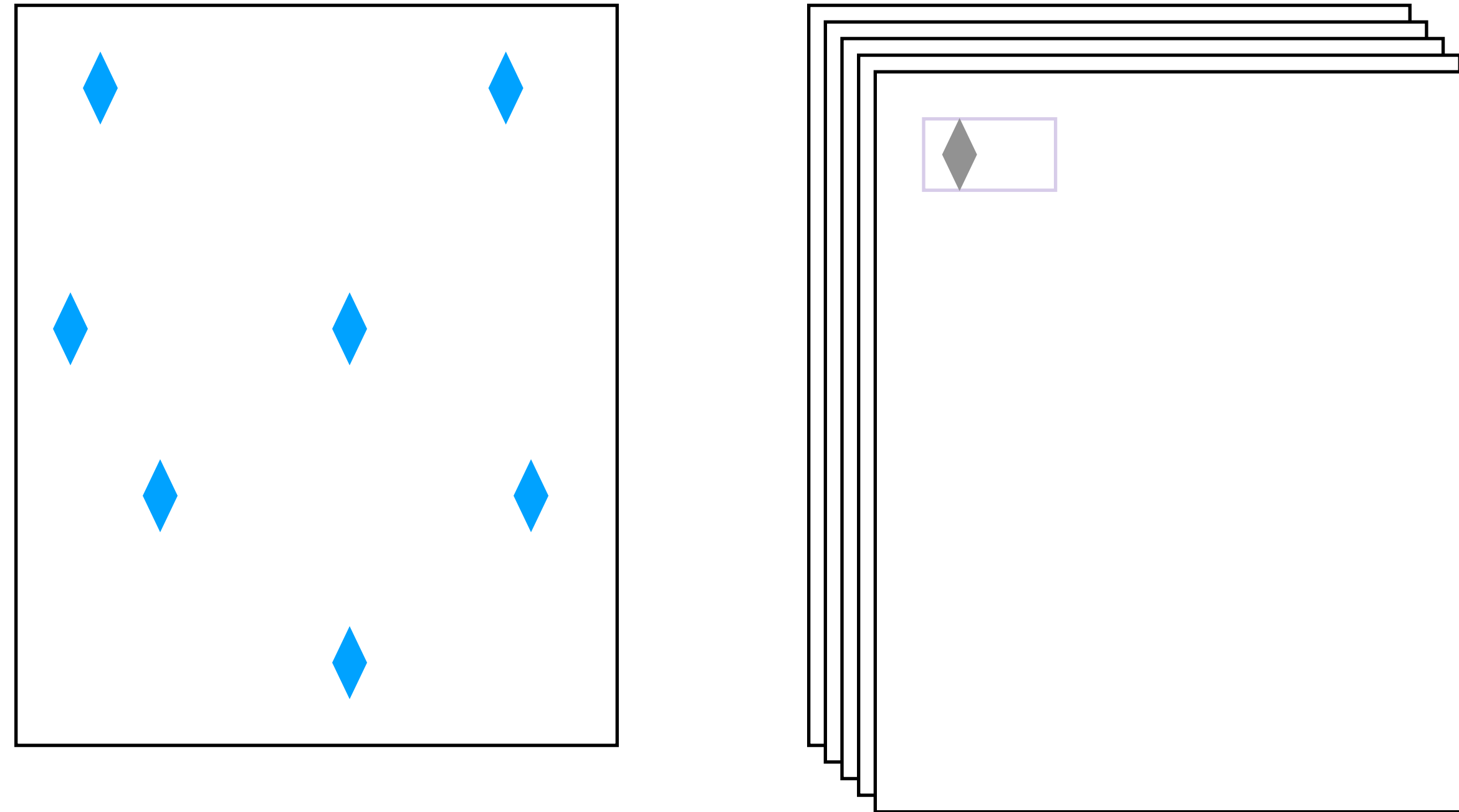
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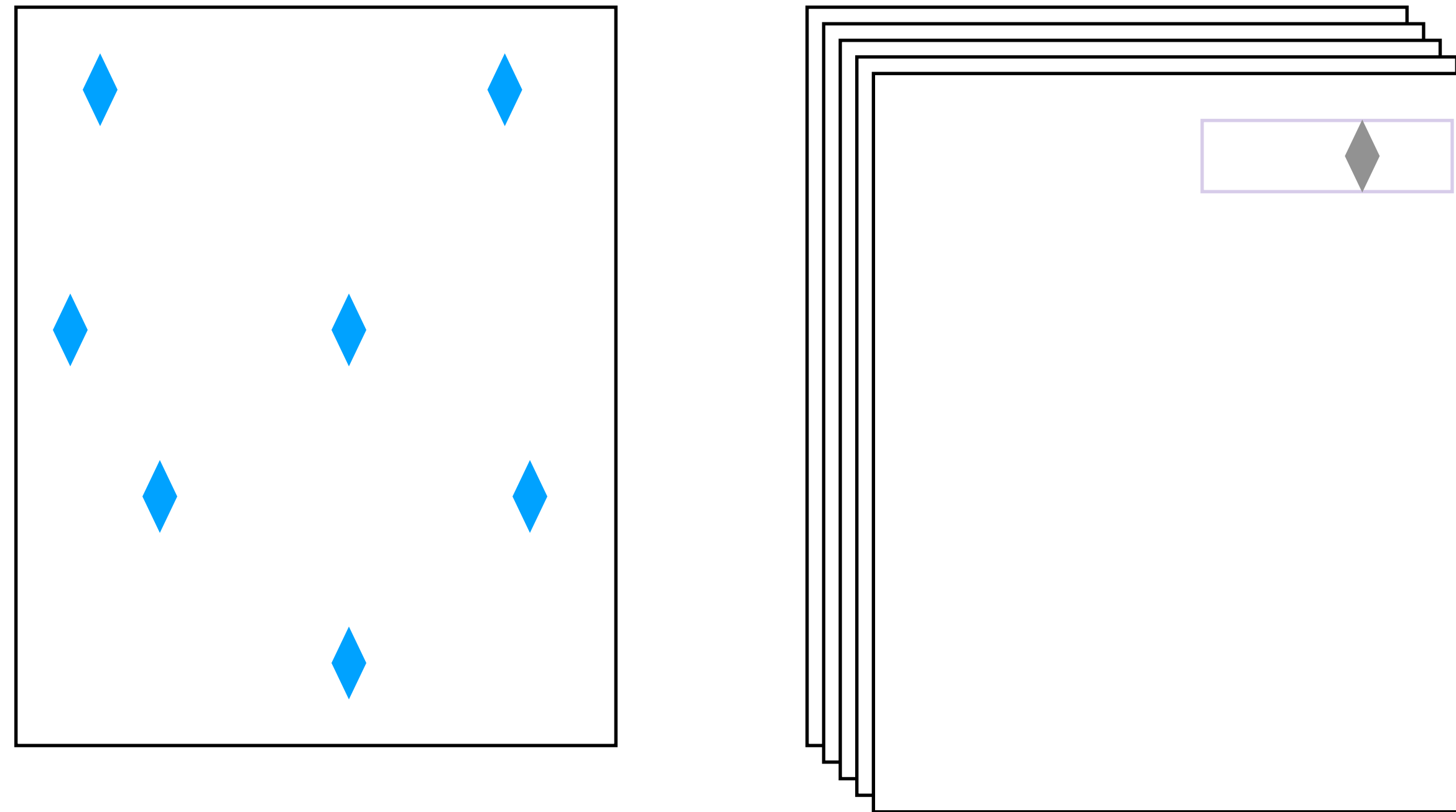
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Span-bound concept annotation can be decomposed to typed position identification and unitization.



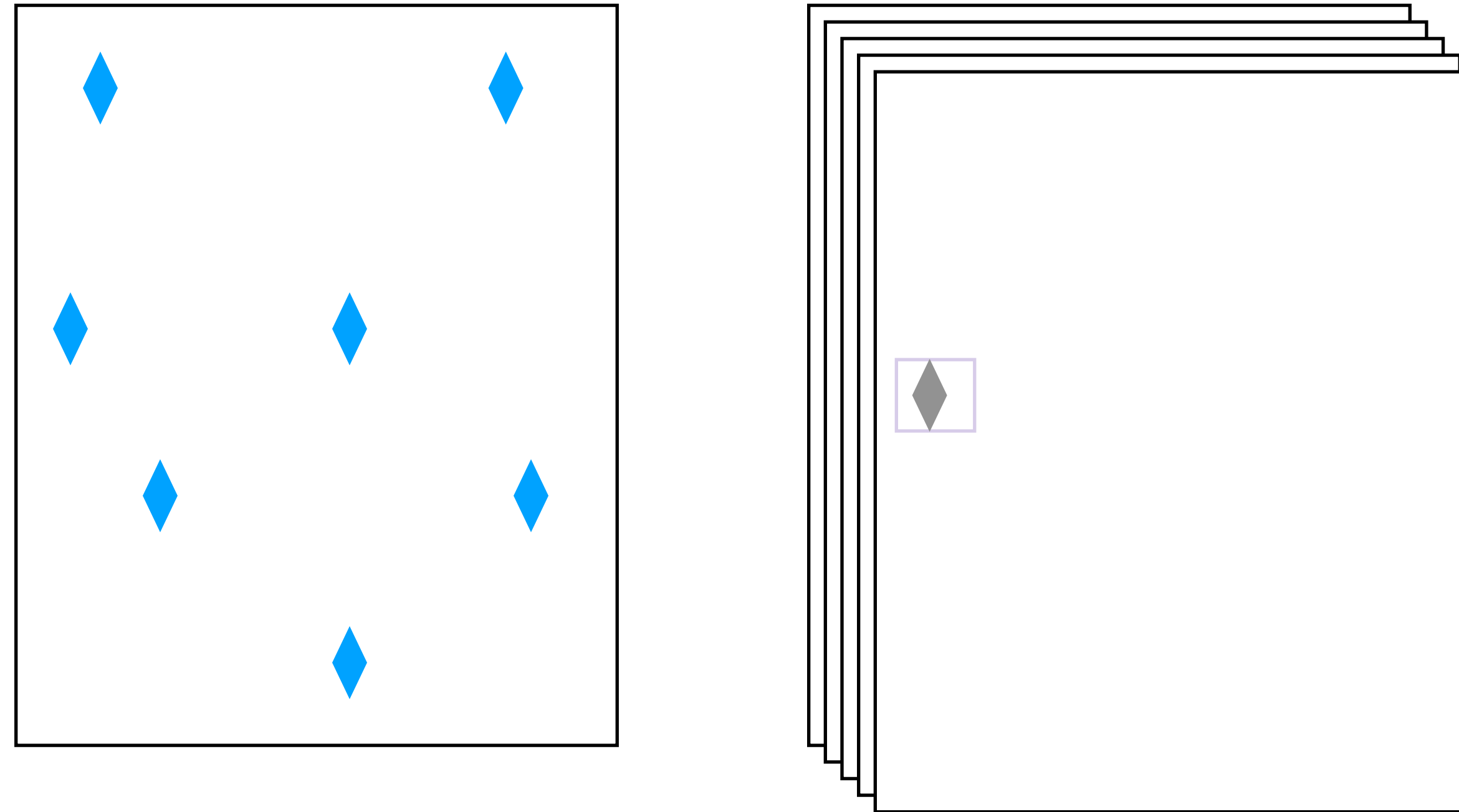
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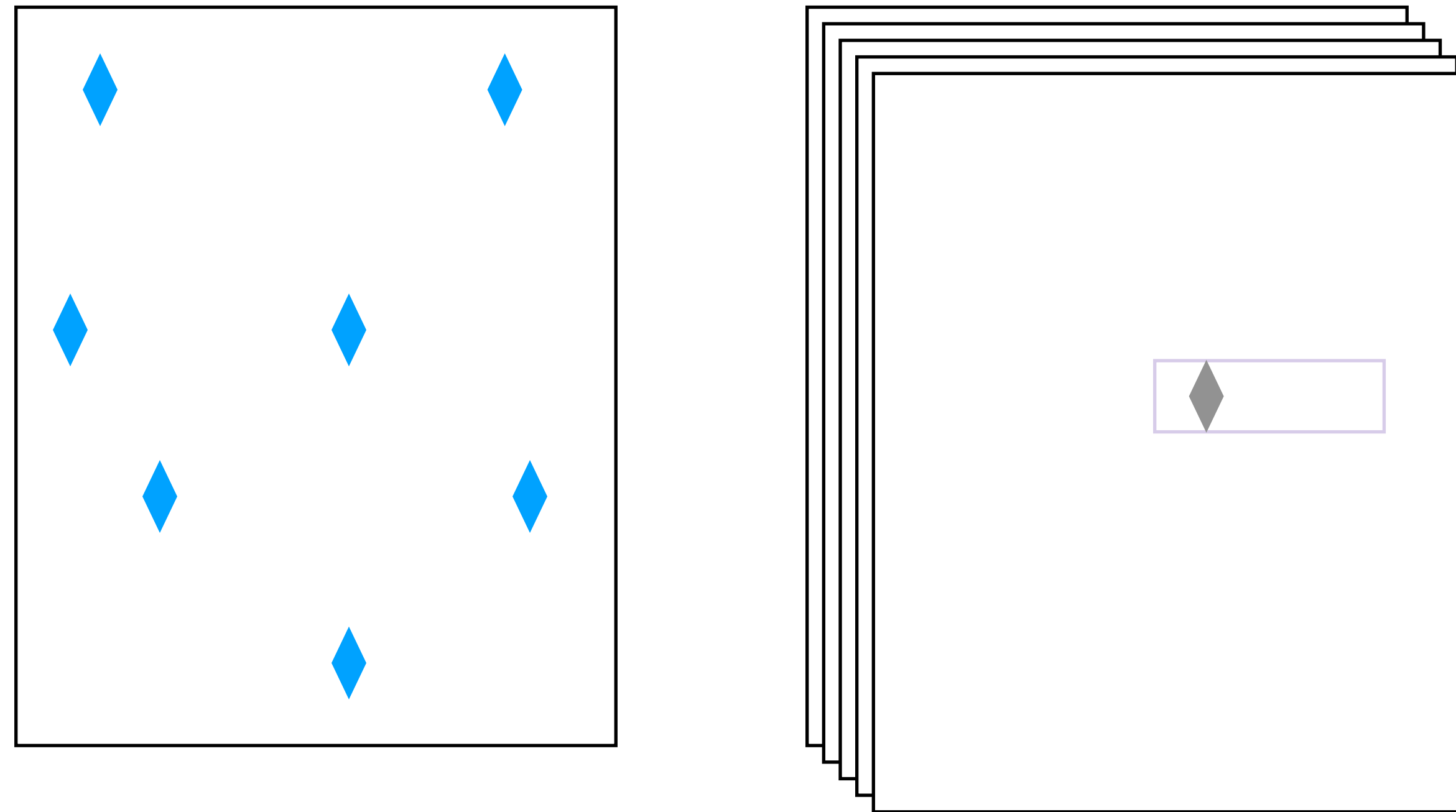
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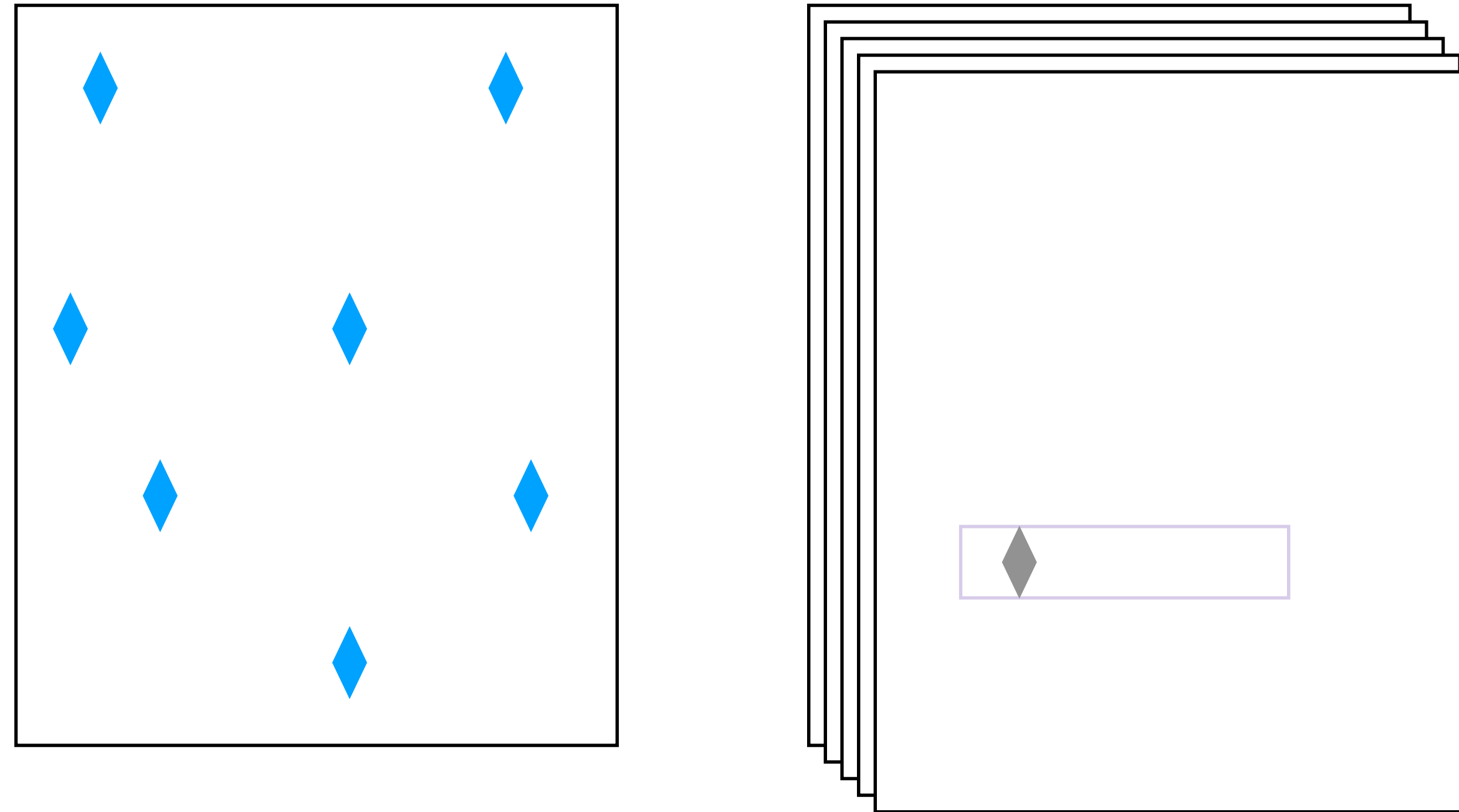
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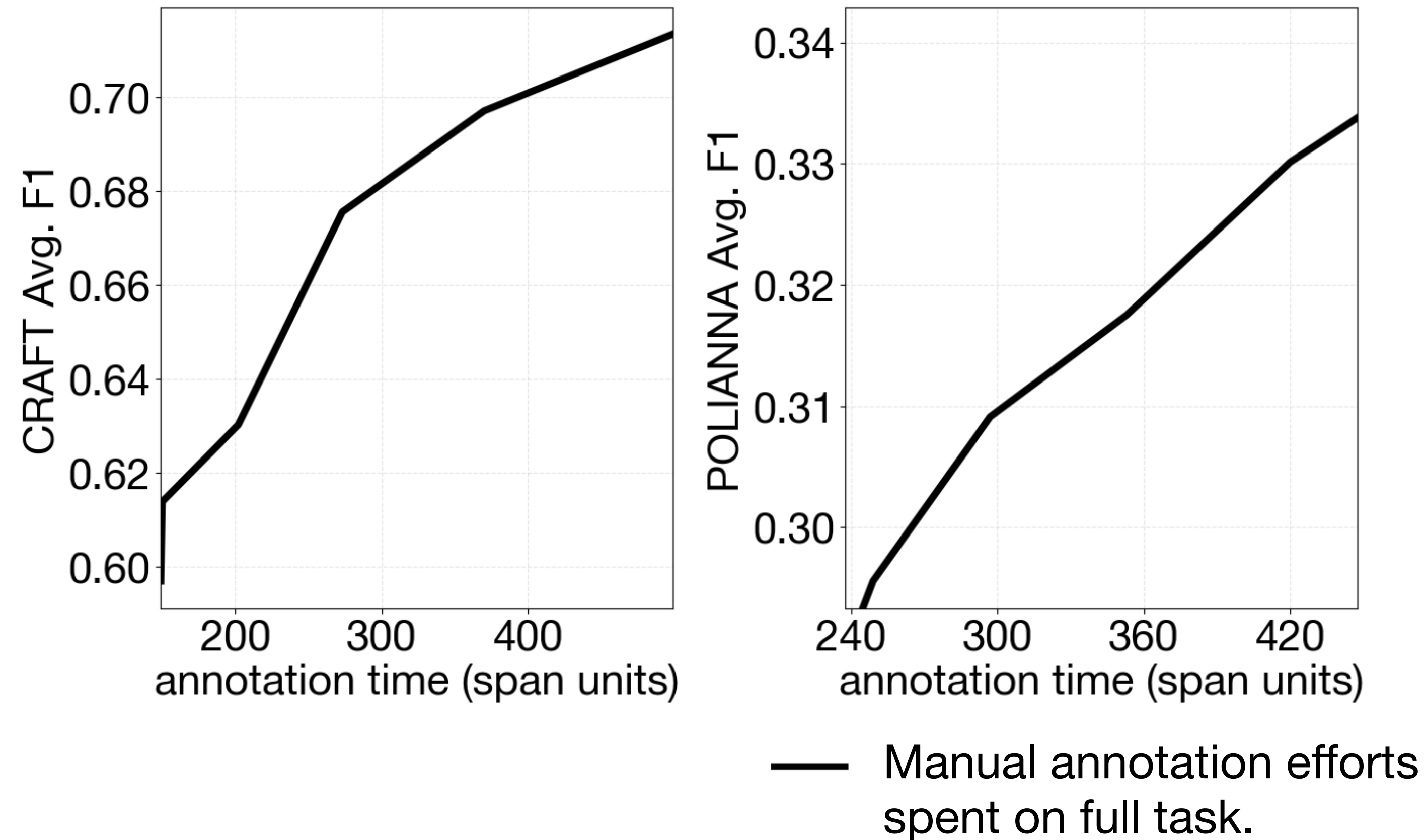
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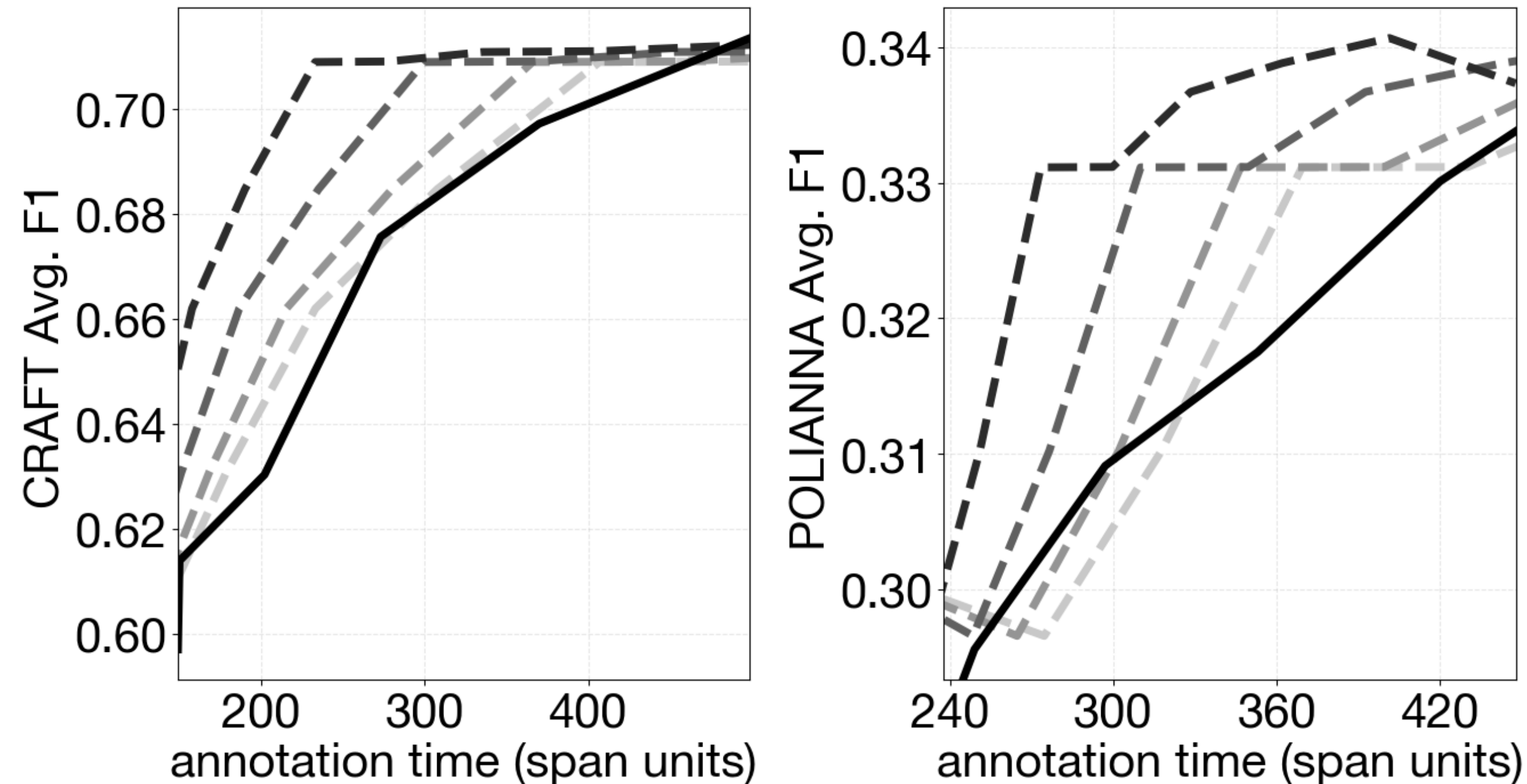
Example: Decomposition for Span-bound Concept Annotation

More efficient to concentrate manual annotation effort on identifying typed positions.



Example: Decomposition for Span-bound Concept Annotation

More efficient to concentrate manual annotation effort on identifying typed positions.



--- Manual annotation efforts spent on **typed positions** or various speed-ups (1.3X, 1.5X, 2X, 3X) from skipping unitization.

— Manual annotation efforts spent on full task.

Reducing the Cost of Annotation

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- We proposed a theoretical model of inferential load, where output space is a key component.

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- We proposed a theoretical model of inferential load, where output space is a key component.
- Grounded in our model of inferential load, we characterized efficient task decompositions as those which isolate center identification and reduce the space of candidate centers.
- We demonstrated efficiency of task decomposition for coreference resolution and span-bound concept annotation.

Roadmap

- Motivating Annotation.
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- Demonstrations of procedure efficiency.
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- Other considerations beyond cost.

When to Abstain from Decomposition

$$\text{INFERENCE LOAD}(\{t_1, \dots, t_k\}) = \sum_{j=1}^k \text{CENTERS INTRODUCED}(t_j) \cdot w_{\text{center}} \\ + \text{OUTPUT SPACE}(t_j) \cdot w_o \\ + \text{INPUT SPACE}(t_j) \cdot w_i \\ + \text{MAP SPACE}(t_j) \cdot w_m$$

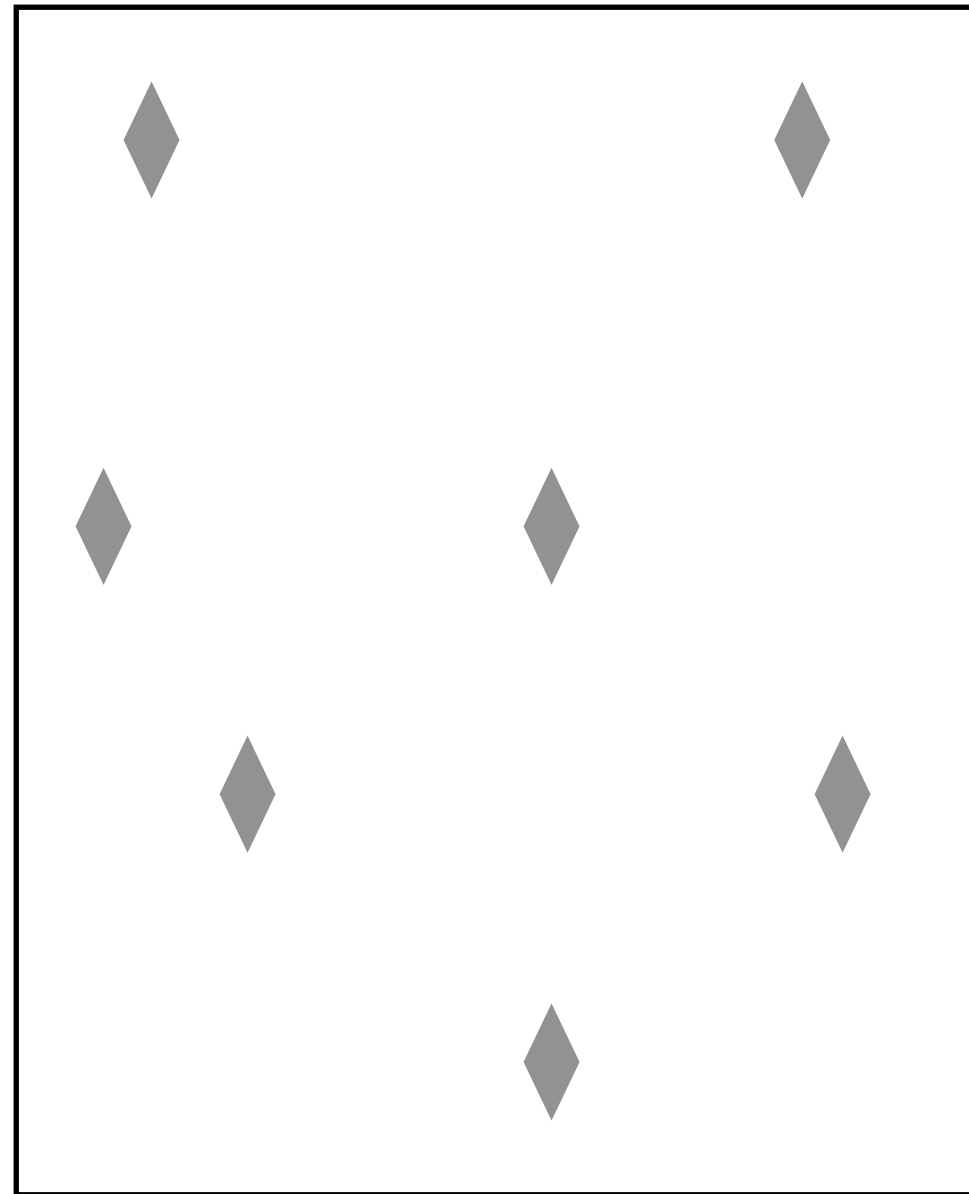
Annotator may be sensitive to the **cost of repeatedly processing information.**

Output space is less significant for **low-complexity structure**

Pipelined systems susceptible to **cascading errors.**

When to Abstain from Decomposition

Position identification t_1



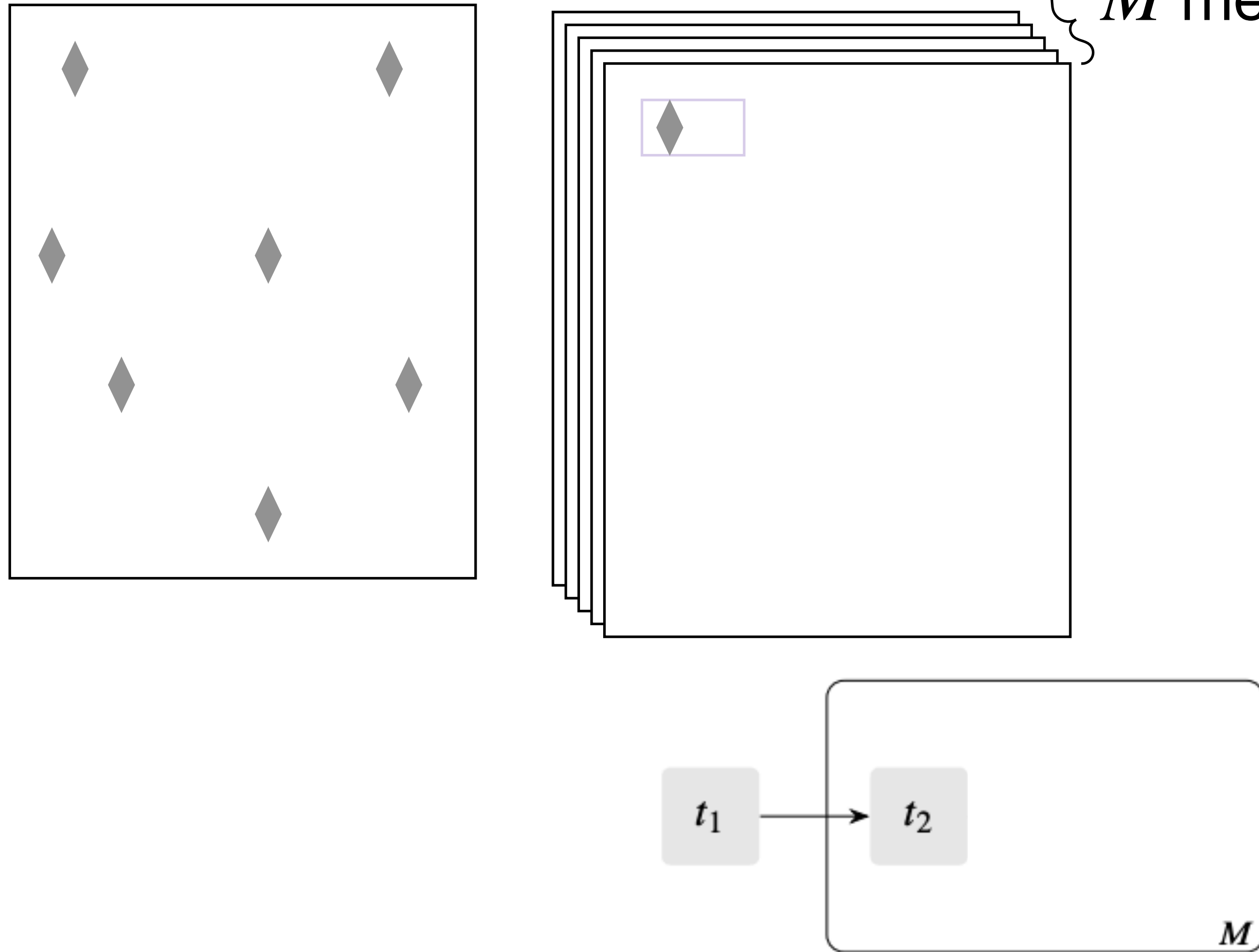
t_1

(Gandhi & Strubell, 2026)

Example of excessively decomposed NER.

When to Abstain from Decomposition

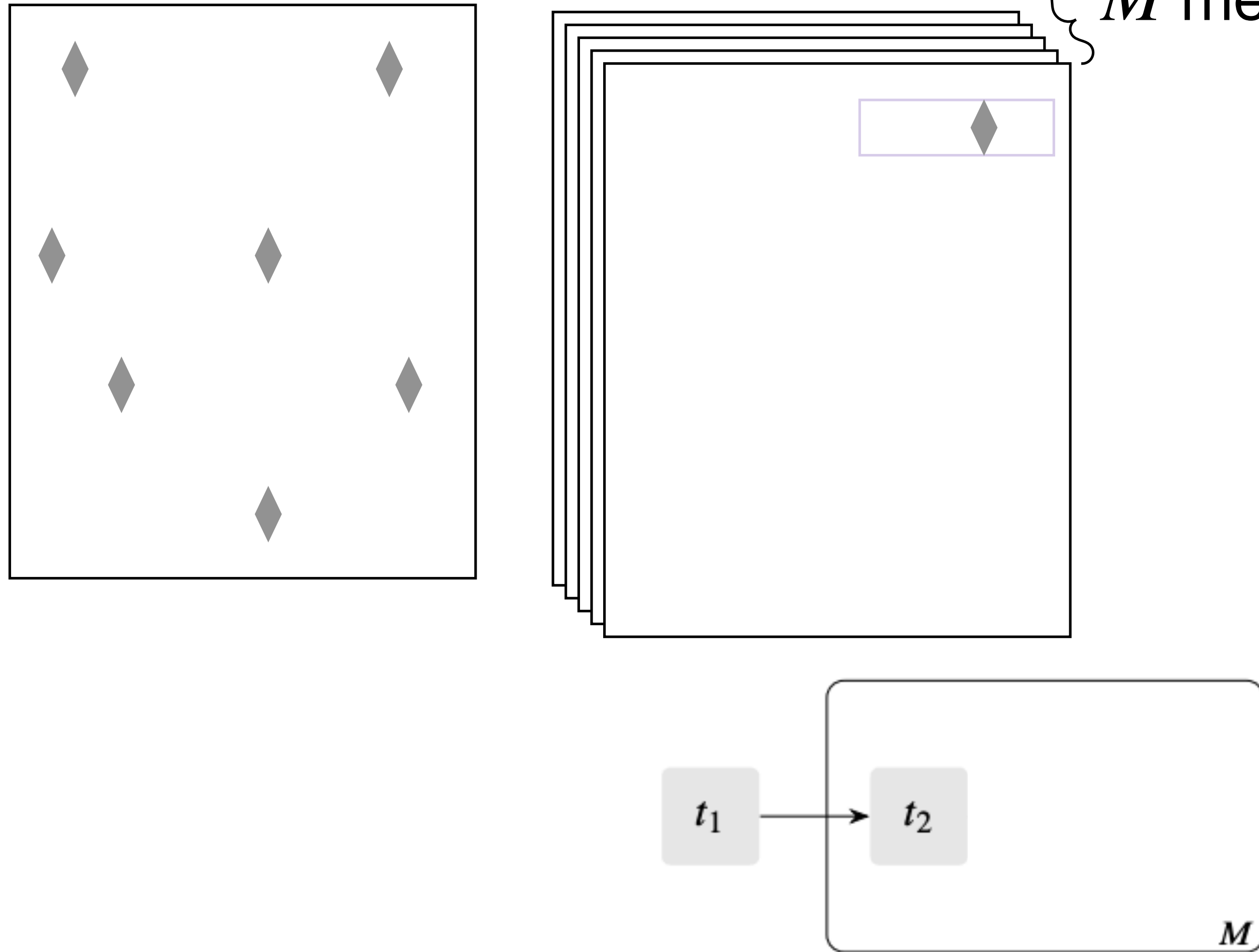
Position identification t_1 Unitization t_2 M mentions



Example of excessively decomposed NER.

When to Abstain from Decomposition

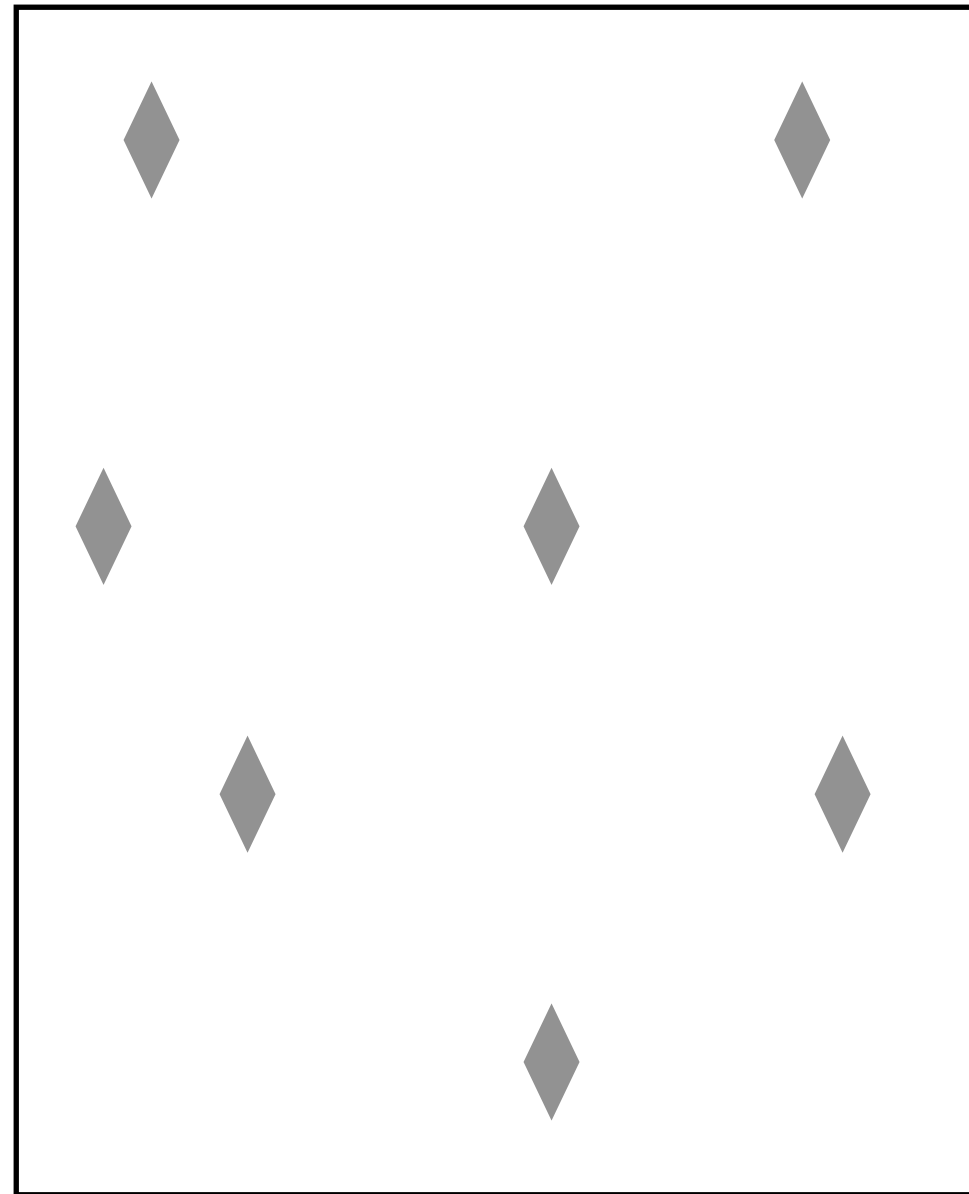
Position identification t_1 Unitization t_2 M mentions



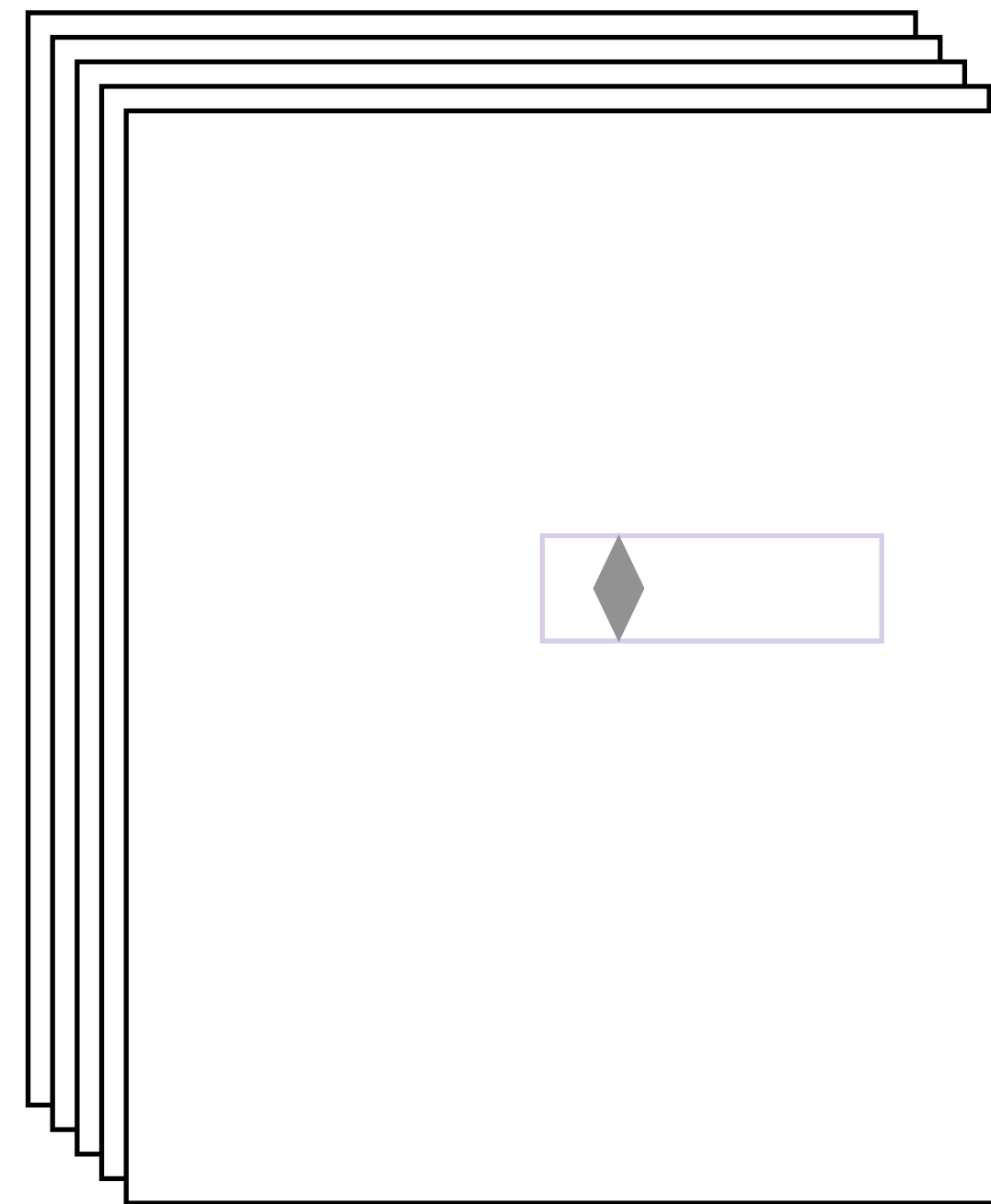
Example of excessively decomposed NER.

When to Abstain from Decomposition

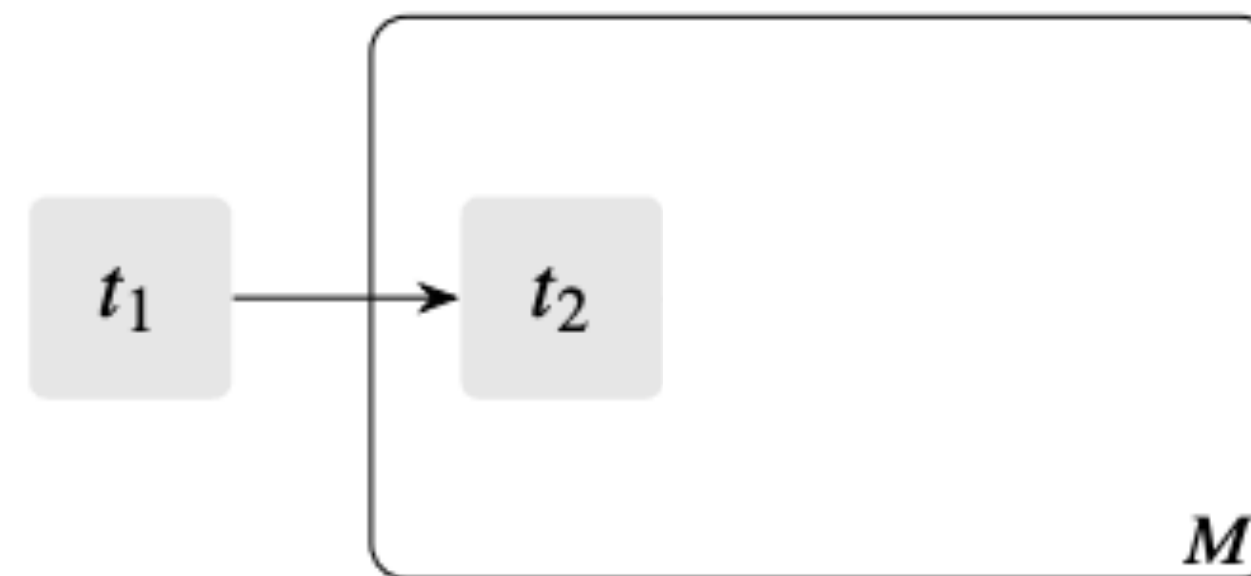
Position identification t_1



Unitization t_2

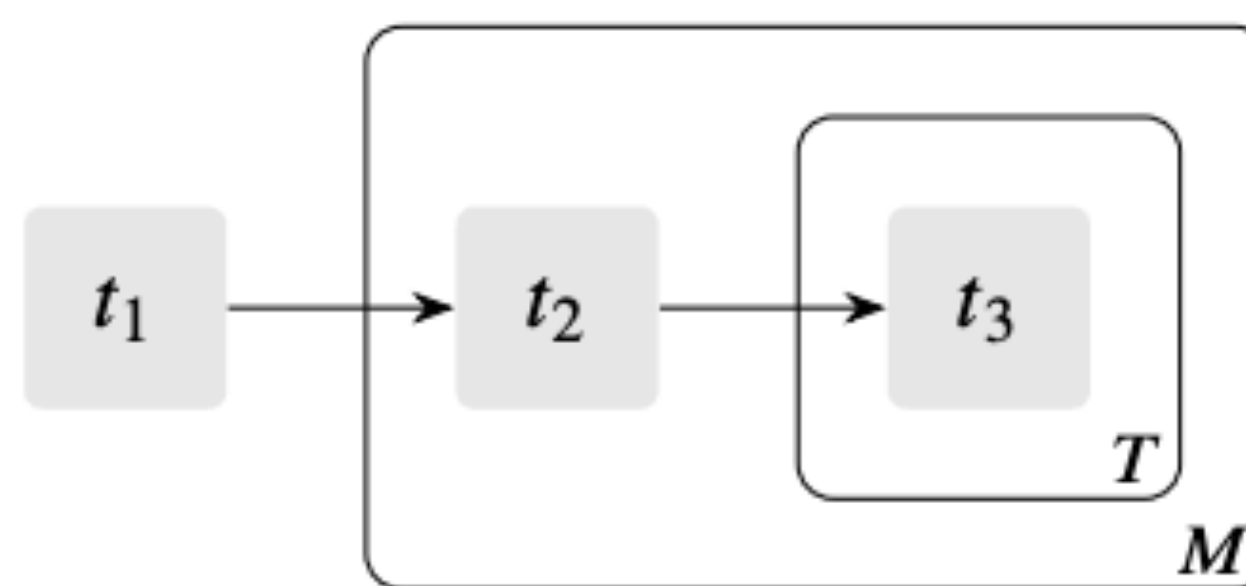
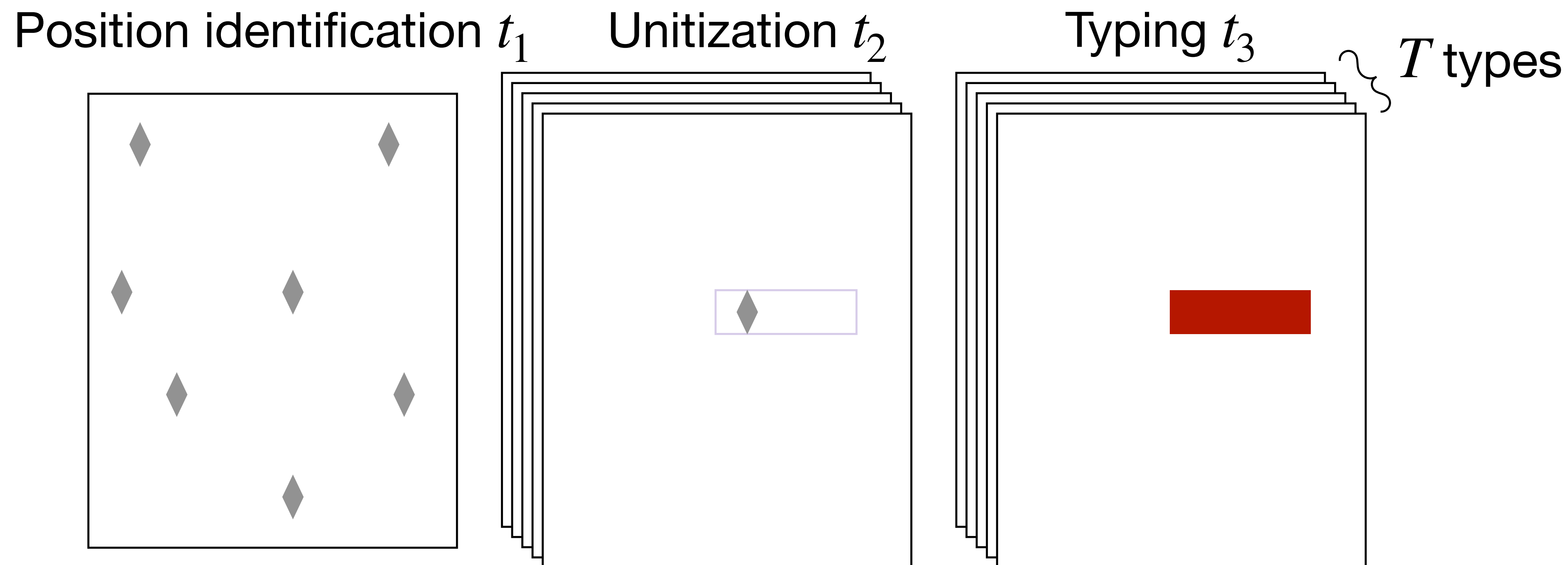


M mentions



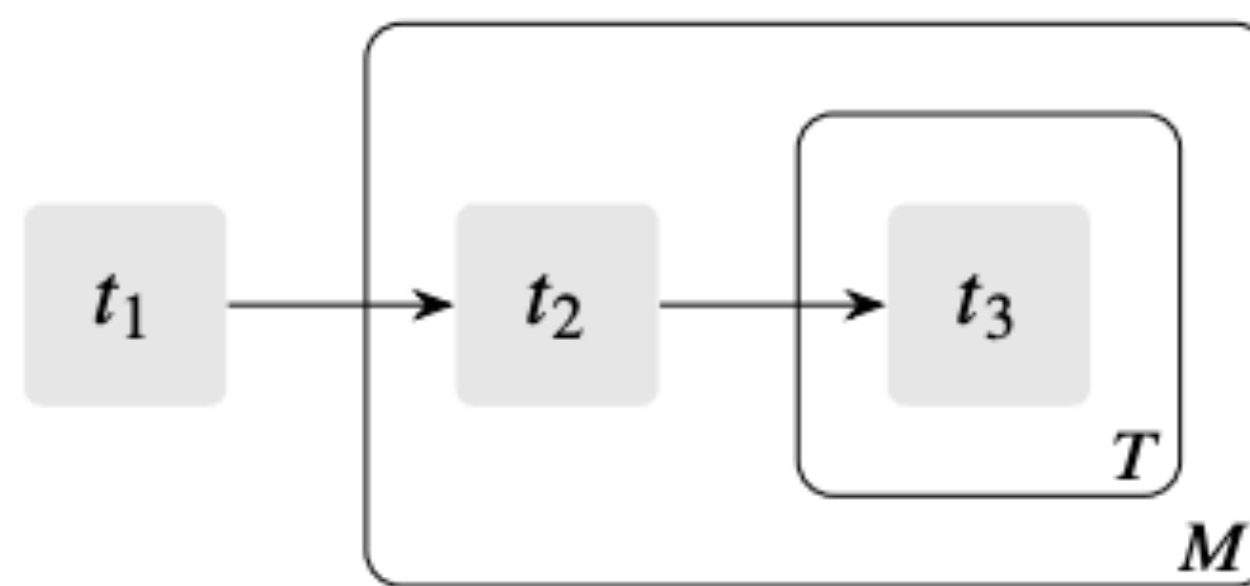
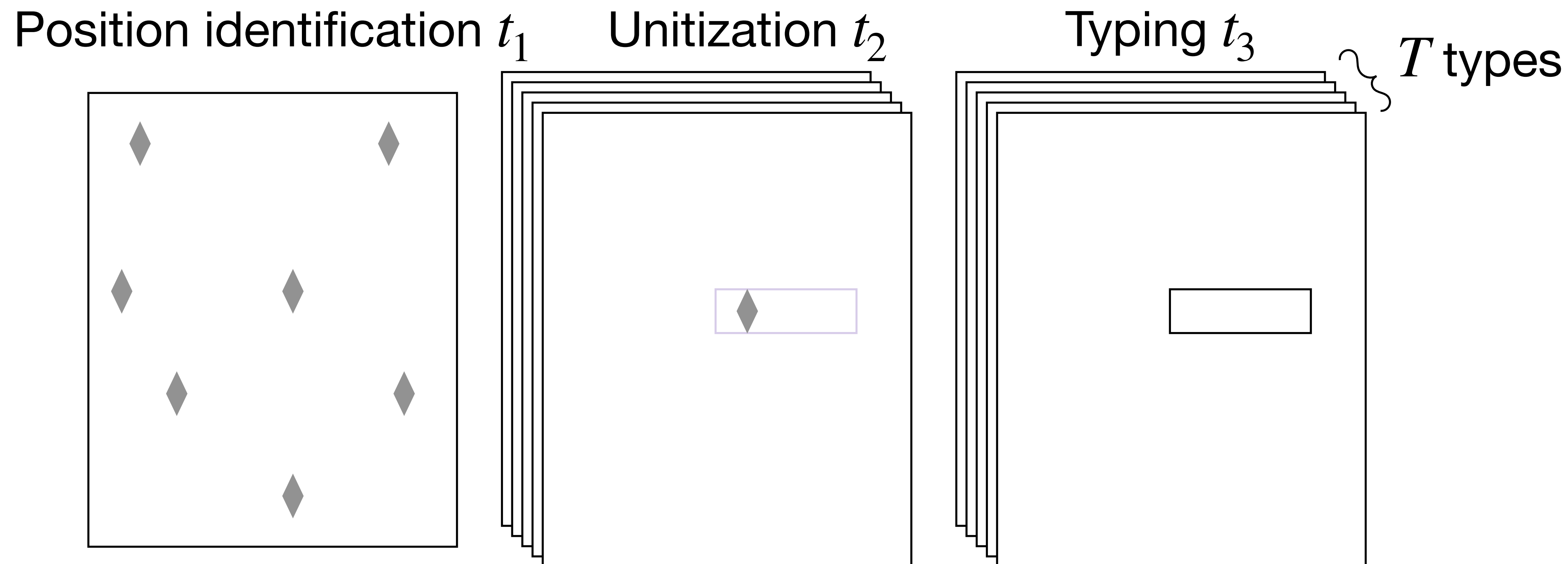
Example of excessively decomposed NER.

When to Abstain from Decomposition



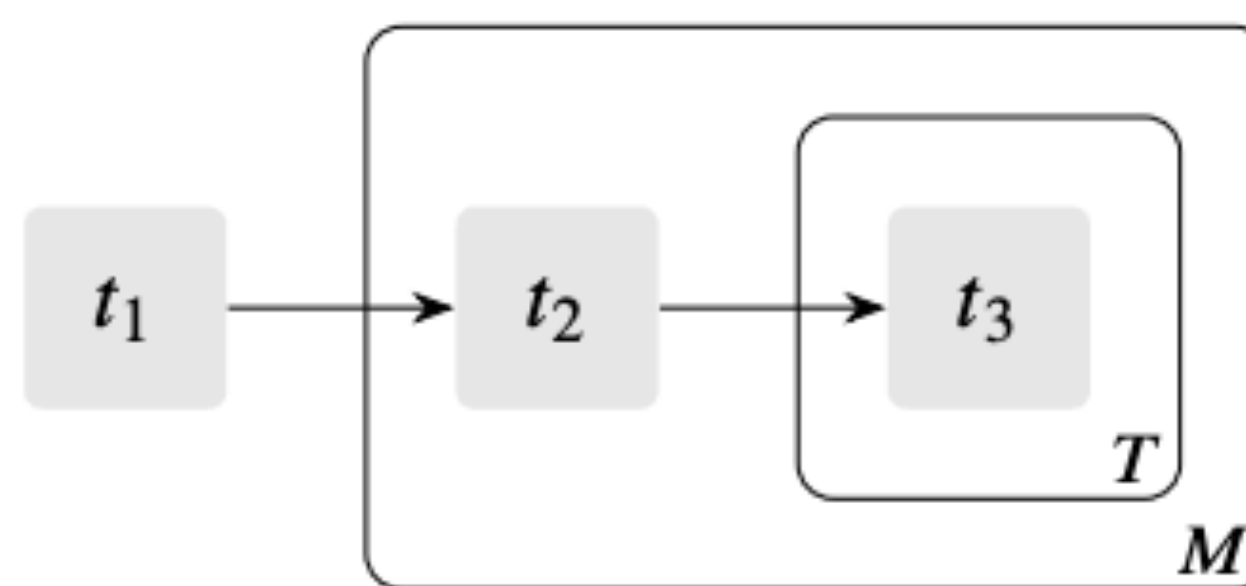
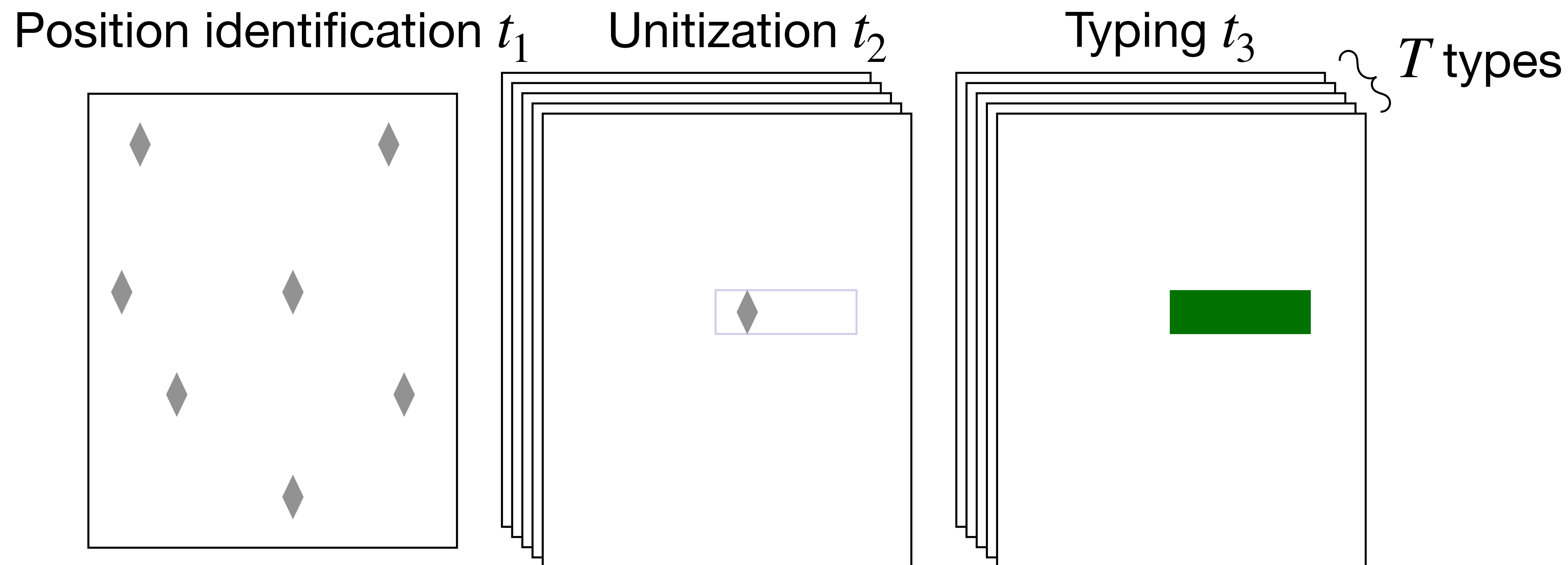
Example of excessively decomposed NER.

When to Abstain from Decomposition



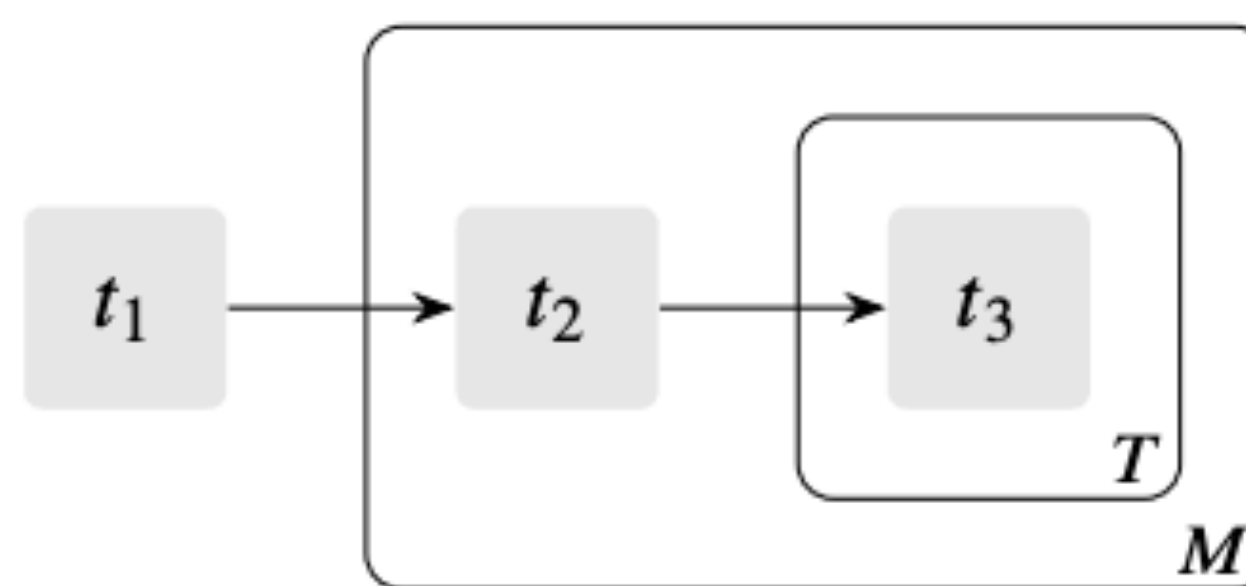
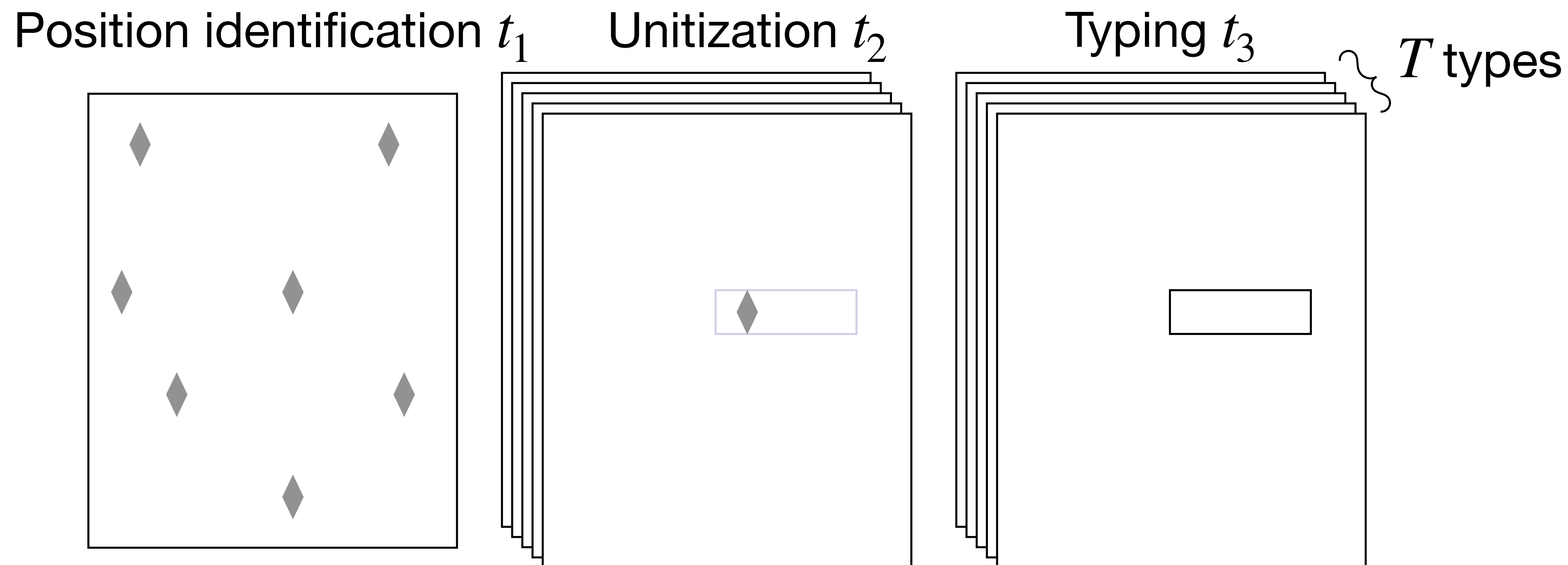
Example of excessively decomposed NER.

When to Abstain from Decomposition



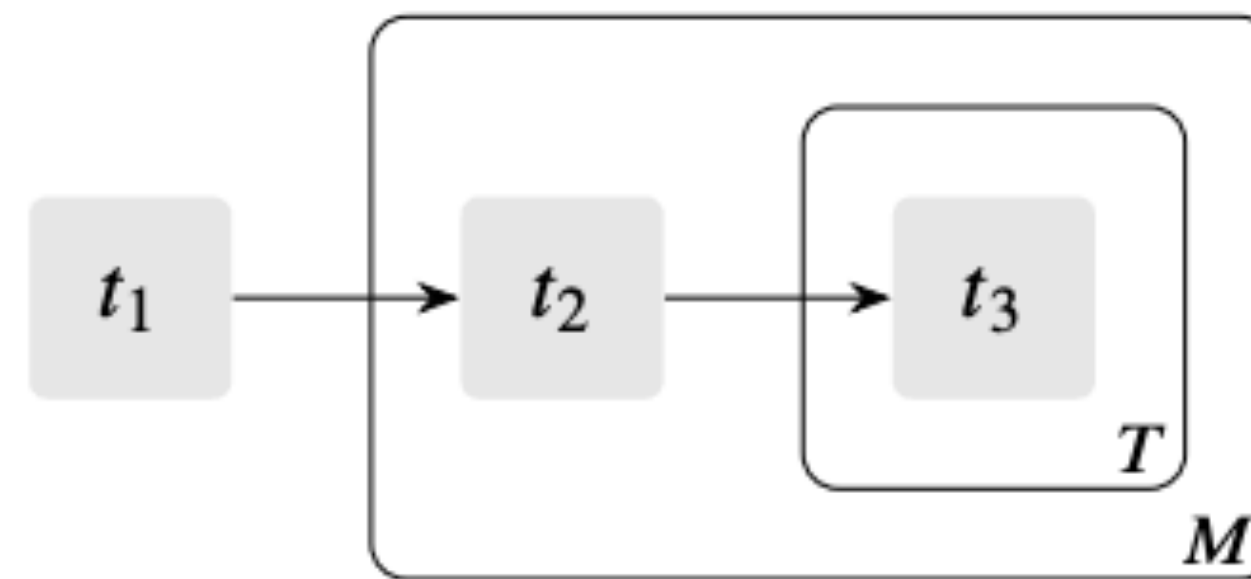
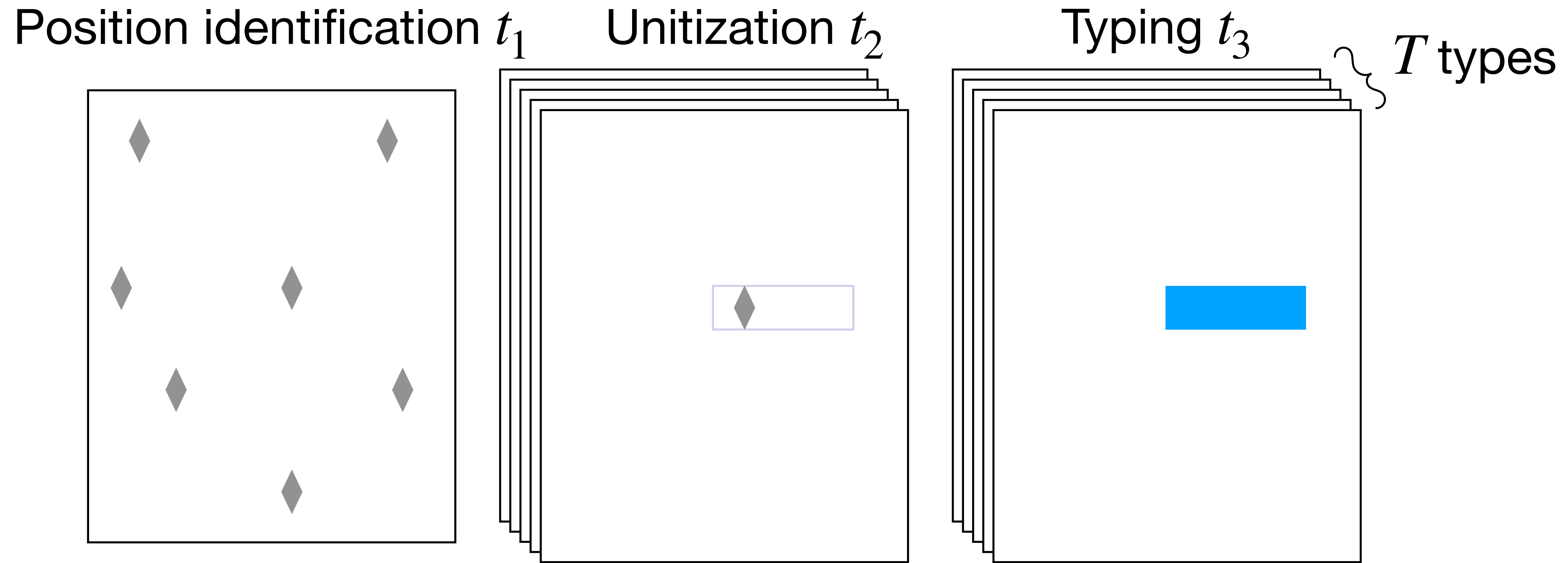
Example of excessively decomposed NER.

When to Abstain from Decomposition



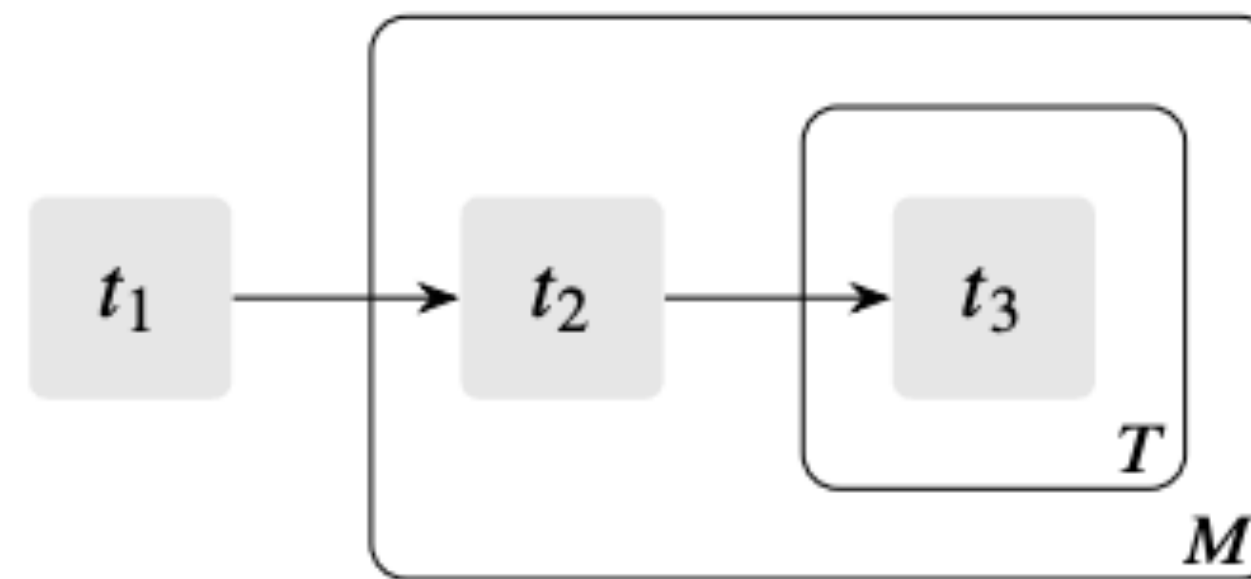
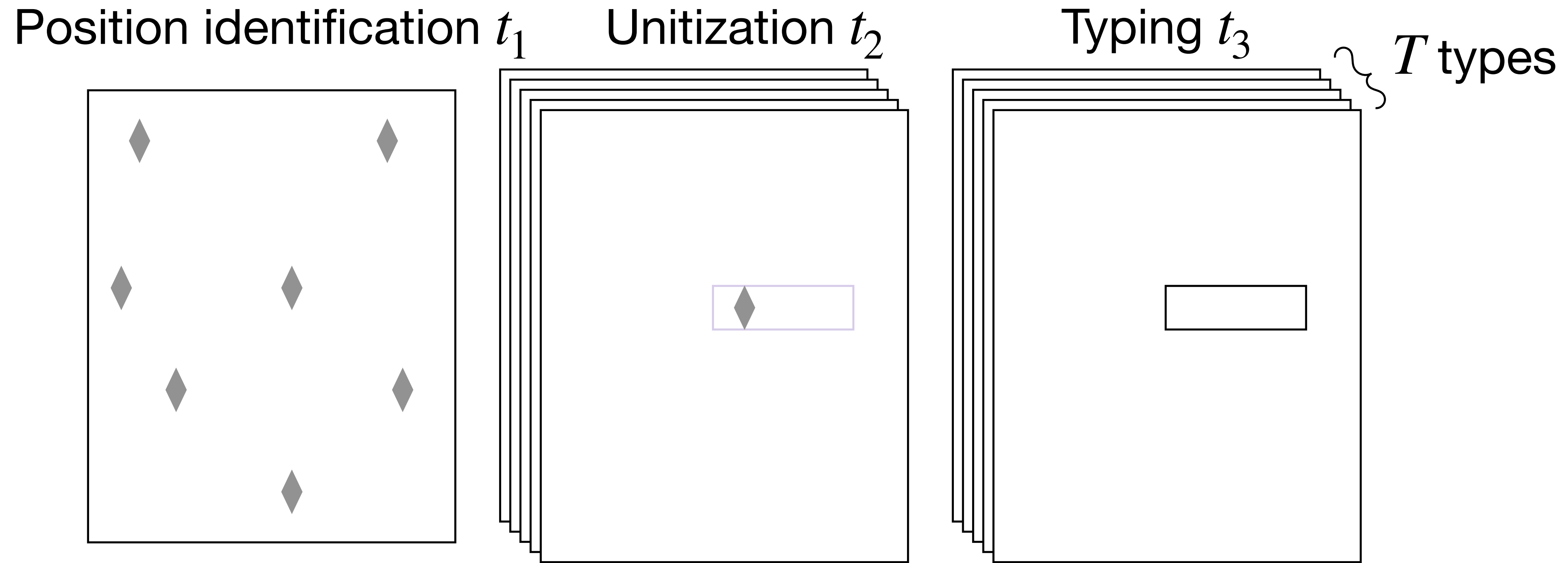
Example of excessively decomposed NER.

When to Abstain from Decomposition



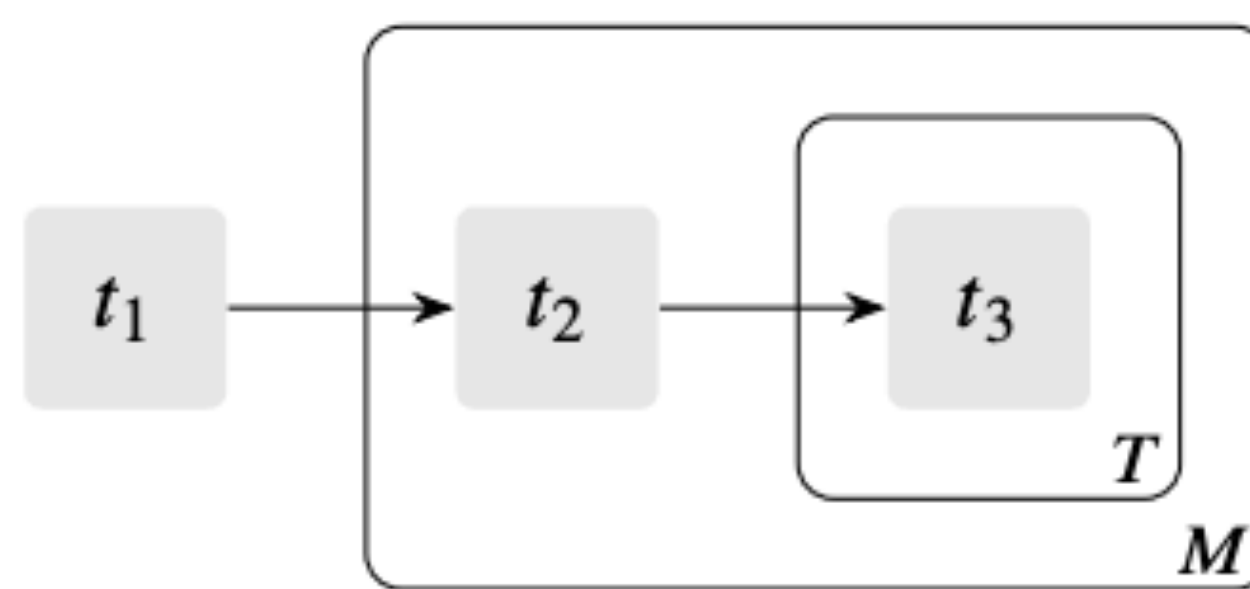
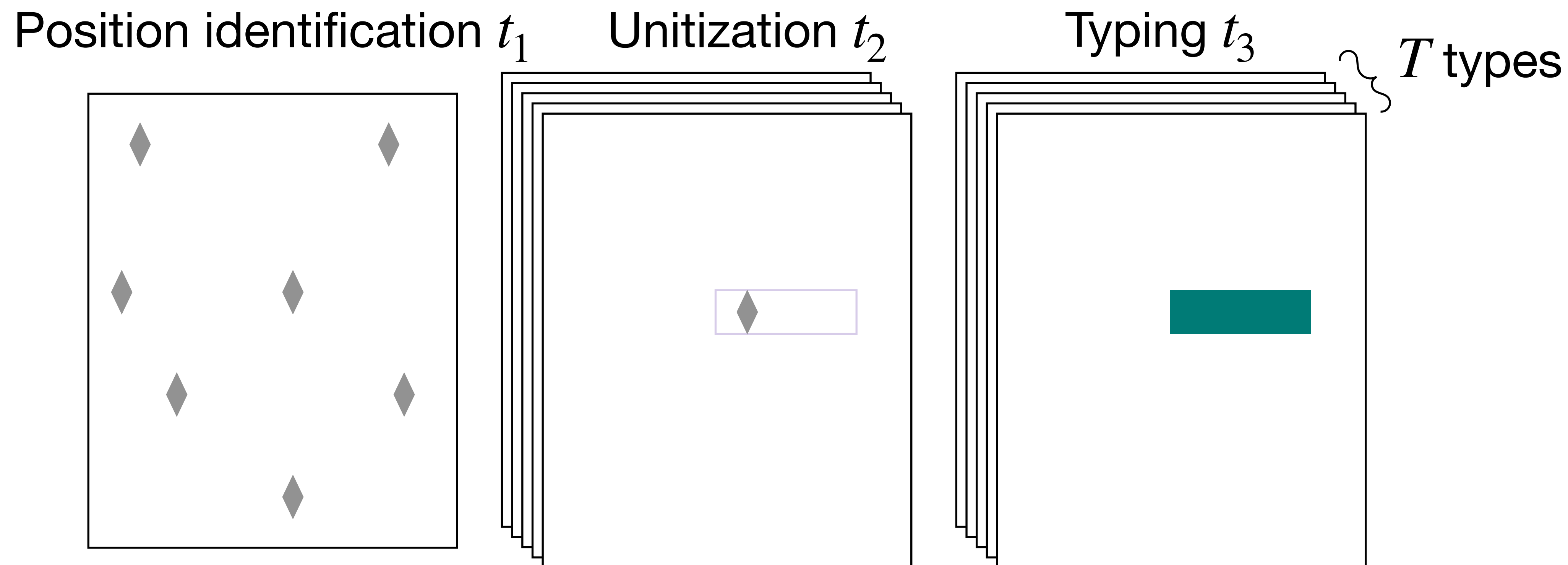
Example of excessively decomposed NER.

When to Abstain from Decomposition



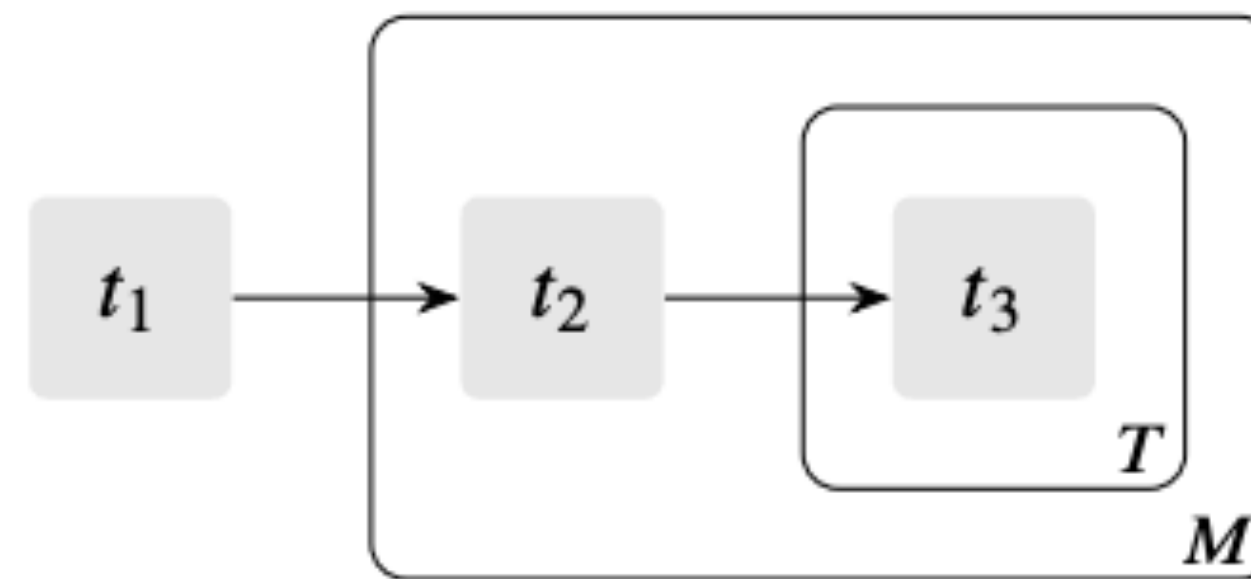
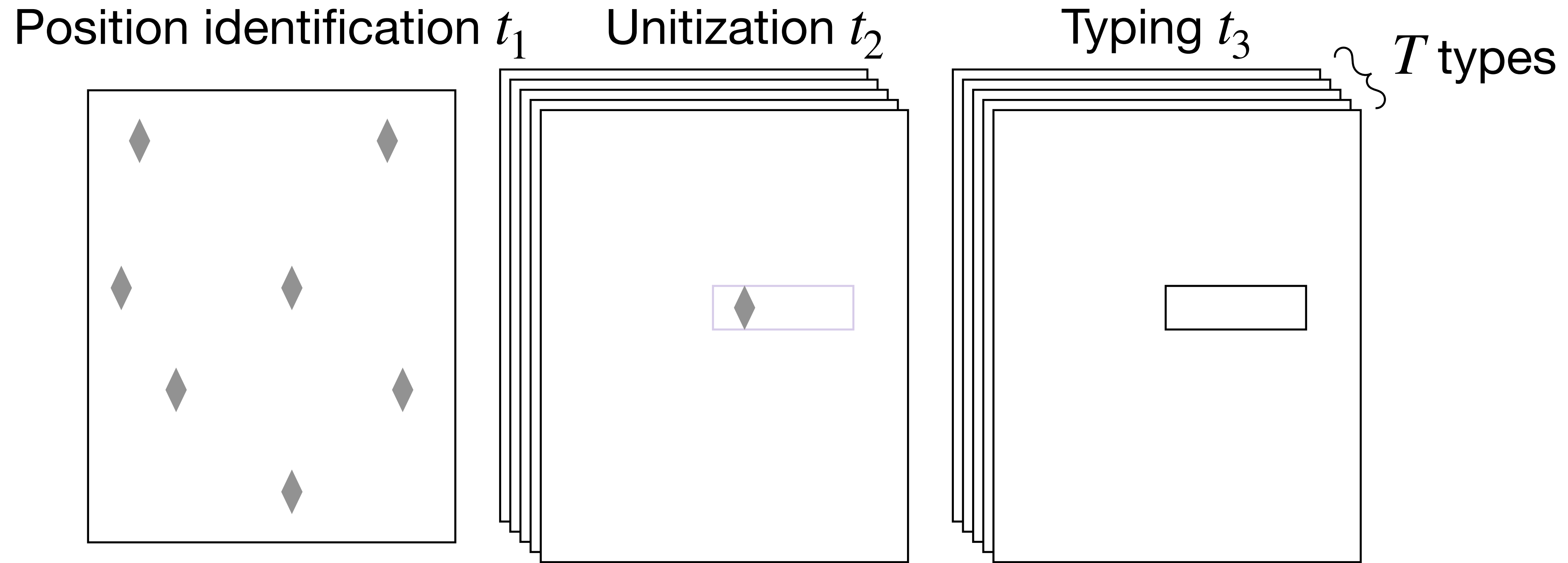
Example of excessively decomposed NER.

When to Abstain from Decomposition



Example of excessively decomposed NER.

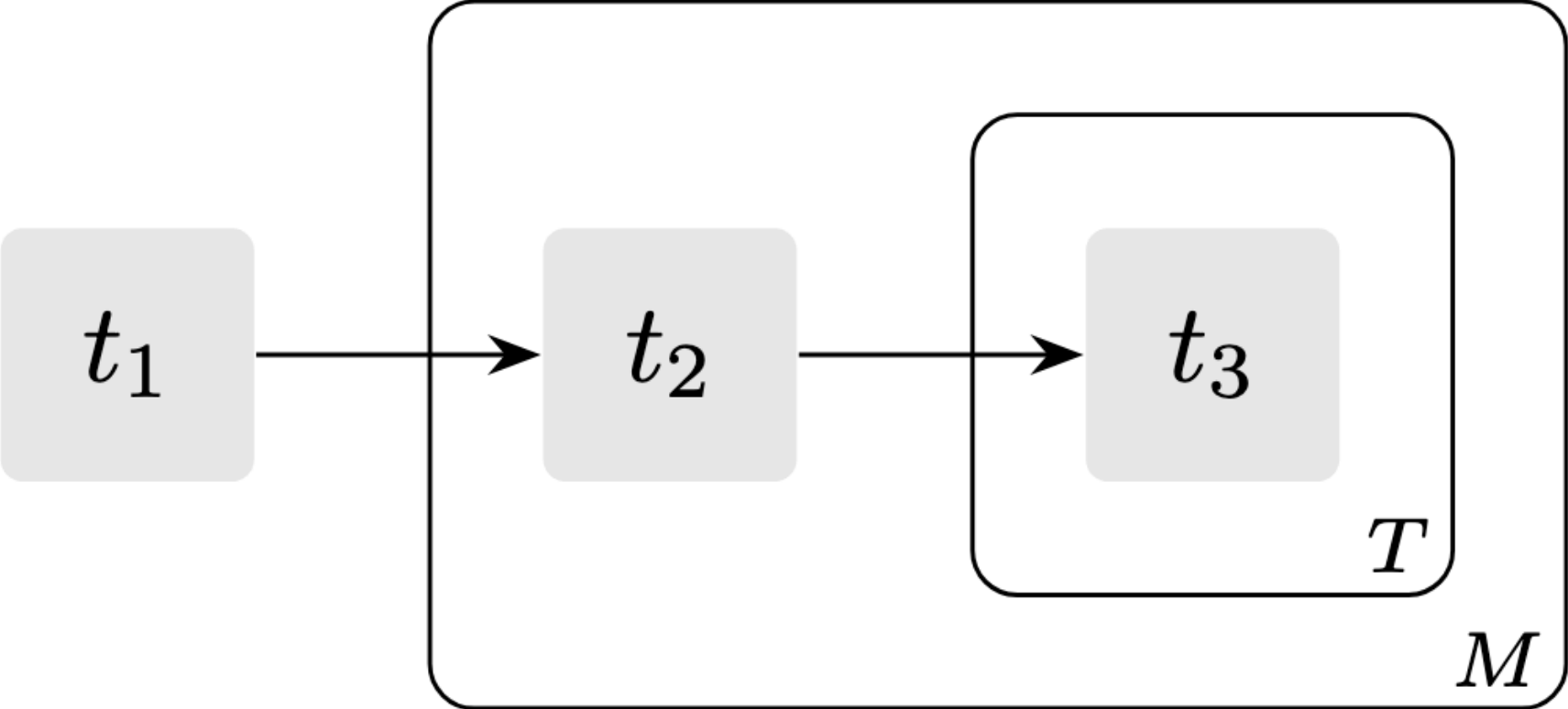
When to Abstain from Decomposition



Example of excessively decomposed NER.

When to Abstain from Decomposition

Example of excessively decomposed NER



Task	Description	Output	Input	Map	Centers	Instances
t	Identify named entity spans	$(T + 1)^S$	N	T	M	1

Space of all possible spans

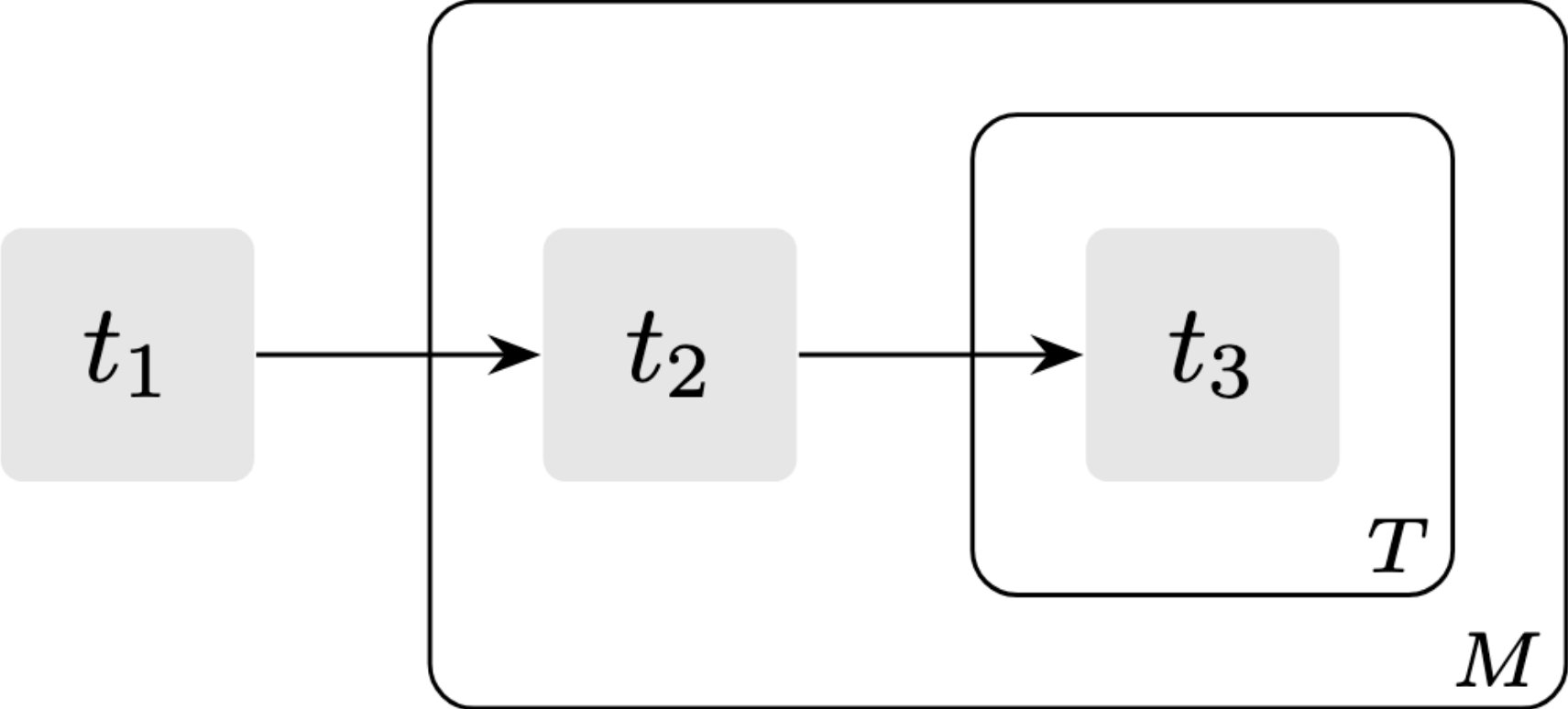
tokens

of types

of mentions

When to Abstain from Decomposition

Example of excessively decomposed NER

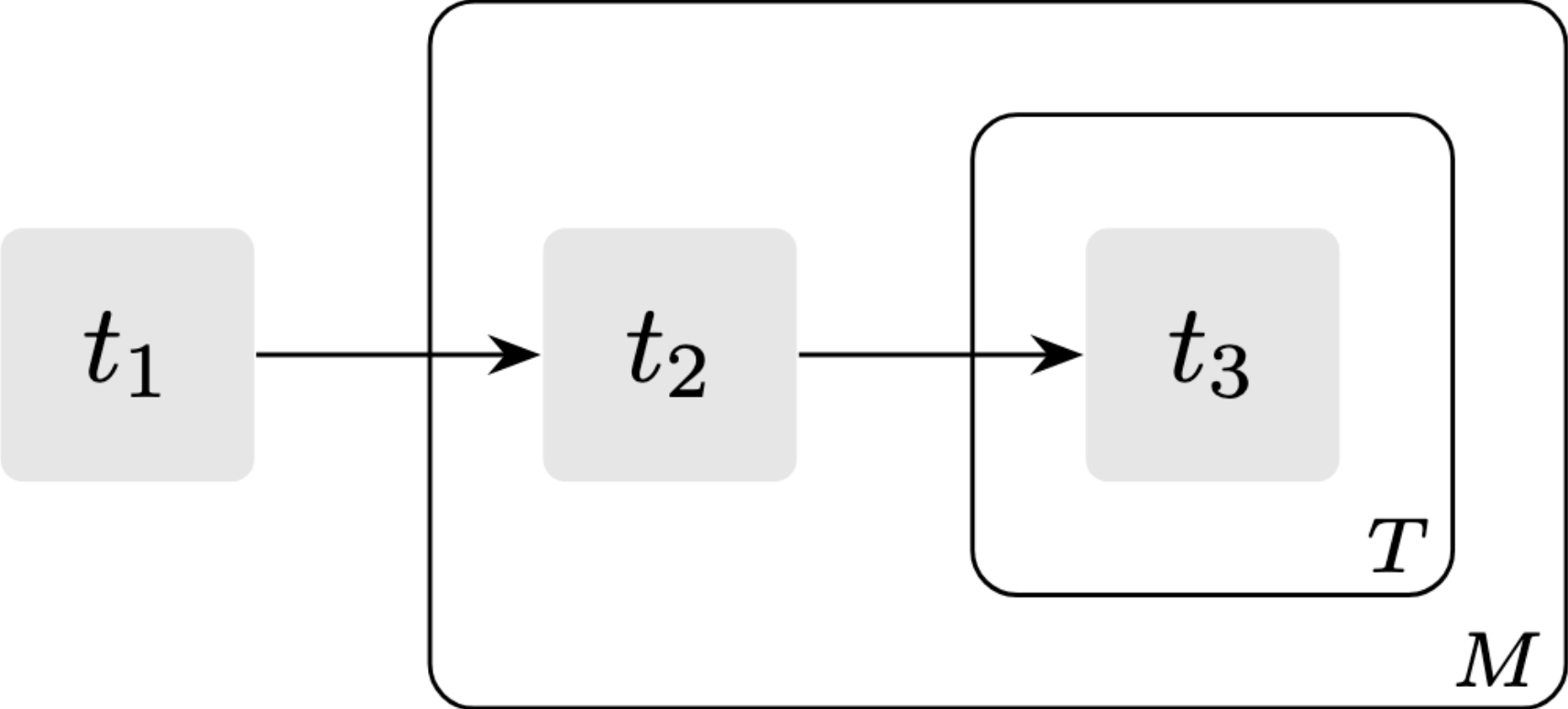


Task	Description	Output	Input	Map	Centers	Instances
t	Identify named entity spans	$(T + 1)^S$	N	T	M	1
t_1	Identify positions	2^C	C	0	M	1

characters

When to Abstain from Decomposition

Example of excessively decomposed NER



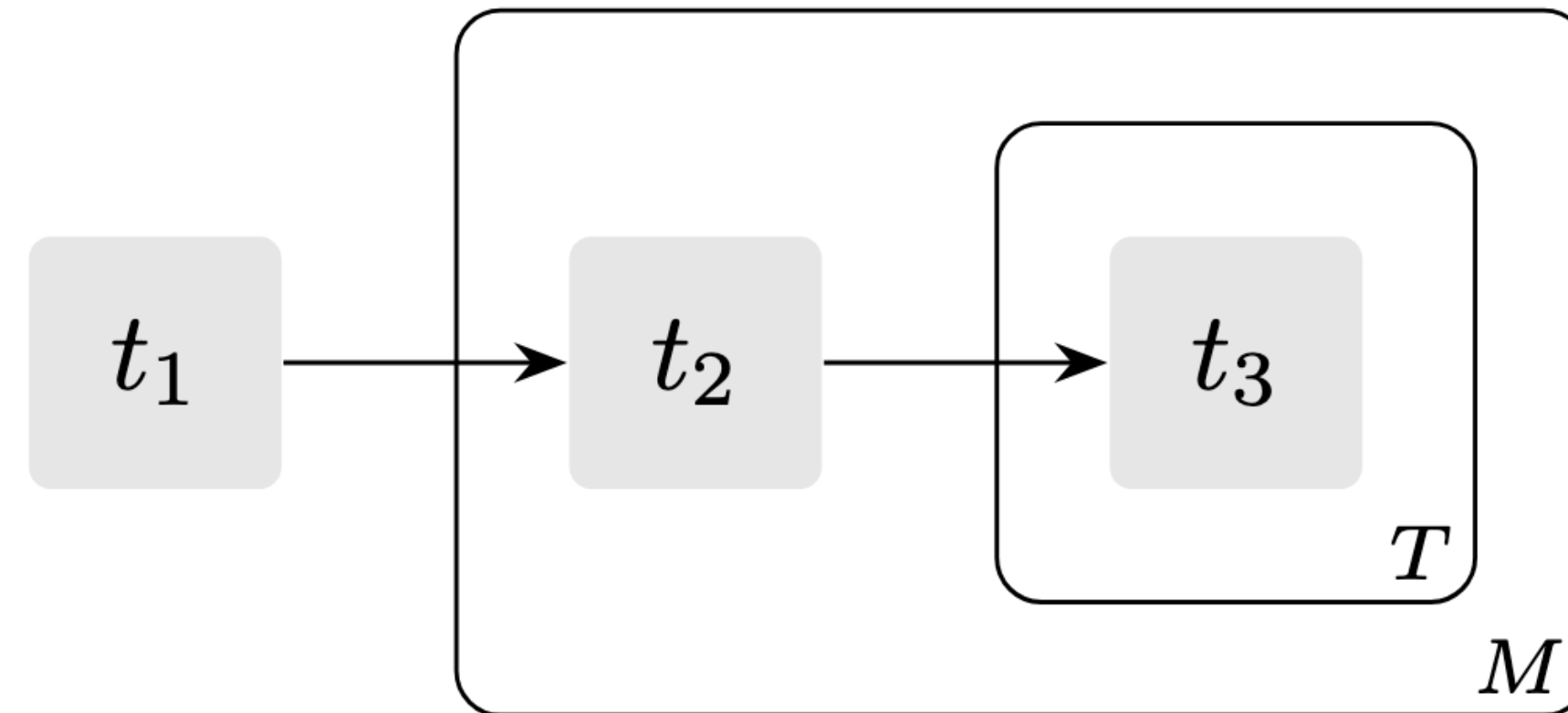
Task	Description	Output	Input	Map	Centers	Instances
t	Identify named entity spans	$(T + 1)^S$	N	T	M	1
t_1	Identify positions	2^C	C	0	M	1
t_2	Unitize position	$\binom{d+1}{2}$	$2d$	0	0	M

max span width

↖ ↗

When to Abstain from Decomposition

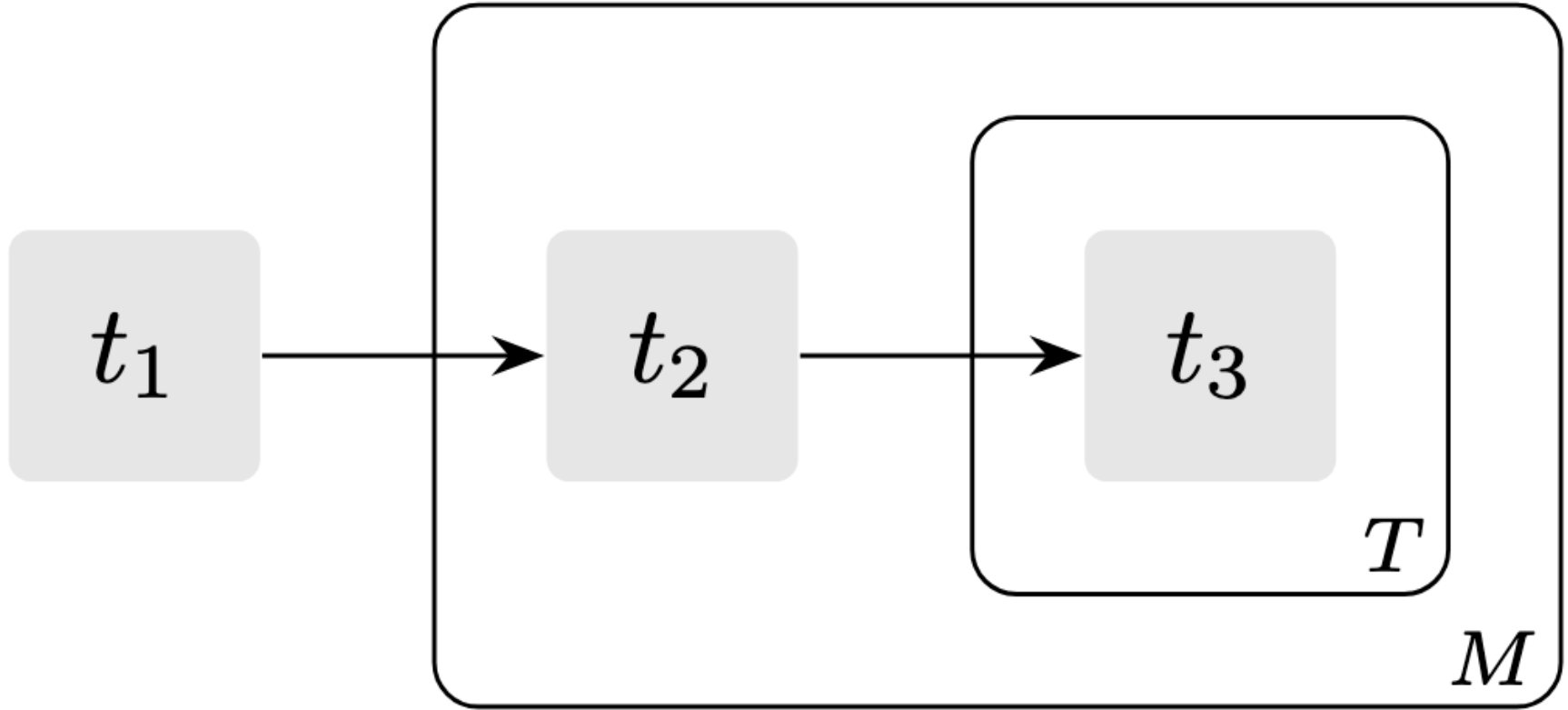
Example of excessively decomposed NER



Task	Description	Output	Input	Map	Centers	Instances
t	Identify named entity spans	$(T + 1)^S$	N	T	M	1
t_1	Identify positions	2^C	C	0	M	1
t_2	Unitize position	$\binom{d+1}{2}$	$2d$	0	0	M
t_3	Binary position typing	2	1	2	0	$M \cdot T$

When to Abstain from Decomposition

Example of excessively decomposed NER



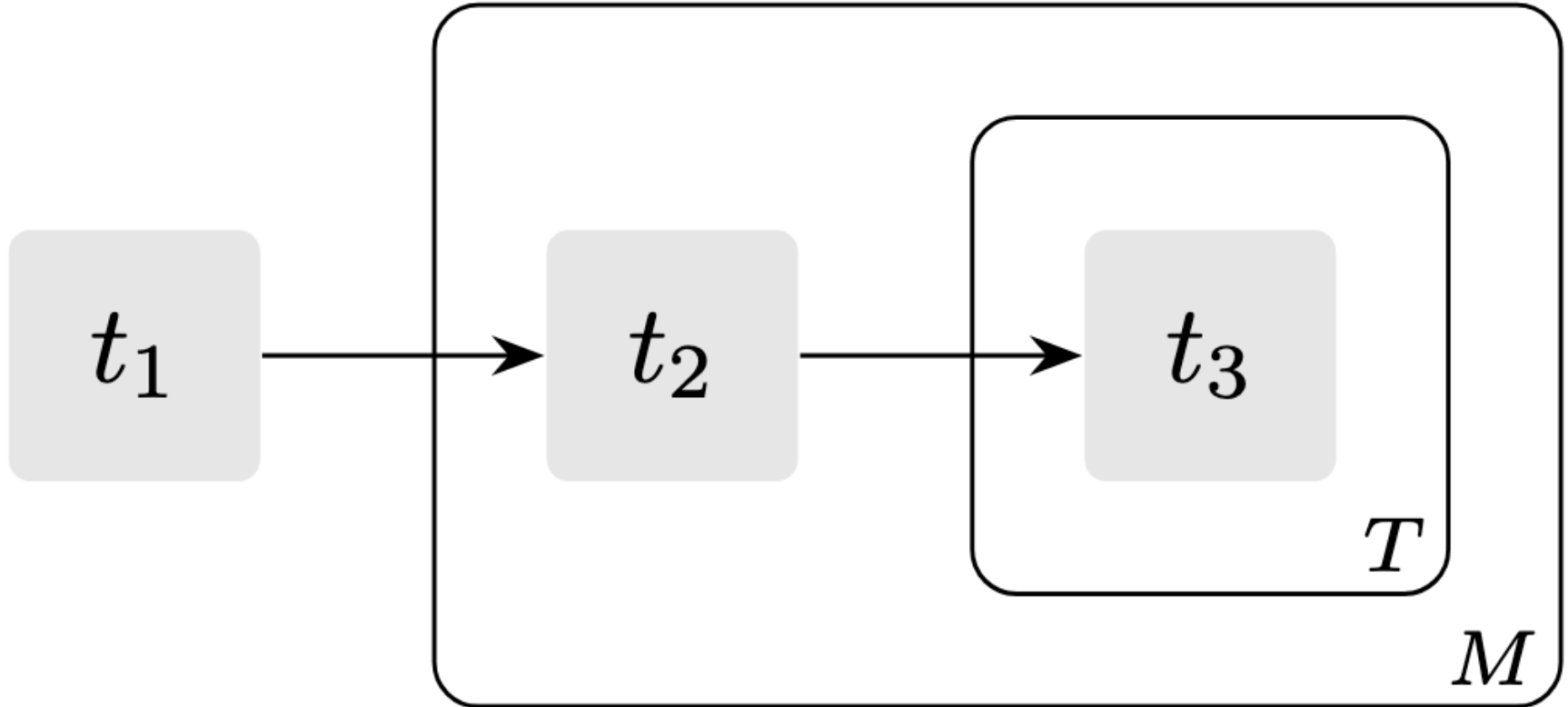
Input space larger under decomposition

Task	Description	Output	Input	Map	Centers	Instances
t	Identify named entity spans	$(T + 1)^S$	N	T	M	1
t_1	Identify positions	2^C	C	0	M	1
t_2	Unitize position	$\binom{d+1}{2}$	$2d$	0	0	M
t_3	Binary position typing	2	1	2	0	$M \cdot T$

(Gandhi & Strubell, 2026)

When to Abstain from Decomposition

Example of excessively decomposed NER



Map space larger under decomposition

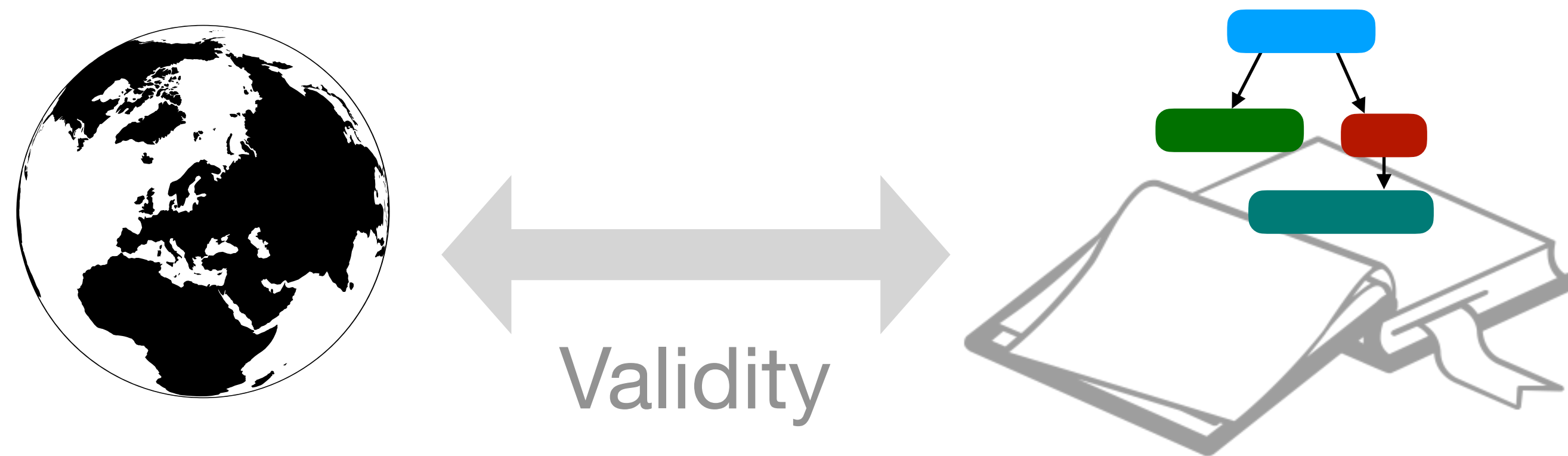
Task	Description	Output	Input	Map	Centers	Instances
t	Identify named entity spans	$(T + 1)^S$	N	T	M	1
t_1	Identify positions	2^C	C	0	M	1
t_2	Unitize position	$\binom{d+1}{2}$	$2d$	0	0	M
t_3	Binary position typing	2	1	2	0	$M \cdot T$

Roadmap

- Motivating Annotation.
- Decomposition-based procedure for reducing the cost of annotation grounded in a model of inferential load.
- Grounding procedure in the literature.
- Demonstrations of procedure efficiency.
- Deciding when to decompose.
- **Other considerations beyond cost.**

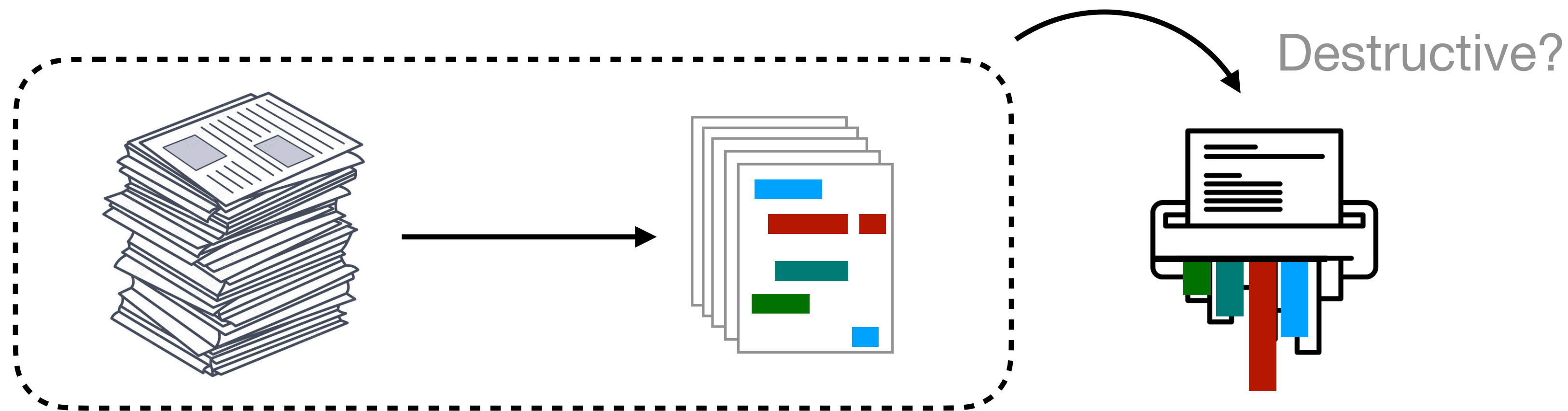
Other Considerations beyond Cost

- Not conventional to assess validity in addition to reliability.



Other Considerations beyond Cost

- Not conventional to assess validity in addition to reliability.
- With respect to structured representations, annotation is inherently reductive.
 - Necessarily discards important pieces of text or the body of text as a whole.



Thank you